

# UNIQUENESS OF GERVER'S SOFA

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**ABSTRACT.** The moving sofa problem asks for the largest area of a closed connected planar set that can be moved around the right-angled corner of a hallway of unit width. Gerver found a set of area  $2.2195\dots$  in 1992 and conjectured that it is optimal; Baek proved this in 2024. We prove that Gerver's sofa is the only optimal set: every moving sofa of the same area is the image of Gerver's sofa under a rotation and a translation, as a set. Baek's proof studies one particular maximizer, produced by a compactness argument: its sofa, after a rotation, turns through a right angle and then satisfies an injectivity condition. A given maximizer need not be that one. We show that every maximizer has these properties, by approximating a maximizing convex cap by polygonal maximizers of a penalized problem, which are almost balanced, and passing to the limit. Equality in Baek's concave upper bound, whose nonlinear terms are halves of integrals of squared lengths of tangent segments, then forces differential equations for the difference of the support functions of the cap and of Gerver's cap, and these equations identify the cap. A last step recovers the set from its cap. The proof, together with Baek's, is formalized in Lean 4 with Mathlib.

## 1. INTRODUCTION

**1.1. The moving sofa problem.** Let  $L$  be the L-shaped hallway of unit width, the union of the horizontal side  $H_L = (-\infty, 1] \times [0, 1]$  and the vertical side  $V_L = [0, 1] \times (-\infty, 1]$ . A *moving sofa* is a closed connected planar set that can be moved rigidly from the horizontal side to the vertical side without leaving  $L$  (Definition 2.2). Moser [8] asked for the largest area of a moving sofa. The unit square is a moving sofa of area 1. Hammersley [5] found a sofa of area  $\pi/2 + 2/\pi = 2.2074\dots$ , and Gerver [3] one of area  $2.21953\dots$ , bounded by 18 analytic arcs and segments, which he conjectured to be optimal. Romik [10] derived Gerver's sofa  $G$  from differential equations that express the local optimality of its sides, and described its boundary explicitly in terms of two angles that solve transcendental equations (Figure 1, Appendix A). Kallus and Romik [6] proved by a computer-assisted method that no sofa has area more than 2.37. Baek [1] proved Gerver's conjecture in a paper of 119 pages: every moving sofa has area at most  $|G| = 2.2195\dots$

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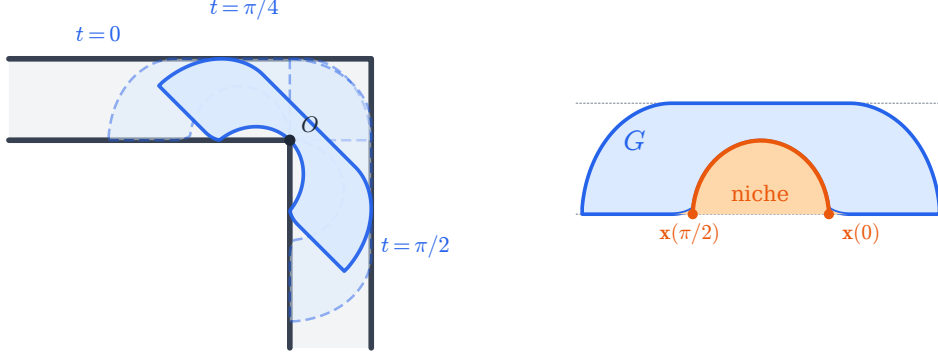


FIGURE 1. Gerver's sofa  $G$  in the hallway  $L$ . Left: it slides along the horizontal side (dashed), turns through a right angle around the inner corner (solid, halfway, turned by  $\pi/4$ ) and leaves along the vertical side (dashed). Right:  $G$  with its niche, the hollow in its lower side that makes room for the corner; the path  $\mathbf{x}$  of the inner corner of the hallway, as seen from the sofa, is drawn in orange.

**1.2. Uniqueness.** We prove that Gerver's sofa is the only optimal sofa, up to a rotation and a translation.

**Theorem 1.1** (uniqueness of Gerver's sofa). *Let  $S$  be a moving sofa with  $|S| = |G|$ . Then there are an angle  $\theta$  and a vector  $v$  such that*

$$R_\theta S + v = G.$$

Equivalently, a moving sofa has maximum area if and only if it is a rotated and translated copy of  $G$  (Corollary 9.4).

The theorem gives an equality of sets. This uses that a moving sofa is closed and that  $G$  is the closure of its interior (Section 9); equality of areas alone would determine a set only up to a null set.

The rotation in Theorem 1.1 can be taken to be trivial. The width of  $G$  exceeds one in every direction other than the vertical (Lemma 9.5), so a copy  $R_\theta G$  is a moving sofa only if  $\theta$  is a multiple of  $\pi$ . The proof of Theorem 1.1 produces  $\theta \in [0, \pi/2 - \omega_0]$ , so that  $\theta = 0$  and every moving sofa of maximal area is a translate of  $G$  (Corollary 9.6).

Baek's paper does not claim uniqueness. At the time of writing, Google DeepMind's formal-conjectures project lists the statement as an open problem [4], and we know of no earlier proof.

The proof also describes the extremal caps: the caps with rotation angle  $\pi/2$  that maximize Baek's sofa area functional (Section 2.2) are the horizontal translates of Gerver's cap (Theorem 8.5).

**1.3. Baek's argument and the strategy of the proof.** Baek's proof has four steps.

- (1) After a translation, a moving sofa lies in a *monotone* sofa, which is a convex body  $K$ , its *cap*, minus a *niche*  $\mathcal{N}(K)$  cut out by the inner corners of the supporting hallways. The problem becomes the maximization of  $\mathcal{A}_\omega(K) = |K| - |\mathcal{N}(K)|$  over caps with rotation angle  $\omega$ .
- (2) For every  $\omega$  he constructs a maximizing cap as a limit of *maximum polygon caps*, which are *balanced*: each side of the cap is as long as the corresponding side of the niche, since otherwise pushing one wall of one hallway would increase the area (Gerver's balancing argument).
- (3) From the balance he shows that, if the area is at least  $11/5$ , a rotated copy of the sofa  $K \setminus \mathcal{N}(K)$  of this cap turns through the full right angle, so that only the rotation angle  $\pi/2$  matters. From the balance he also shows that for the rotation angle  $\pi/2$  the cap satisfies the *injectivity condition*: the inner corner of the hallway, as seen from the sofa, moves steadily to the left and never crosses its own path.
- (4) For caps with the injectivity condition, a concave quadratic functional  $\mathcal{Q}$  bounds the area, and Gerver's cap maximizes it.

Suppose now that  $S$  has area  $|G|$ . Every inequality in the chain of Baek's estimates becomes an equality, but this chain compares  $|S|$  with the areas of other sets, namely its monotone and Baek's special balanced maximum sofa. A maximizing cap given in advance need not be a limit of maximum polygon caps, and then the properties of steps (2) and (3) are known only for the cap that Baek constructed.

We replace the balance by bounds that we prove for every maximizing cap  $K_*$  of positive sofa area. We approximate  $K_*$  by polygon caps  $K_n$  that maximize the polygon sofa area minus a penalty for straying from  $K_*$  (Section 4). The penalty is a weighted sum of squared differences of support values at dyadic normals, with weights that persist from one approximation to the next and add up to at most 1. Pushing one side of  $K_n$  as in Gerver's balancing argument shows that  $K_n$  is almost balanced: at each normal, the difference between the length  $\sigma$  of the side of the cap and the length  $\tau$  of the corresponding side of the niche is at most a multiple of the distance from  $K_n$  to  $K_*$  at the sampled normals, and by the choice of the weights these bounds add up to  $O(d_H(K_n, K_*))$  (Section 5).

In the limit this gives two inequalities between wedge gaps and side lengths of  $K_*$ , the pinned bounds. They suffice for Baek's rotation argument (Section 6), which produces a second monotone sofa  $U$  of area  $|G|$  with rotation angle  $\pi/2$  that contains a rigid image of  $S$ . For the maximizing cap  $K$  of  $U$ , the same selection gives the curvature bounds  $\sigma_K \leq k_0(g_K^+(t))dt$  on  $[0, \pi/2)$  and  $\sigma_K \leq k_0(f_K^-(t - \pi/2))dt$  on  $(\pi/2, \pi]$ , from which the injectivity condition follows (Section 7).

Then  $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq |G|$  is an equality, and equality in  $\mathcal{Q}$  forces equality in each of its convex Mamikon terms. By Mamikon's theorem, each of these is half the integral of the squared length  $\varrho$  of a tangent segment,  $\frac{1}{2} \int \varrho^2$ . Equality in each term gives differential equations for the difference of

the support functions of  $K$  and of Gerver’s cap, whose solutions are  $s \cos t$ , so that  $K$ , and with it  $U$ , is a horizontal translate of Gerver’s cap and sofa (Section 8). Finally  $G$  is the closure of its interior, which turns the inclusion of a rigid image of  $S$  in  $G$  into an equality (Section 9).

What is new compared with [1] is Theorem 1.1 and its proof: the penalized selection with persistent samples, which gives the defect bounds for an arbitrary maximizer; the use of these bounds in place of the balance in Baek’s arguments of his Chapters 4 and 6; the equality case of Baek’s bound, with the description of the extremal caps (Theorem 8.5); and the regularity of  $G$  that recovers the set. The formalization of Theorem 1.1 is new. Baek’s theorem is also formalized in [9, 12], and twelve findings of the audit of [1] that accompanies our formalization [14] come from the notes of [9].

**1.4. Formal verification and the use of AI.** The proof, together with Baek’s, is formalized in Lean 4 with Mathlib [2, 13], in the repository [14]: every result of [1] used here, with the corrections of Appendix B, and the proof of Theorem 1.1. The statement of Theorem 1.1 is `Baek.gerver_sofa_unique` in `Challenge.lean`, and Appendix C lists the Lean declarations behind each result of this paper. Three of the Facts of Section 2 (Fact 2.15(a), Fact 2.16 and Fact 2.17) are known to us only through the formalization.

At the author’s request, AI systems wrote the first version of the proof, the formalization and this text. Section 10 names them, describes what each did, and says what has and has not been checked. Lean’s kernel has checked the formalization; the argument has not been peer reviewed, and no person has checked it independently of the formalization.

**1.5. Organization.** Section 2 recalls the definitions and the results of [1] that we use. Section 3 reduces the theorem to caps and outlines the rest; Sections 4 to 9 carry out the strategy described above; Section 10 describes the formalization and the use of AI, and lists open questions. The appendices give the explicit description of Gerver’s sofa, the corrections to [1], and the dictionary to the Lean declarations.

## 2. SETTING, AND WHAT WE USE FROM BAEK’S PAPER

This section fixes the notation and records the results that the proof uses and does not prove; we call them *Facts*. Most are results of Baek’s paper [1], quoted in the form in which we use them and with the numbering of version 1 of the preprint (chapter.section.number). A few printed statements need a correction. Each such place is noted with a label  $En$ , the number of the correction in the list of Appendix B. Three Facts are known to us only through the formalization, because the printed proof is missing or invalid, or because the statement is not in [1]: Fact 2.15(a) (E21), Fact 2.16 (E12, E24) and Fact 2.17; they are proved in [14], and Section 10 says what the machine check does and does not establish. The reader who knows [1] can skim this section and return to it for the notation.

**2.1. Notation and convex bodies.** The plane is  $\mathbb{R}^2$ , with points  $p = (x, y)$ , inner product  $\langle p, q \rangle$  and norm  $\|p\|$ ;  $|X|$  is the area (Lebesgue measure) of  $X \subset \mathbb{R}^2$ . For an angle  $t \in \mathbb{R}$  put

$$u_t = (\cos t, \sin t), \quad v_t = (-\sin t, \cos t) = u_{t+\pi/2},$$

and let  $R_t$  be the counterclockwise rotation by  $t$ . For  $c \in \mathbb{R}$  let  $l(t, c) = \{p : \langle p, u_t \rangle = c\}$ ,  $H_-(t, c) = \{p : \langle p, u_t \rangle \leq c\}$ ,  $H_+(t, c) = \{p : \langle p, u_t \rangle \geq c\}$ , and let  $H_\pm^\circ$  be the open half-planes. The half-plane  $H_-(t, c)$  has *normal angle*  $t$ .

A *convex body* is a nonempty compact convex subset of  $\mathbb{R}^2$ ; it may be a segment or a point. Its *support function* is  $h_K(t) = \max_{p \in K} \langle p, u_t \rangle$ , its *supporting line* is  $l_K(t) = l(t, h_K(t))$ , and  $H_K(t) = H_-(t, h_K(t))$ . Then  $K = \bigcap_t H_K(t)$ , so  $h_K$  determines  $K$ ;  $h_{K+w}(t) = h_K(t) + \langle w, u_t \rangle$  and  $h_{(1-\lambda)K+\lambda L} = (1-\lambda)h_K + \lambda h_L$  for Minkowski combinations with  $\lambda \in [0, 1]$ . The Hausdorff distance is  $d_H(K, L) = \sup_t |h_K(t) - h_L(t)|$ , and  $K_n \rightarrow K$  means  $d_H(K_n, K) \rightarrow 0$ . The *edge* of  $K$  with normal angle  $t$  is the segment  $e_K(t) = K \cap l_K(t) = [v_K^-(t), v_K^+(t)]$ , where  $v_K^+(t)$  is its endpoint farthest in the direction  $v_t$ ; it may be a point. For angles  $a, b$  with  $\sin(b-a) \neq 0$  the supporting lines  $l_K(a)$  and  $l_K(b)$  meet at

$$v_K(a, b) = h_K(a) u_a + \frac{h_K(b) - h_K(a) \cos(b-a)}{\sin(b-a)} v_a.$$

**Fact 2.1** (convex bodies). *Let  $K$  be a convex body.*

- (a)  $h_K$  has the one-sided derivatives  $h'_K(t^\pm) = \langle v_K^\pm(t), v_t \rangle$ ;  $v_K^\pm(s) \rightarrow v_K^\pm(t)$  as  $s \downarrow t$  and  $v_K^\pm(s) \rightarrow v_K^\mp(t)$  as  $s \uparrow t$ ;  $v_K(t, s) \rightarrow v_K^+(t)$  as  $s \downarrow t$  and  $v_K(s, t) \rightarrow v_K^-(t)$  as  $s \uparrow t$ . [1, Thm. 2.1.3]
- (b) The surface area measure  $\sigma_K$  is the Lebesgue–Stieltjes measure on  $\mathbb{R}$  of  $G_K(t) = \langle v_K^+(t), v_t \rangle + \int_0^t h_K$  (formally  $\sigma_K = h_K'' + h_K$ ); it is  $2\pi$ -periodic. We write  $\sigma_K(t) = \sigma_K(\{t\})$ . Its atoms are the lengths of the edges:  $\sigma_K(t) = \|v_K^+(t) - v_K^-(t)\|$  and  $v_K^+(t) = v_K^-(t) + \sigma_K(t)v_t$ . Moreover  $v_K^+(d) - v_K^+(c) = \int_{[c,d]} v_s d\sigma_K(s)$  for  $c \leq d$ , and  $|K| = \frac{1}{2} \int_{[0,2\pi)} h_K d\sigma_K$ . [1, Prop. 2.1.2, Thm. 5.2.2, Thm. 7.1.3], [11, Rem. 5.1.2]
- (c) Every sequence of convex bodies in a bounded set has a subsequence that converges to a convex body (Blaschke's selection theorem), and  $K \mapsto |K|$  is continuous for  $d_H$ . [11, Section 1.8, in particular Thm. 1.8.20]

Baek's Thm. 2.1.3 states the limits in (a), and the printed statement has a misprint (E27). The formulas for the one-sided derivatives follow from these limits and from  $\langle v_K(t, s), v_t \rangle = \langle v_K(s, t), v_t \rangle = (h_K(s) - h_K(t) \cos(s-t)) / \sin(s-t)$ . For  $a < b$ , (b) gives  $\sigma_K((a, b)) = G_K(b^-) - G_K(a)$ , and  $G_K(b^-) = \langle v_K^-(b), v_b \rangle + \int_0^b h_K$  since  $v_K^+(s) \rightarrow v_K^-(b)$  as  $s \uparrow b$ ; so

$$\sigma_K((a, b)) = \langle v_K^-(b), v_b \rangle - \langle v_K^+(a), v_a \rangle + \int_a^b h_K. \quad (2.1)$$

**2.2. Moving sofas, monotonization, caps and niches.** The *hallway* is  $L = H_L \cup V_L$ , the union of its horizontal side  $H_L = (-\infty, 1] \times [0, 1]$  and its vertical side  $V_L = [0, 1] \times (-\infty, 1]$ ; its inner corner is the origin  $O$  and its outer corner is  $(1, 1)$ .

**Definition 2.2** (moving sofa). A *movement* of a set  $S \subset \mathbb{R}^2$  is a pair of continuous maps  $\theta : [0, 1] \rightarrow \mathbb{R}$ ,  $c : [0, 1] \rightarrow \mathbb{R}^2$  with  $\theta(0) = 0$  such that the rigid motions  $\Phi_s(p) = R_{\theta(s)}p + c(s)$  satisfy

$$\Phi_0(S) \subset H_L, \quad \Phi_s(S) \subset L \quad (0 \leq s \leq 1), \quad \Phi_1(S) \subset V_L.$$

Its *rotation angle* is  $\omega = -\theta(1)$ . A *moving sofa* is a nonempty, closed and connected set that has a movement; it is a moving sofa *with rotation angle*  $\omega$  if it has a movement with that rotation angle.

This is [1, Def. 1.1.2]: Baek moves the set along a continuous curve of orientation-preserving isometries that starts at a translation, and the angle of such a curve lifts to a continuous real function. A moving sofa  $S$  is bounded, hence compact. Indeed, the start position  $\Phi_0(S) \subset H_L$  bounds the ordinate on  $S$  and bounds the abscissa from above. If  $\theta(1) = 0$ , the end position  $\Phi_1(S) \subset V_L$  bounds the abscissa from below. Otherwise, by the intermediate value theorem,  $0 < |\theta(s)| \leq 1/2$  for some  $s \in [0, 1]$ , with  $\theta(s)$  of the sign of  $\theta(1)$ , and  $\Phi_s(S) \subset L$  bounds the abscissa from below: through the bound  $y \leq 1$  on  $L$  if  $\theta(s) < 0$ , and because  $L$  does not meet the open quarter-plane  $\{x < 0, y < 0\}$  if  $\theta(s) > 0$ . A translate of a moving sofa with rotation angle  $\omega$  is again one. Let  $\omega_0 = \sec^{-1}(11/5) = \arccos(5/11) = 1.0989 \dots$  ( $63^\circ$  approximately).

For  $\omega \in (0, \pi/2]$  let  $H = \mathbb{R} \times [0, 1]$  be the horizontal strip,  $V_\omega = \{p : 0 \leq \langle p, u_\omega \rangle \leq 1\}$ ,  $P_\omega = H \cap V_\omega$  the parallelogram (a strip if  $\omega = \pi/2$ ), and  $F_\omega = \{p : y \geq 0, \langle p, u_\omega \rangle \geq 0\}$  the *fan*. If  $\omega < \pi/2$  the two upper sides of  $P_\omega$  meet at  $o_\omega = (c_\omega, 1)$ ,  $c_\omega = \sec \omega - \tan \omega$ , and the two lower sides at  $O$ . A moving sofa  $S$  with rotation angle  $\omega$  is in *standard position* if  $h_S(\omega) = h_S(\pi/2) = 1$ .

**Fact 2.3** (rotation angle and standard position). (a) *Every moving sofa  $S$  with  $|S| \geq 11/5$  is a moving sofa with some rotation angle  $\omega \in [\omega_0, \pi/2]$ . [1, Thm. 1.5.1]*

(b) *If  $S$  is a moving sofa with rotation angle  $\omega \in (0, \pi/2]$ , some translate  $S + v$  is in standard position, and a moving sofa with rotation angle  $\omega$  in standard position lies in  $P_\omega$ . [1, Prop. 1.2.1, Prop. 2.3.1]*

*Supporting hallways.* For a compact set  $S$  and an angle  $t$  let

$$f_{S,t}(p) = R_t p + (h_S(t) - 1)u_t + (h_S(t + \pi/2) - 1)v_t,$$

and let  $L_S(t) = f_{S,t}(L)$  be the *supporting hallway* of  $S$  at  $t$ : the translate of  $R_t L$  whose two outer walls  $a_S(t) = l_S(t)$  and  $c_S(t) = l_S(t + \pi/2)$  are

supporting lines of  $S$ . Its inner corner and outer corner are

$$\begin{aligned}\mathbf{x}_S(t) &= (h_S(t) - 1)u_t + (h_S(t + \pi/2) - 1)v_t, \\ \mathbf{y}_S(t) &= h_S(t)u_t + h_S(t + \pi/2)v_t = \mathbf{x}_S(t) + u_t + v_t,\end{aligned}$$

its inner walls lie on the lines  $b_S(t) = l(t, h_S(t) - 1)$  and  $d_S(t) = l(t + \pi/2, h_S(t + \pi/2) - 1)$  (the half-lines from  $\mathbf{x}_S(t)$  in the directions  $-v_t$  and  $-u_t$  are  $\vec{b}_S(t)$  and  $\vec{d}_S(t)$ ), and

$$\begin{aligned}L_S(t) &= Q_S^+(t) \setminus Q_S^-(t), & Q_S^+(t) &= H_S(t) \cap H_S(t + \frac{\pi}{2}), \\ Q_S^-(t) &= H_-^\circ(t, h_S(t) - 1) \cap H_-^\circ(t + \frac{\pi}{2}, h_S(t + \frac{\pi}{2}) - 1).\end{aligned}$$

$Q_S^-(t)$  is the open quarter-plane below the inner corner; the printed [1, Prop. 2.2.2] lacks the  $-1$  in its second half-plane (E27). A moving sofa in standard position lies in  $L_S(t)$  for every  $t \in [0, \omega]$ , and the *monotonization* of  $S$  is

$$\mathcal{I}(S) = P_\omega \cap \bigcap_{t \in [0, \omega]} L_S(t).$$

A *monotone sofa* with rotation angle  $\omega$  is a set  $\mathcal{I}(S')$  for a moving sofa  $S'$  in standard position with rotation angle  $\omega$ .

*Caps and niches.* Let  $J_\omega = [0, \omega] \cup [\pi/2, \omega + \pi/2]$ . A *cap* with rotation angle  $\omega \in (0, \pi/2]$  is a convex body  $K$  with

$$h_K(\omega) = h_K(\pi/2) = 1, \quad h_K(\omega + \pi) = h_K(3\pi/2) = 0,$$

that is an intersection of closed half-planes with normal angles in  $J_\omega \cup \{\omega + \pi, 3\pi/2\}$ . The caps with rotation angle  $\omega$  form the set  $\mathcal{K}_\omega^c$ . A cap lies in  $P_\omega$ , and

$$K = F_\omega \cap \bigcap_{s \in J_\omega} H_K(s). \quad (2.2)$$

The *niche* of a cap  $K$  and its *sofa area* are

$$\mathcal{N}(K) = F_\omega \cap \bigcup_{t \in (0, \omega)} Q_K^-(t), \quad \mathcal{A}_\omega(K) = |K| - |\mathcal{N}(K)|.$$

For  $S$  in standard position, the *cap* of  $S$  is  $\mathcal{C}(S) = P_\omega \cap \bigcap_{t \in [0, \omega]} Q_S^+(t)$ . The reflection  $M_\omega(x, y) = (-x \sin \omega + y \cos \omega, x \cos \omega + y \sin \omega)$  in the line through  $O$  at the angle  $\pi/4 + \omega/2$  (the line through  $O$  and  $o_\omega$  if  $\omega < \pi/2$ ) maps  $F_\omega$  to itself; for  $\omega = \pi/2$  it is  $(x, y) \mapsto (-x, y)$ . The *mirror image* of a cap  $K$  is  $K^m = M_\omega(K)$ . For a convex body  $K$  we write  $A_K^\pm(t) = v_K^\pm(t)$  and  $C_K^\pm(t) = v_K^\pm(t + \pi/2)$  [1, Def. 2.5.1].

**Fact 2.4** (monotone sofas and caps). *Let  $\omega \in (0, \pi/2]$ .*

- (a) *If  $S$  is a moving sofa with rotation angle  $\omega$  in standard position, then  $\mathcal{I}(S)$  is a moving sofa with rotation angle  $\omega$  in standard position and  $S \subset \mathcal{I}(S)$ . [1, Thm. 2.3.2]*
- (b) *For such an  $S$ ,  $K = \mathcal{C}(S)$  is a cap with  $h_K = h_S$  on  $J_\omega$ , and  $\mathcal{I}(S) = K \setminus \mathcal{N}(K)$ . [1, Lemma 2.3.5, Thm. 2.4.1, Thm. 2.4.2]*

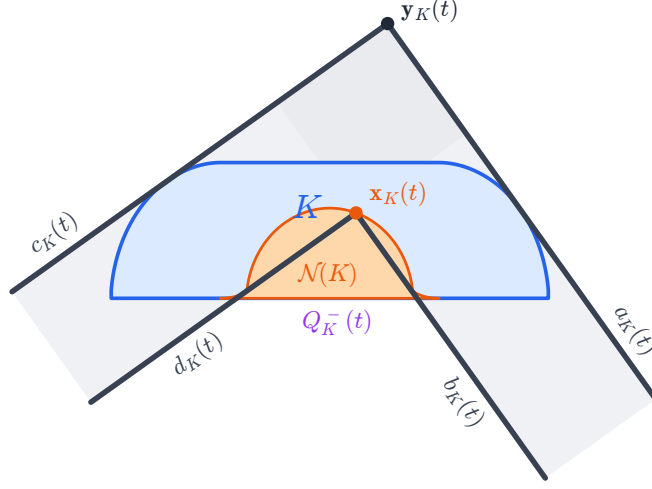


FIGURE 2. A cap, its niche and a supporting hallway: Gerver's cap  $K = K_G$  (blue), its niche  $\mathcal{N}(K)$  (orange), and the supporting hallway  $L_K(t)$  at  $t = 0.62$  (grey). The outer walls  $a_K(t)$  and  $c_K(t)$  of the hallway touch  $K$  and meet at the outer corner  $\mathbf{y}_K(t)$ ; its inner walls  $b_K(t)$  and  $d_K(t)$  meet at the inner corner  $\mathbf{x}_K(t)$ , on the boundary of the niche.  $Q_K^-(t)$  is the open quarter-plane below the inner corner; the niche is the union of these quarter-planes over  $t \in (0, \pi/2)$ , cut at the  $x$ -axis.

- (c) If  $T$  is a monotone sofa with rotation angle  $\omega$  and  $K = \mathcal{C}(T)$ , then  $T = K \setminus \mathcal{N}(K)$ ,  $\mathcal{N}(K) \subset K$  and  $|T| = \mathcal{A}_\omega(K)$ . [1, Thm. 2.4.3, Thm. 2.5.9, Thm. 2.5.10]
- (d) If  $K$  is a cap, so is  $K^m$ , with  $h_{K^m}(t) = h_K(\omega + \pi/2 - t)$ ,  $C_{K^m}^+(t) = M_\omega(A_K^-(\omega - t))$ ,  $\mathcal{N}(K^m) = M_\omega(\mathcal{N}(K))$ ,  $\mathcal{A}_\omega(K^m) = \mathcal{A}_\omega(K)$  and  $\sigma_{K^m}(E) = \sigma_K(\omega + \pi/2 - E)$  for Borel sets  $E$ . [1, Prop. 2.5.4]

The proof of (a) uses Def. 2.3.9, which defines the left and the right side of a line for lines not parallel to the  $y$ -axis, where it needs the  $x$ -axis (E1). In (b), the union over  $t \in [0, \omega]$  in Thm. 2.4.2 and the union over  $(0, \omega)$  in the definition of the niche agree after intersection with  $F_\omega$ , and  $\mathcal{C}(S)$  is bounded also for  $\omega = \pi/2$ , which Baek does not show (E2). In (d), Prop. 2.5.4 prints  $?_{K^m}(t) = M_\omega(?_K(\omega - t))$  for each of the walls  $a, b, c, d$  and the ends  $W, Z$ , whereas  $M_\omega$  exchanges  $a \leftrightarrow c$ ,  $b \leftrightarrow d$  and  $W \leftrightarrow Z$ , and the identity for the vertices has  $K^m$  in place of  $K$  on the right (E3). Part (d) states the corrected identity  $C_{K^m}^+(t) = v_{K^m}^+(t + \pi/2) = M_\omega(A_K^-(\omega - t))$ .

Two more facts about caps are used without proof in Baek's [1, Lemma 2.5.6], [1, Lemma 3.4.8] and [1, Thm. 2.5.8] (E4): that  $O$  and

$o_\omega$  lie in every cap if  $\omega < \pi/2$ , and where the ends of the *upper boundary*  $\bigcup_{t \in [0, \omega + \pi/2]} e_K(t)$  of a cap  $K$  [1, Def. 2.5.2] are.

**Lemma 2.5** (every cap contains  $O$  and  $o_\omega$ ). *If  $\omega < \pi/2$ , every cap  $K \in \mathcal{K}_\omega^c$  contains  $O$  and  $o_\omega$ .*

**Lemma 2.6** (the ends of the upper boundary). *For every cap  $K \in \mathcal{K}_\omega^c$ ,*

$$A_K^-(0) = (h_K(0), 0) \quad \text{and} \quad C_K^+(\omega) = h_K(\omega + \pi/2) v_\omega. \quad (2.3)$$

*Proof of Lemma 2.6.* Put  $m = h_K(\omega + \pi/2)$  and  $p = mv_\omega$ , so that  $\langle p, u_\omega \rangle = 0$  and  $\langle p, v_\omega \rangle = m$ . We show that  $p \in K$ , by (2.2). Let  $q \in K \cap l(\pi/2, 1)$ . As  $u_{\pi/2} = \sin \omega u_\omega + \cos \omega v_\omega$ ,  $\langle q, u_\omega \rangle \leq h_K(\omega) = 1$  and  $\langle q, v_\omega \rangle \leq m$ ,

$$1 = \langle q, u_{\pi/2} \rangle = \sin \omega \langle q, u_\omega \rangle + \cos \omega \langle q, v_\omega \rangle \leq \sin \omega + m \cos \omega,$$

so  $p_y = m \cos \omega \geq 1 - \sin \omega \geq 0$ , and  $p \in F_\omega$ . Let  $r \in e_K(\omega + \pi/2)$ ; then  $\langle r, v_\omega \rangle = m$ , and  $\langle r, u_\omega \rangle \geq 0$  as  $K \subset P_\omega$ . For  $s \in J_\omega$ ,  $\cos(s - \omega) \geq 0$ , so

$$\langle p, u_s \rangle = m \sin(s - \omega) = \langle r, u_s \rangle - \langle r, u_\omega \rangle \cos(s - \omega) \leq \langle r, u_s \rangle \leq h_K(s).$$

Hence  $p \in K$  by (2.2), and  $p \in e_K(\omega + \pi/2)$ . The end  $C = C_K^+(\omega)$  of this edge is its point farthest in the direction  $v_{\omega + \pi/2} = -u_\omega$ , so  $\langle C, u_\omega \rangle \leq \langle p, u_\omega \rangle = 0$ , while  $\langle C, u_\omega \rangle \geq 0$  as  $C \in K$ . With  $\langle C, v_\omega \rangle = m$ , this gives  $C = mv_\omega$ .

By Fact 2.4(d), the mirror image  $K^m$  is a cap, with  $h_{K^m}(\omega + \pi/2) = h_K(0)$  and  $C_{K^m}^+(\omega) = M_\omega(A_K^-(0))$  (E3). The first part, applied to  $K^m$ , gives  $M_\omega(A_K^-(0)) = h_K(0)v_\omega$ . As  $M_\omega$  is a linear involution and  $M_\omega(v_\omega) = u_0$ ,  $A_K^-(0) = h_K(0)u_0 = (h_K(0), 0)$ .  $\square$

*Proof of Lemma 2.5.* Let  $\omega < \pi/2$ , so that  $\cos \omega > 0$ . By Lemma 2.6,  $A = A_K^-(0) = (h_K(0), 0)$  and  $C = C_K^+(\omega) = h_K(\omega + \pi/2)v_\omega$ ; they are ends of edges of  $K$ , so they lie in  $K \subset P_\omega$ . Hence  $h_K(0) \cos \omega = \langle A, u_\omega \rangle \geq 0$  and  $h_K(\omega + \pi/2) \cos \omega = C_y \geq 0$ , that is,  $h_K(0) \geq 0$  and  $h_K(\omega + \pi/2) \geq 0$ . For  $s \in [0, \omega]$ ,  $h_K(s) \geq \langle A, u_s \rangle = h_K(0) \cos s \geq 0$ , and for  $s \in [\pi/2, \omega + \pi/2]$ ,  $h_K(s) \geq \langle C, u_s \rangle = h_K(\omega + \pi/2) \sin(s - \omega) \geq 0$ . So  $O \in F_\omega$  satisfies  $\langle O, u_s \rangle = 0 \leq h_K(s)$  for  $s \in J_\omega$ , and  $O \in K$  by (2.2).

Let  $q \in K \cap l(\pi/2, 1)$  and  $q' \in K \cap l(\omega, 1)$ . As  $K \subset P_\omega$ ,  $q = o_\omega - \lambda u_0$  and  $q' = o_\omega - \mu v_\omega$  with  $\lambda, \mu \geq 0$ . For  $s \in [0, \omega]$ ,  $\langle o_\omega, u_s \rangle = \langle q', u_s \rangle + \mu \sin(s - \omega) \leq \langle q', u_s \rangle \leq h_K(s)$ , and for  $s \in [\pi/2, \omega + \pi/2]$ ,  $\langle o_\omega, u_s \rangle = \langle q, u_s \rangle + \lambda \cos s \leq \langle q, u_s \rangle \leq h_K(s)$ . As  $o_\omega \in F_\omega$ , (2.2) gives  $o_\omega \in K$ .  $\square$

The next fact is the main theorem of [1], in the form for caps. It assembles Baek's construction of a balanced maximum cap  $K_\omega$  for every  $\omega$  ([1, Thm. 3.5.2, Thm. 3.5.4], whose proofs have minor gaps, E8), its maximality ([1, Thm. 3.5.5]), and the optimality of Gerver's sofa  $G$  ([1, Thm. 1.1.1]), which is recalled in Section 2.6.

**Fact 2.7** (optimality). *Every moving sofa  $S$  has  $|S| \leq |G|$ , and  $|G| > 11/5$ . For every  $\omega \in (0, \pi/2]$  and every cap  $K \in \mathcal{K}_\omega^c$ ,  $\mathcal{A}_\omega(K) \leq |G|$ .*

For the last claim, [1, Thm. 3.5.6] gives a monotone sofa  $S_\omega = K_\omega \setminus \mathcal{N}(K_\omega)$  with rotation angle  $\omega$ , whose cap  $K_\omega$  satisfies  $\mathcal{A}_\omega(K) \leq \mathcal{A}_\omega(K_\omega)$  for every cap  $K$  [1, Thm. 3.5.5]. Then  $\mathcal{A}_\omega(K_\omega) = |S_\omega|$  by Fact 2.4(c), and  $S_\omega$  is a moving sofa by Fact 2.4(a), so that  $|S_\omega| \leq |G|$  by [1, Thm. 1.1.1]. The bound  $|G| > 11/5$  holds because  $|G| \geq 2.2192$  (Fact 2.16(a)).

**2.3. Polygon caps.** Let  $\omega \in (0, \pi/2]$ . An *angle set* is a finite nonempty  $\Theta \subset (0, \omega)$ , and  $\Theta^\diamond = \Theta \cup (\Theta + \pi/2) \cup \{\omega, \pi/2\} \subset (0, \pi)$  is its set of *defining normals*; the normals  $\omega$  and  $\pi/2$ , which carry the strips of  $P_\omega$ , are *pinned*, the others *floating*. A *polygon cap* with angle set  $\Theta$  is a cap that is an intersection of closed half-planes with normal angles in  $\Theta^\diamond \cup \{\omega + \pi, 3\pi/2\}$ ; they form the set  $\mathcal{K}_\Theta^c \subset \mathcal{K}_\omega^c$ . A polygon cap is the intersection of its supporting half-planes  $H_K(s)$  at these normal angles. For a cap  $K$  put

$$\begin{aligned} \mathcal{C}_\Theta(K) &= P_\omega \cap \bigcap_{t \in \Theta} Q_K^+(t), & \mathcal{N}_\Theta(K) &= F_\omega \cap \bigcup_{t \in \Theta} Q_K^-(t), \\ \mathcal{A}_\Theta(K) &= |\mathcal{C}_\Theta(K)| - |\mathcal{N}_\Theta(K)|, \end{aligned}$$

with the fan  $F_\omega$  in the polygon niche: the printed [1, Def. 3.2.5] has the parallelogram  $P_\omega$  in its place, and with that definition [1, Prop. 3.3.5] and [1, Lemma 3.4.2] are false, whereas with  $F_\omega$  they hold (E6). [1, Prop. 3.3.1 (2)], which characterizes the translates of polygon caps, omits the bottom normals  $\omega + \pi$  and  $3\pi/2$  from their normal angles (E7).

**Fact 2.8** (the polygon bound). *For every cap  $K$  and angle set  $\Theta$ ,  $\mathcal{C}_\Theta(K)$  is a polygon cap that contains  $K$  and has the same support values as  $K$  on  $\Theta^\diamond$ , and  $\mathcal{C}_\Theta(K) = K$  if  $K \in \mathcal{K}_\Theta^c$ ;  $\mathcal{N}_\Theta(K) = \mathcal{N}_\Theta(\mathcal{C}_\Theta(K)) \subset \mathcal{N}(K)$ . Hence  $\mathcal{A}_\omega(K) \leq \mathcal{A}_\Theta(K) = \mathcal{A}_\Theta(\mathcal{C}_\Theta(K))$ , and  $\mathcal{A}_\Theta(K) = |K| - |\mathcal{N}_\Theta(K)|$  for a polygon cap. [1, Prop. 3.2.1, Prop. 3.2.2, Thm. 3.2.3]*

**Fact 2.9** (bounded width). *For  $t \in (0, \omega)$  there is a constant  $c_{\omega, t} > 0$  such that every polygon cap  $K$  with an angle set  $\Theta \ni t$  and  $\mathcal{A}_\Theta(K) > 0$  has width  $h_K(0) + h_K(\pi) \leq c_{\omega, t}$ . [1, Lemma 3.4.2], with  $F_\omega$  (E6)*

*Sides of the polygon niche.* For a polygon cap  $K$  with angle set  $\Theta$ , the boundary of  $F_\omega \setminus \mathcal{N}_\Theta(K)$  consists of an open half-line, a polyline  $\mathbf{p}_K$  from  $C_K^+(\omega)$  to  $A_K^-(0)$  (see (2.3)), along which the abscissa increases and whose segments have normal angles in  $\Theta^\diamond$ , and an open half-line [1, Thm. 3.4.4]. For  $t \in \Theta^\diamond$  let  $\tau_K(t)$  be the total length of the segments of  $\mathbf{p}_K$  with normal angle  $t$ ; recall that  $\sigma_K(t)$  is the length of the edge  $e_K(t)$  of  $K$ .

**Fact 2.10** (sides of the polygon niche). *Let  $K \in \mathcal{K}_\Theta^c$ .*

- (a) *For  $t \in \Theta$ ,  $\mathcal{H}^1(\partial \mathcal{N}_\Theta(K) \cap \vec{b}_K(t)) = \tau_K(t)$  and  $\mathcal{H}^1(\partial \mathcal{N}_\Theta(K) \cap \vec{d}_K(t)) = \tau_K(t + \pi/2)$ . [1, Lemma 3.4.5 (1)]*
- (b) *For  $t \in \{\omega, \pi/2\}$ ,  $\mathcal{H}^1(\mathcal{N}_\Theta(K) \cap l(t, 0)) = \sigma_K(t + \pi) - \tau_K(t)$ . [1, Lemma 3.4.5 (2)]*

$$(c) \ C_K^+(\omega) - A_K^-(0) = \sum_{t \in \Theta^\diamond} \sigma_K(t) v_t, \text{ and}$$

$$\sum_{t \in \Theta^\diamond} \sigma_K(t) \sin t = A_K^-(0)_x - C_K^+(\omega)_x = \sum_{t \in \Theta^\diamond} \tau_K(t) \sin t.$$

[1, Proof of Lemma 3.4.6]

In (c), the vector identity is a sum along the upper boundary of  $K$ : taken in the increasing order of  $t \in \Theta^\diamond$ , the edges  $e_K(t)$  go from  $v_K^-(t)$  to  $v_K^+(t) = v_K^-(t) + \sigma_K(t)v_t$ , the first starts at  $A_K^-(0)$ , each ends where the next starts, and the last ends at  $C_K^+(\omega)$ . Its first coordinate is the first of the two equalities, as  $\langle v_t, u_0 \rangle = -\sin t$ . In the second,  $\tau_K(t) \sin t$  is the total length of the projections to the  $x$ -axis of the segments of  $\mathbf{p}_K$  with normal angle  $t$ ; since the abscissa increases along  $\mathbf{p}_K$ , the projections of all its segments cover  $[C_K^+(\omega)_x, A_K^-(0)_x]$  and meet only at their ends.

*Support values and first variations.* For  $h : \Theta^\diamond \rightarrow \mathbb{R}$  let

$$\begin{aligned} P_h &= \bigcap_{s \in \{\omega, \pi/2\}} (H_-(s, h(s)) \cap H_+(s, h(s) - 1)), \\ \mathcal{C}_\Theta(h) &= P_h \cap \bigcap_{s \in \Theta^\diamond \setminus \{\omega, \pi/2\}} H_-(s, h(s)), \\ F_h &= \bigcap_{s \in \{\omega, \pi/2\}} H_+(s, h(s) - 1), \\ \mathcal{N}_\Theta(h) &= F_h \cap \bigcup_{t \in \Theta} (H_-^\circ(t, h(t) - 1) \cap H_-^\circ(t + \frac{\pi}{2}, h(t + \frac{\pi}{2}) - 1)), \end{aligned}$$

and  $\mathcal{A}_\Theta(h) = |\mathcal{C}_\Theta(h)| - |\mathcal{N}_\Theta(h)|$  ([1, Def. 3.3.3]). For a polygon cap  $K$  and  $h = h_K|_{\Theta^\diamond}$ ,  $\mathcal{C}_\Theta(h) = K$ ,  $\mathcal{N}_\Theta(h) = \mathcal{N}_\Theta(K)$  and  $\mathcal{A}_\Theta(h) = \mathcal{A}_\Theta(K)$  ([1, Prop. 3.3.4, Prop. 3.3.5], with  $F_\omega$ , E6). For  $\varepsilon \geq 0$  and  $t \in \Theta^\diamond$  let  $h_K^{t, \varepsilon}$  be  $h_K$  with the value at  $t$  replaced by  $h_K(t) + \varepsilon$ .

**Fact 2.11** (pushing one side). *Let  $K \in \mathcal{K}_\Theta^c$  and  $t \in \Theta^\diamond$ .*

(a) *There are  $\varepsilon_0 > 0$  and  $M \in \mathbb{R}$  such that for  $0 < \varepsilon \leq \varepsilon_0$*

$$|\mathcal{A}_\Theta(h_K^{t, \varepsilon}) - \mathcal{A}_\Theta(K) - (\sigma_K(t) - \tau_K(t))\varepsilon| \leq M\varepsilon^2.$$

[1, Lemma 3.4.7] (E5)

(b) *If  $\sigma_K(t) > 0$ , then for small  $\varepsilon > 0$  the set  $\mathcal{C}_\Theta(h_K^{t, \varepsilon})$  is a translate  $C + v$  of a polygon cap  $C \in \mathcal{K}_\Theta^c$ . [1, Lemma 3.4.8, Prop. 3.3.1] (E7)*

(c) *If  $\mathcal{C}_\Theta(h_K^{t, \varepsilon}) = C + v$  with  $C \in \mathcal{K}_\Theta^c$ , then  $\mathcal{A}_\Theta(h_K^{t, \varepsilon}) \leq \mathcal{A}_\Theta(C)$ . [1, Prop. 3.3.7, Thm. 3.3.6]*

Part (a) rests on [1, Thm. 3.1.2], which is about simple Nef polygons, the sets  $\{p : \mathcal{E}(p \in H_1, \dots, p \in H_n)\}$  for closed or open half-planes  $H_1, \dots, H_n$  with distinct boundary lines and a monotone Boolean function  $\mathcal{E}$ , and needs them to be bounded; every use is to bounded sets (E5). In the proof of [1, Lemma 3.4.7] the sign of the change of  $|\mathcal{N}_\Theta|$  is misprinted (E27).

**2.4. The injectivity condition and arm lengths.** The rest of this section, and Sections 7 and 8, concern caps with rotation angle  $\pi/2$ , and  $\mathcal{A} = \mathcal{A}_{\pi/2}$ . For a convex body  $K$  the *arm lengths* are

$$f_K^\pm(t) = \langle \mathbf{y}_K(t) - A_K^\pm(t), v_t \rangle, \quad g_K^\pm(t) = \langle \mathbf{y}_K(t) - C_K^\pm(t), u_t \rangle,$$

with  $\mathbf{y}_K(t) = h_K(t)u_t + h_K(t + \pi/2)v_t$  the outer corner of the supporting hallway and  $A_K^\pm, C_K^\pm$  as in Section 2.2. Thus  $\mathbf{y}_K(t) = A_K^\pm(t) + f_K^\pm(t)v_t = C_K^\pm(t) + g_K^\pm(t)u_t$  [1, Prop. 6.2.1]. The arm lengths are nonnegative, since  $A_K^\pm(t)$  and  $C_K^\pm(t)$  lie in  $K$ , so that  $\langle A_K^\pm(t), v_t \rangle \leq h_K(t + \pi/2)$  and  $\langle C_K^\pm(t), u_t \rangle \leq h_K(t)$ ; they are the distances from the outer corner to the points where  $K$  touches the two outer walls. Moreover  $g_K^+(t) - g_K^-(t) = \sigma_K(t + \pi/2) \geq 0$ , by Fact 2.1(b) and  $v_{t+\pi/2} = -u_t$ .

**Definition 2.12** (injectivity condition). A cap  $K \in \mathcal{K}_{\pi/2}^c$  satisfies the *injectivity condition* if

- (1)  $\sigma_K$  has a density on  $[0, \pi/2)$  and on  $(\pi/2, \pi]$  (the top edge, at  $\pi/2$ , may carry an atom);
- (2) the inner corner  $\mathbf{x}_K$  is continuously differentiable on  $[0, \pi/2]$ ;
- (3)  $\langle \mathbf{x}'_K(t), u_t \rangle < 0 < \langle \mathbf{x}'_K(t), v_t \rangle$  for  $0 < t < \pi/2$ .

We write  $\mathcal{K}^i$  for the set of caps in  $\mathcal{K}_{\pi/2}^c$  that satisfy the injectivity condition and have  $|K| \geq 11/5$ .

**Fact 2.13** (arms). Let  $K \in \mathcal{K}_{\pi/2}^c$ .

- (a) For every  $t$ ,  $\mathbf{x}_K$  has the right derivative  $-(f_K^+(t) - 1)u_t + (g_K^+(t) - 1)v_t$ , and the left derivative given by the same formula with  $f_K^-, g_K^-$ . [1, Thm. 6.2.3]
- (b)  $f_K^+(b) - f_K^+(a) = \int_a^b g_K^+ - \sigma_K((a, b])$  for  $0 \leq a \leq b \leq \pi/2$ . [1, Thm. 6.2.5]
- (c) For the reflection  $K^m$  in  $x \mapsto -x$ ,  $f_{K^m}^\pm(t) = g_K^\mp(\pi/2 - t)$  and  $g_{K^m}^\pm(t) = f_K^\mp(\pi/2 - t)$ . [1, Prop. 6.2.2]
- (d) If  $K$  satisfies (1), then  $A_K^+ = A_K^-$  and  $f_K^+ = f_K^-$  on  $[0, \pi/2)$ ,  $C_K^+ = C_K^-$  and  $g_K^+ = g_K^-$  on  $(0, \pi/2]$ . With  $A_K = A_K^+$ ,  $C_K = C_K^+$ ,  $f_K = f_K^+$  and  $g_K = g_K^+$  these four functions are continuous on  $[0, \pi/2]$ ;  $f_K(0) = g_K(\pi/2) = 1$ , and  $\mathbf{x}'_K(t) = -(f_K(t) - 1)u_t + (g_K(t) - 1)v_t$  for  $t \in [0, \pi/2]$ , the derivative being one-sided at 0 and  $\pi/2$ . In particular (2) holds, and (3) holds if  $f_K, g_K > 1$  on  $(0, \pi/2)$ . [1, Def. 6.4.1, Prop. 6.4.5, Prop. 6.4.6]

In (c) the printed statement omits that the reflection reverses the angle (E13). In (a) the right derivative is called the left one, and the second display of the printed theorem is about the left derivative (E14). In (d), the printed [1, Def. 6.4.1] lets  $f_K$  and  $g_K$  map  $[0, \pi/2]$  to  $\mathbb{R}^2$  instead of  $\mathbb{R}$  (E16). The end values  $f_K(0) = g_K(\pi/2) = 1$  in (d) are not stated in [1] (E4, E17), and they hold for every  $K \in \mathcal{K}_{\pi/2}^c$ . The half-planes whose intersection is  $K$  have normal angles in  $[0, \pi] \cup \{3\pi/2\}$ , none of

them strictly between  $-\pi/2$  and 0 modulo  $2\pi$ ; so the supporting lines  $l_K(-\pi/2)$  and  $l_K(0)$  meet at a point of  $K$ , the vertex  $v_K^-(0)$ , and this point is  $v_K(-\pi/2, 0) = (h_K(0), 0)$ , as  $h_K(-\pi/2) = h_K(3\pi/2) = 0$ . Hence  $f_K(0) = f_K^-(0) = h_K(\pi/2) - \langle v_K^-(0), v_0 \rangle = 1$ . The same holds for the cap  $K^m$  (Fact 2.4(d)), and (c) gives  $g_K(\pi/2) = g_K^+(\pi/2) = f_{K^m}^-(0) = 1$ . Let

$$k_0(x) = \max\left(|x - 1|, \frac{|x-1|+1}{2}\right), \quad m_0(x) = x - k_0(x).$$

The function  $k_0$  is 1-Lipschitz, and  $m_0$  is nondecreasing, with  $m_0(x) = \frac{3}{2}x - 1$  on  $[0, 1]$ . For a continuous  $f$  on  $[0, \pi/2]$  let  $\mathcal{F}f(x) = 1 + \int_0^x m_0(f(\pi/2 - u))du$ , and let  $f_0 = 0$ ,  $f_{j+1} = \max(f_j, \mathcal{F}f_j)$ ; each  $f_j$  is continuous.

**Fact 2.14** (convergence of arms and the iteration). *(a) Let  $K_n$  be polygon caps, with any angle sets, and  $K \in \mathcal{K}_{\pi/2}^c$  a cap with  $d_H(K_n, K) \rightarrow 0$ .*

*Then  $\int_0^{\pi/2} |g_{K_n}^+ - g_K^+| \rightarrow 0$ . [1, Lemma 6.4.2]*

*(b)  $f_{11}(x) > 1$  for  $x \in (0, \pi/2]$ . [1, Lemma 6.5.5], which is printed for  $x \in (0, 1]$  and used on  $(0, \pi/2]$  (E17).*

**2.5. Baek's upper bound.** Let  $\varphi = \varphi_G = 0.03917\dots$  be Gerver's angle (Section 2.6),  $\varphi^R = \varphi$  and  $\varphi^L = \pi/2 - \varphi$ . For  $K \in \mathcal{K}_{\pi/2}^c$  let  $H_K^b(t) = \{\langle p, u_t \rangle \geq h_K(t) - 1\}$  and  $H_K^d(t) = \{\langle p, v_t \rangle \geq h_K(t + \pi/2) - 1\}$  be the closed half-planes bounded by the inner walls  $b_K(t)$  and  $d_K(t)$ , on the side away from  $Q_K^-(t)$ , and

$$B_K = K \cap \bigcap_{t \in [\varphi^R, \pi/2]} H_K^b(t), \quad D_K = K \cap \bigcap_{t \in [0, \varphi^L]} H_K^d(t).$$

Baek's upper bound is the functional  $\mathcal{Q}$  of [1, Def. 8.2.2] on the set  $\mathcal{T}$  of triples  $(K, B, D)$  of convex bodies with  $K \in \mathcal{K}^i$ ,  $B, D \subset K$ ,  $h_K(t) + h_B(\pi + t) \leq 1$  on  $[\varphi^R, \pi/2]$  with equality at the ends, and  $h_K(\pi/2 + t) + h_D(3\pi/2 + t) \leq 1$  on  $[0, \varphi^L]$  with equality at the ends. The sets  $\mathcal{K}^i$  and  $\mathcal{T}$  are closed under Minkowski combinations  $c_\lambda((K, B, D), (K', B', D')) = ((1 - \lambda)K + \lambda K', \dots)$ ,  $\lambda \in [0, 1]$  [1, Thm. 8.1.1 (1), Prop. 8.1.2]. A functional on such a set is *convex-linear* if it commutes with  $c_\lambda$ , and *convex* or *concave* if it satisfies the corresponding inequality for  $c_\lambda$ . The *canonical triple* of  $K \in \mathcal{K}^i$  is  $\xi_K = (K, B_K, D_K)$ . For a convex body  $K$ ,  $a < b < a + \pi$ , and a continuous curve  $\mathbf{z} : [a, b] \rightarrow \mathbb{R}^2$  of bounded variation with  $\mathbf{z}(t) \in l_K(t)$ , [1, Def. 7.4.2] defines the signed combination of curve areas

$$\mathcal{M}_K(a, b; \mathbf{z}) = \mathcal{J}([v_K^+(a), \mathbf{z}(a)]) + \mathcal{J}(\mathbf{z}) + \mathcal{J}([\mathbf{z}(b), v_K^-(b)]) - \frac{1}{2} \int_{(a,b)} h_K d\sigma_K,$$

where  $\mathcal{J}(\gamma) = \frac{1}{2} \int \gamma \times d\gamma$  for an oriented curve  $\gamma$ , with  $p \times q = p_x q_y - p_y q_x$  [1, Def. 7.2.6, Def. 7.2.8], and  $[p, q]$  is the segment from  $p$  to  $q$ ; the last term is  $\mathcal{J}$  of the arc  $\mathbf{u}_K^{a,b}$  of the boundary of  $K$  from  $v_K^+(a)$  to  $v_K^-(b)$  [1, Def. 7.3.1, Thm. 7.3.2]. Mamikon's theorem, in Baek's form [1, Thm. 7.4.1] (see [7]),

evaluates it:

$$\mathcal{M}_K(a, b; \mathbf{z}) = \frac{1}{2} \int_a^b \varrho_{K, \mathbf{z}}(t)^2 dt, \quad \varrho_{K, \mathbf{z}}(t) = \langle \mathbf{z}(t) - v_K^+(t), v_t \rangle.$$

We use  $\mathcal{M}$  only through this formula, which makes  $\mathcal{M}_K(a, b; \mathbf{z})$  a convex function of  $\varrho_{K, \mathbf{z}}$ . The curves that occur are the outer corner  $\mathbf{y}_K$  and the tangent line parametrizations  $\mathbf{l}_K^T(t) = v_K(t, T)$  for  $T - \pi < t < T$  (the point where  $l_K(t)$  meets  $l_K(T)$ ) and  $\mathbf{l}_K^T(T) = v_K^-(T)$ . Let  $\mathcal{S}_K, \mathcal{R}_B, \mathcal{L}_D$  be the *Mamikon terms*

$$\begin{aligned} \mathcal{S}_K &= \mathcal{M}_K(0, \varphi^R; \mathbf{l}_K^{\pi/2}) + \mathcal{M}_K(\varphi^R, \varphi^L; \mathbf{y}_K) \\ &\quad + \mathcal{M}_K(\varphi^L, \frac{\pi}{2}; \mathbf{l}_K^{\pi-\varphi}) + \mathcal{M}_K(\frac{\pi}{2}, \pi; \mathbf{l}_K^\pi), \\ \mathcal{R}_B &= \mathcal{M}_B(\pi + \varphi^R, \frac{3\pi}{2}; \mathbf{l}_B^{3\pi/2}), \quad \mathcal{L}_D = \mathcal{M}_D(\frac{3\pi}{2}, \frac{3\pi}{2} + \varphi^L; \mathbf{l}_D^{3\pi/2+\varphi^L}). \end{aligned}$$

(In the third term  $\pi/2 + \varphi^L = \pi - \varphi$ .) The printed  $\mathcal{R}_B$  has  $\pi/2 + \varphi^R$  in place of  $\pi + \varphi^R$  (E27).

**Fact 2.15** (the upper bound). *There is a functional  $\mathcal{Q}$  on  $\mathcal{T}$  such that:*

- (a)  $\xi_K \in \mathcal{T}$  for  $K \in \mathcal{K}^i$ , and  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$ . [1, Thm. 8.1.8, Thm. 8.2.4]
- (b)  $\mathcal{Q}(K, B, D) = \Lambda(K) - \mathcal{S}_K - \mathcal{R}_B - \mathcal{L}_D$ , where  $\Lambda$  (Baek's  $\mathcal{P}_K + \mathcal{S}_K$ , [1, Def. 8.3.4]) is convex-linear on  $\mathcal{K}^i$ . For each of the four terms of  $\mathcal{S}_K$  (resp. for  $\mathcal{R}_B$ , for  $\mathcal{L}_D$ ) the curve  $\mathbf{z}$  above is convex-linear in  $K$  (resp. in  $B$ , in  $D$ ), so that  $\varrho(t)$  is convex-linear for every  $t$ , and each of the terms  $\mathcal{M}$ , being  $\frac{1}{2} \int \varrho^2$ , is convex. [1, Lemma 8.3.3, Lemma 8.3.4, Lemma 8.3.7, Thm. 8.3.1, Thm. 8.3.2, Thm. 7.1.2, Thm. 7.4.1, Thm. 7.4.2] (E20, E23)
- (c)  $\mathcal{Q} \leq \mathcal{Q}(\xi_G) = |G|$  on  $\mathcal{T}$ , where  $\xi_G = \xi_{K_G}$  is the canonical triple of Gerver's cap. [1, Cor. 8.5.8, Thm. 8.4.6] (E25, E26)

Part (a) is stated in [1] with a proof that uses  $\mathcal{N}(K) \subset K$ , which  $\mathcal{K}^i$  does not provide. The statement is true, and [14] proves it without this inclusion (E21). The proofs of [1, Thm. 8.1.8] and [1, Thm. 8.2.4] use [1, Lemma 8.1.5], whose proof deduces that the bottom edge  $e_K(3\pi/2)$  has length  $\sigma_K(3\pi/2) \geq 2.2$  from  $|K| \geq 2.2$  without a reason; this holds, since the horizontal width of a cap with rotation angle  $\pi/2$  does not increase with the height, so that  $|K| \leq \sigma_K(3\pi/2)$  (E19). In (b), the proof of [1, Lemma 8.3.7] uses [1, Lemma 8.3.5] and [1, Lemma 8.3.6 (3)], which need the corrections E20 and E23. In (c), the proof of [1, Thm. 8.4.6] uses [1, Thm. 8.4.3 (2)], and that of [1, Cor. 8.5.8] uses [1, Thm. 8.4.3 (2)] and, through [1, Thm. 8.4.5], [1, Prop. 8.4.4 (4)]; these need E25 and E26.

**2.6. Gerver's sofa.** Seen from a sofa that turns clockwise by  $t \in [0, \pi/2]$ , the hallway occupies  $\mathbf{x}(t) + R_t L$ ; the curve  $\mathbf{x} : [0, \pi/2] \rightarrow \mathbb{R}^2$  is the *rotation path*, and its *shape* is

$$\text{shape}(\mathbf{x}) = H_L \cap \bigcap_{t \in [0, \pi/2]} (\mathbf{x}(t) + R_t L) \cap (\mathbf{x}(\pi/2) + R_{\pi/2} V_L).$$

*Gerver's sofa*  $G$  is the shape of Romik's rotation path [10], which is glued from five explicit curves, on  $[0, \varphi]$ ,  $[\varphi, \theta]$ ,  $[\theta, \pi/2 - \theta]$ ,  $[\pi/2 - \theta, \pi/2 - \varphi]$  and  $[\pi/2 - \varphi, \pi/2]$ , with  $\varphi = 0.03917\dots$  and  $\theta = 0.68130\dots$  (the formulas and Romik's equations are in Appendix A). Write  $t_0, \dots, t_5 = 0, \varphi, \theta, \pi/2 - \theta, \pi/2 - \varphi, \pi/2$ ,  $\alpha = \langle \mathbf{x}', u_t \rangle$ ,  $\beta = \langle \mathbf{x}', v_t \rangle$ , and

$$\mathbf{A} = \mathbf{x} + \alpha v_t + u_t, \quad \mathbf{B} = \mathbf{x} + \alpha v_t, \quad \mathbf{C} = \mathbf{x} - \beta u_t + v_t, \quad \mathbf{D} = \mathbf{x} - \beta u_t$$

for the contact points of the sofa with the walls of the hallway.

**Fact 2.16** (Gerver's sofa). *(a) Romik's equations have exactly one solution with  $\varphi \in [0.039, 0.04]$ ,  $\theta \in [0.68, 0.69]$ , so  $G$  is well defined, and  $2.2192 \leq |G| \leq 2.2199$ . (Romik asserts, without proof, the uniqueness for  $0 < \varphi < \theta < \pi/4$ ; uniqueness in this box is proved in [14].)*

*(b)  $G$  is a monotone sofa with rotation angle  $\pi/2$ . Its cap  $K_G = \mathcal{C}(G)$  satisfies  $G = K_G \setminus \mathcal{N}(K_G)$  (Fact 2.4(c)) and has  $A_{K_G} = \mathbf{A}$ ,  $C_{K_G} = \mathbf{C}$  and  $\mathbf{x}_{K_G} = \mathbf{x}$  on  $[0, \pi/2]$ ; and  $K_G$  satisfies the injectivity condition; as  $|K_G| \geq \mathcal{A}(K_G) = |G| > 11/5$ ,  $K_G \in \mathcal{K}^i$ . [1, Thm. 8.4.1 (1), Thm. 6.1.2, Thm. 8.1.1 (3)]*

*(c) Let  $\Gamma = \mathbf{D}([t_0, t_2]) \cup \mathbf{x}([t_1, t_4]) \cup \mathbf{B}([t_3, t_5])$ . Then  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$  lie on the  $x$ -axis,  $\mathbf{D}(t_2) = \mathbf{x}(t_4)$  and  $\mathbf{B}(t_3) = \mathbf{x}(t_1)$ ; the abscissa is strictly increasing along  $\mathbf{D}$  on  $[t_0, t_2]$ , strictly decreasing along  $\mathbf{x}$  on  $[t_1, t_4]$  and strictly increasing along  $\mathbf{B}$  on  $[t_3, t_5]$ ; at every point of  $[t_0, t_1]$  and of  $[t_1, t_2]$  the curve  $\mathbf{D}$  has a derivative within that interval that is a positive multiple of  $u_t$ , and at every point of  $[t_3, t_4]$  and of  $[t_4, t_5]$  the curve  $\mathbf{B}$  has a derivative within that interval that is a negative multiple of  $v_t$  (one-sided at the ends of these intervals); and*

$$\mathcal{N}(K_G) = \{q : q_y \geq 0, q_y < \gamma_y \text{ for some } \gamma \in \Gamma \text{ with } \gamma_x = q_x\}.$$

[1, Thm. 8.4.1 (2), (4)]

*(d)  $\mathbf{x}(\pi/2)_y = 0$  and  $\mathbf{C}(\pi/2) = (\mathbf{x}(\pi/2)_x - 1, 0)$ .*

Baek states [1, Thm. 8.4.1] without proof (his Remark 8.4.1: the properties are “easy to verify numerically”), and proves [1, Thm. 6.1.2] by a misreading of a theorem of Gerver (E12, E24). The statements are true, with the niche of [1, Thm. 8.4.1 (2)] described as in (c), and the formalization proves them from Romik's equations [14]. Baek's [1, Thm. 8.4.1 (2)] describes the niche as the region enclosed by  $\mathbf{B}$  reversed,  $\mathbf{x}|_{[t_1, t_4]}$ ,  $\mathbf{D}$  reversed and the segment of the  $x$ -axis from  $\mathbf{D}(t_0)$  to  $\mathbf{B}(t_5)$ . That open region omits the open segment of the  $x$ -axis between  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$ , which lies in  $\mathcal{N}(K_G)$ : each of its points lies strictly below a point of  $\Gamma$ . Part (d) is not in [1]; the formalization proves it from Romik's equations too, the second identity as  $\mathbf{C}(\pi/2) = \mathbf{x}(\pi/2) - \beta(\pi/2)u_{\pi/2} + v_{\pi/2}$  with  $\beta(\pi/2) = 0$ . We also use one numerical fact that is not in [1]:

**Fact 2.17** (height of the rotation path).  $\mathbf{x}(t)_y < 1$  for every  $t \in [0, \pi/2]$ , and  $\mathbf{x}(t)_y > 0$  for  $t \in [t_1, t_4]$ .

### 3. FROM A MAXIMAL SOFA TO A MAXIMIZING CAP

Throughout this section  $S$  is a moving sofa with  $|S| = |G|$ . We reduce the problem to caps and then outline the rest of the proof.

**Definition 3.1** (maximizing cap). Let  $\omega \in (0, \pi/2]$ . A cap  $K \in \mathcal{K}_\omega^c$  is *maximizing* if  $\mathcal{A}_\omega(C) \leq \mathcal{A}_\omega(K)$  for every  $C \in \mathcal{K}_\omega^c$ .

By Fact 2.7, every cap with  $\mathcal{A}_\omega(K) = |G|$  is maximizing. A maximizing cap maximizes  $\mathcal{A}_\omega$  over all caps, including those that do not contain their niche. Baek constructs a particular maximizing cap, a limit of maximum polygon caps. A given maximizing cap need not be such a limit.

**Proposition 3.2** (the monotonization of a maximal sofa). *Let  $S'$  be a moving sofa with rotation angle  $\omega \in (0, \pi/2]$  and  $|S'| = |G|$ . Then there are  $w \in \mathbb{R}^2$  and a monotone sofa  $T$  with rotation angle  $\omega$  such that*

$$S' + w \subset T, \quad |T| = |G|,$$

*and the cap  $K = \mathcal{C}(T)$  is a maximizing cap, with  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |G|$ . In particular, if  $S$  is any moving sofa with  $|S| = |G|$ , there is such a  $T$  for  $S' = S$  and some  $\omega \in [\omega_0, \pi/2]$ .*

*Proof.* By Fact 2.3(b),  $S' + w$  is in standard position for some  $w$ , and is a moving sofa with rotation angle  $\omega$ . By Fact 2.4(a),  $T = \mathcal{I}(S' + w)$  is a moving sofa in standard position containing  $S' + w$ , and it is a monotone sofa. So  $|G| = |S' + w| \leq |T| \leq |G|$  by Fact 2.7. By Fact 2.4(b),  $K = \mathcal{C}(T)$  is a cap, and Fact 2.4(c) gives  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |T| = |G|$ ;  $K$  is maximizing by Fact 2.7. Finally, if  $|S| = |G|$  then  $|S| > 11/5$ , so  $S$  has a rotation angle  $\omega \in [\omega_0, \pi/2]$  by Fact 2.3(a).  $\square$

So the problem is to show that  $S$  is a rigid image of  $G$ . The proof follows  $S$  through a chain of sets, each of which contains a rigid image of  $S$  and has area  $|G|$  (Figure 3). Steps (1) and (2) are used twice: first for the maximizing cap  $K$  of Proposition 3.2, which has the rotation angle  $\omega$ , to obtain step (3), and then, after step (3), for the maximizing cap of the new monotone sofa  $U$  and for its mirror image, which have the rotation angle  $\pi/2$ , to obtain step (4).

- (1) *Selection* (Section 4). A given maximizing cap  $K_*$  with  $\mathcal{A}_\omega(K_*) > 0$  is the Hausdorff limit of polygon caps  $K_n$  that maximize the polygon sofa area minus a penalty for straying from  $K_*$ .
- (2) *Variations* (Section 5). Pushing one side of  $K_n$  shows that  $K_n$  is almost balanced: the defect  $\sigma_{K_n} - \tau_{K_n}$  at a normal is bounded by a multiple of the distance from  $K_n$  to  $K_*$  at the sampled normals, and the weights are chosen so that these bounds add up.
- (3) *The right angle* (Section 6). If  $\omega < \pi/2$ , the limit of the defect bounds gives the two *pinned bounds*  $w_{K_*}^\circ \leq \sigma_{K_*}(\pi/2)$  and  $z_{K_*}^\circ \leq \sigma_{K_*}(\omega)$  between wedge gaps and side lengths of  $K_*$ . They give  $T$  width at most one in the directions  $u_t$ ,  $t \in [\omega, \pi/2]$ , so that a rotated copy  $R_\psi T$ ,  $\psi = \pi/2 - \omega$ ,

turns through  $\pi/2$ . Monotonizing  $R_\psi T$  gives a monotone sofa  $U$  with rotation angle  $\pi/2$  and  $|U| = |G|$  that contains a rigid image of  $S$ .

- (4) *Injectivity* (Section 7). For the cap  $K$  of  $U$ , the defect bounds become the curvature bounds  $\sigma_K \leq k_0(g_K^+(t))dt$  on  $[0, \pi/2)$  and  $\sigma_K \leq k_0(f_K^-(t - \pi/2))dt$  on  $(\pi/2, \pi]$ , and Baek's iteration gives the injectivity condition.
- (5) *Equality* (Section 8). Then  $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq \mathcal{Q}(\xi_G) = |G|$ , and equality in the concave functional  $\mathcal{Q}$  forces equality in each Mamikon term, a convex integral  $\frac{1}{2} \int \varrho^2$ . This gives differential equations for the difference of the support functions of  $K$  and of Gerver's cap, whose solutions make  $K$  a horizontal translate of Gerver's cap and  $U$  a horizontal translate of  $G$ .
- (6) *Recovery* (Section 9). A closed subset of  $G$  with the area of  $G$  is  $G$ , since  $G$  is the closure of its interior.

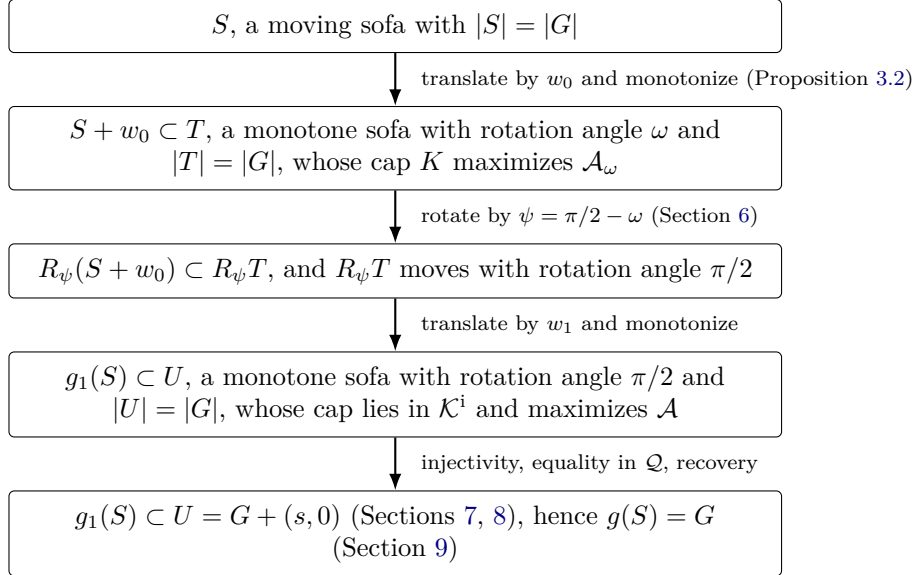


FIGURE 3. The containment chain of the proof. Every set contains the image of  $S$  under the maps so far, and all the sets have area  $|G|$ ; the first three arrows are a translation followed by a monotonization, a rotation, and a translation followed by a monotonization. Here  $g_1(p) = R_\psi(p + w_0) + w_1$  and  $g(p) = g_1(p) - (s, 0)$ .

#### 4. APPROXIMATING A MAXIMIZING CAP BY POLYGON CAPS

In this section and in Section 5 we fix  $\omega \in (0, \pi/2]$  and a cap  $K_* \in \mathcal{K}_\omega^c$  with  $\mathcal{A}_\omega(K_*) > 0$ , and write  $h_* = h_{K_*}$ ; Proposition 4.8 assumes in addition that  $K_*$  maximizes  $\mathcal{A}_\omega$ . Section 6 applies these results, when  $\omega < \pi/2$ , to the cap of Proposition 3.2, a maximizing cap with sofa area  $|G| > 11/5$ .

Section 7 applies them, with  $\omega = \pi/2$ , to the cap of the monotone sofa  $U$  of Proposition 6.9 and to the mirror image of that cap; both are maximizing, with sofa area  $|G| > 0$ .

Baek obtains his cap as a limit of maximum polygon caps, which he proves to be balanced. This procedure need not reach a given maximizing cap: if  $F_n \downarrow F$  uniformly on  $[0, 1]$ , a maximizer of  $F$  need not be a limit of maximizers of the  $F_n$  (take  $F_n(x) = x/n$  and  $F = 0$ , whose maximizers are all of  $[0, 1]$  while those of  $F_n$  are  $\{1\}$ ). We therefore subtract a penalty that vanishes at the chosen maximizer, as in the example  $F_n(x) - (x - x_*)^2$ , whose maximizers converge to  $x_*$ . The penalty is a weighted sum of squared differences of support values at dyadic normals. Section 5 bounds the resulting departure from balance.

**4.1. Dyadic angles and the penalty.** For  $k \geq 0$  let  $\Theta_k = \{i\omega/2^{k+1} : 0 < i < 2^{k+1}\}$ , an angle set with  $2^{k+1}$  intervals. Then  $\Theta_k \subset \Theta_{k+1}$ ,  $\omega/2 \in \Theta_k$ , and  $\bigcup_k \Theta_k$  is dense in  $[0, \omega]$ . Write  $\mathcal{K}_k^c = \mathcal{K}_{\Theta_k}^c$  for the polygon caps with angle set  $\Theta_k$  and  $\Theta_k^\circ$  for the defining normals.

For  $m \geq 0$  let  $\nu_m = 2^{-m-1}/(2|\Theta_m|)$ . For a convex body  $K$  and  $k \geq 0$  let

$$\Pi_k(K) = \sum_{m=0}^k \nu_m \sum_{t \in \Theta_m} \left[ (h_K(t) - h_*(t))^2 + (h_K(t + \frac{\pi}{2}) - h_*(t + \frac{\pi}{2}))^2 \right].$$

The terms are the *samples*; an angle of level  $m$  also lies in every later level, so it carries one sample from each level, from the first level that contains it up to  $k$ . For  $s \in \mathbb{R}$  let  $\nu_k(s) \geq 0$  be the total weight of the samples taken at the normal  $s$ , so that  $\nu_k(s) = 0$  unless  $s \in \Theta_k \cup (\Theta_k + \pi/2)$ . The samples of level  $m$  have total weight  $2|\Theta_m|\nu_m = 2^{-m-1}$ , hence

$$\sum_s \nu_k(s) = \sum_{m=0}^k 2^{-m-1} = 1 - 2^{-k-1} \leq 1. \quad (4.1)$$

Since every term of  $\Pi_k$  is nonnegative, each sample is at most the penalty (*persistence*): if  $m \leq k$  and  $t \in \Theta_m$ , then

$$\nu_m(h_K(s) - h_*(s))^2 \leq \Pi_k(K) \quad (s = t \text{ and } s = t + \pi/2). \quad (4.2)$$

Let  $K$  and  $K'$  be convex bodies and  $\eta \geq 0$  with  $|h_K(s) - h_*(s)| \leq \eta$  at every sampled normal  $s$ . For reals  $a, b, c, r$  with  $|a - c| \leq \eta$  and  $|b - a| \leq r$ , the identity  $(b - c)^2 - (a - c)^2 = 2(a - c)(b - a) + (b - a)^2$  gives  $|(b - c)^2 - (a - c)^2| \leq 2\eta r + r^2$ ; we apply it to each sample, with  $a = h_K(s)$ ,  $b = h_{K'}(s)$  and  $c = h_*(s)$ . If  $|h_{K'}(s) - h_K(s)| \leq r$  at every sampled normal  $s$ , this gives

$$|\Pi_k(K') - \Pi_k(K)| \leq \left( \sum_s \nu_k(s) \right) (2\eta r + r^2). \quad (4.3)$$

If  $h_{K'}(s) = h_K(s)$  at every sampled normal  $s \neq s_0$  and  $|h_{K'}(s_0) - h_K(s_0)| \leq r$ , only the samples at  $s_0$ , of total weight  $\nu_k(s_0)$ , contribute, and

$$|\Pi_k(K') - \Pi_k(K)| \leq \nu_k(s_0) (2\eta r + r^2). \quad (4.4)$$

**Definition 4.1** (penalized maximizer). A polygon cap  $K \in \mathcal{K}_k^c$  is a *penalized maximizer at stage  $k$*  if  $\mathcal{A}_{\Theta_k}(C) - \Pi_k(C) \leq \mathcal{A}_{\Theta_k}(K) - \Pi_k(K)$  for every  $C \in \mathcal{K}_k^c$ .

**Lemma 4.2** (recovery polygon).  $K_k^r = \mathcal{C}_{\Theta_k}(K_*)$  lies in  $\mathcal{K}_k^c$  and has  $\Pi_k(K_k^r) = 0$  and  $\mathcal{A}_{\Theta_k}(K_k^r) - \Pi_k(K_k^r) \geq \mathcal{A}_\omega(K_*)$ .

*Proof.* By Fact 2.8,  $K_k^r$  is a polygon cap with the support values of  $K_*$  on  $\Theta_k^\circ$ , which contains all the sampled normals; so  $\Pi_k(K_k^r) = 0$ . Also  $\mathcal{A}_{\Theta_k}(K_k^r) = \mathcal{A}_{\Theta_k}(K_*) \geq \mathcal{A}_\omega(K_*)$ , by Fact 2.8.  $\square$

**Lemma 4.3** (two samples bound a cap). Let  $\Theta$  be an angle set,  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \Theta$ , and suppose that  $q(h_K(s) - h_*(s))^2 \leq c$  for  $s = t$  and  $s = t + \pi/2$ , with  $q > 0$ ,  $c \geq 0$ . Then  $K \subset [-R, R] \times [0, 1]$  with  $R = H/\sin t + H/\cos t$ ,  $H = |h_*(t)| + |h_*(t + \pi/2)| + c/q + 1$ .

*Proof.* A real number  $a$  with  $a^2 \leq c/q$  has  $|a| \leq c/q + 1$ , as  $2|a| \leq a^2 + 1$ . So  $h_K(t) \leq h_*(t) + c/q + 1 \leq H$ , and likewise  $h_K(t + \pi/2) \leq H$ . A point  $(x, y) \in K$  has  $0 \leq y \leq 1$ , since  $K \subset P_\omega$ . As  $t \in (0, \pi/2)$ ,  $x \cos t + y \sin t \leq H$  gives  $x \leq H/\cos t$ , and  $-x \sin t + y \cos t \leq H$  gives  $-x \leq H/\sin t$ ; both bounds are at most  $R$ , since  $H \geq 0$ .  $\square$

**Lemma 4.4** (the niche lies in a box). Let  $R \geq 0$ . If  $K \in \mathcal{K}_\omega^c$  and  $K \subset [-R, R] \times [0, 1]$ , then  $\mathcal{N}(K) \subset [-R, R] \times [0, 2R]$ ; the same holds for  $\mathcal{N}_\Theta(K) \subset \mathcal{N}(K)$ , for every angle set  $\Theta$ .

*Proof.* Let  $p \in \mathcal{N}(K)$ , so  $p_y \geq 0$  and  $p \in Q_K^-(t)$  for some  $t \in (0, \omega)$ . As  $K \subset [-R, R] \times [0, 1]$ ,  $h_K(t) \leq R \cos t + \sin t \leq R \cos t + 1$  and  $h_K(t + \pi/2) \leq R \sin t + \cos t \leq R \sin t + 1$ . So  $\langle p, u_t \rangle < R \cos t$  and  $\langle p, v_t \rangle < R \sin t$ . Since  $p_y \geq 0$  and  $\langle p, u_t \rangle = p_x \cos t + p_y \sin t$ , we get  $p_x < R$ ; likewise  $\langle p, v_t \rangle = -p_x \sin t + p_y \cos t$  gives  $p_x > -R$ . Finally  $p_y = \sin t \langle p, u_t \rangle + \cos t \langle p, v_t \rangle < 2R \sin t \cos t \leq 2R$ , as  $R \geq 0$ .  $\square$

**Proposition 4.5** (penalized maximizers exist). For every  $k \geq 0$  there is a penalized maximizer at stage  $k$ , and every penalized maximizer  $K$  at stage  $k$  satisfies

$$\mathcal{A}_{\Theta_k}(K) - \Pi_k(K) \geq \mathcal{A}_\omega(K_*).$$

If  $K^{(k)}$  is a penalized maximizer at stage  $k$  for every  $k \geq 0$ , there is  $R_0 \geq 0$  such that  $K^{(k)} \subset [-R_0, R_0] \times [0, 1]$  for every  $k$ .

*Proof.* Let  $\Psi_k = \mathcal{A}_{\Theta_k} - \Pi_k$  on  $\mathcal{K}_k^c$ , and let  $\bar{t} = \omega/2$ , the angle of  $\Theta_0$ , which lies in every  $\Theta_k$ . Let  $c = c_{\omega, \bar{t}} > 0$  be the constant of Fact 2.9; it does not depend on  $k$ .

The polygon caps with  $\Psi_k > 0$  lie in one box. Let  $C \in \mathcal{K}_k^c$  with  $\Psi_k(C) > 0$ . As  $\Pi_k(C) \geq 0$ ,  $\mathcal{A}_{\Theta_k}(C) > 0$ , so  $C$  has width at most  $c$  and, as it lies in the strip  $0 \leq y \leq 1$ ,  $|C| \leq c$ . By Fact 2.8,  $\Pi_k(C) < \mathcal{A}_{\Theta_k}(C) = |C| - |\mathcal{N}_{\Theta_k}(C)| \leq c$ . The samples of level 0 lie at  $\bar{t}$  and  $\bar{t} + \pi/2$ , so (4.2) gives  $\nu_0(h_C(s) - h_*(s))^2 \leq$

$\Pi_k(C) \leq c$  for  $s = \bar{t}, \bar{t} + \pi/2$ , and Lemma 4.3 with  $q = \nu_0$  puts  $C$  in  $[-R_0, R_0] \times [0, 1]$ , where

$$R_0 = \frac{H}{\sin \bar{t}} + \frac{H}{\cos \bar{t}} \geq 0, \quad H = |h_*(\bar{t})| + |h_*(\bar{t} + \frac{\pi}{2})| + \frac{c}{\nu_0} + 1,$$

do not depend on  $k$  or  $C$ . Moreover  $\Psi_k(C) \leq \mathcal{A}_{\Theta_k}(C) \leq c$ ; hence  $\Psi_k \leq c$  on  $\mathcal{K}_k^c$ , as  $c > 0$ .

*A maximizing sequence and its limit cap.* By Lemma 4.2,  $M_k = \sup \Psi_k \geq \Psi_k(K_k^r) \geq \mathcal{A}_\omega(K_*) > 0$ , and  $M_k \leq c$ . Take  $C_j \in \mathcal{K}_k^c$  with  $\Psi_k(C_j) > 0$  and  $\Psi_k(C_j) \rightarrow M_k$ . They lie in  $[-R_0, R_0] \times [0, 1]$ , so  $|h_{C_j}| \leq R_0 + 1$ , and as  $\Theta_k^\diamond$  is finite, a subsequence, still written  $(C_j)$ , has  $h_{C_j}(s) \rightarrow h_\infty(s)$  for every  $s \in \Theta_k^\diamond$ . Let  $I = \Theta_k^\diamond \cup \{\omega + \pi, 3\pi/2\}$ , the normal angles of the polygon caps in  $\mathcal{K}_k^c$ , and put  $h_\infty(\omega + \pi) = h_\infty(3\pi/2) = 0$ ; then  $h_{C_j}(s) \rightarrow h_\infty(s)$  for every  $s \in I$ , and  $h_\infty(\omega) = h_\infty(\pi/2) = 1$ . For  $\delta \geq 0$  let  $K_\infty^\delta$  be the set of points  $p$  with  $\langle p, u_s \rangle \leq h_\infty(s) + \delta$  for every  $s \in I$ . It is closed and convex, and bounded, since it lies in the strip  $-\delta \leq y \leq 1 + \delta$  and in  $H_-(t, h_\infty(t) + \delta) \cap H_-(t + \pi/2, h_\infty(t + \pi/2) + \delta)$  for  $t \in \Theta_k \subset (0, \pi/2)$ . Let  $K_\infty = K_\infty^0$ ; it is the set  $\mathcal{C}_{\Theta_k}(h_\infty)$  of Section 2.3. Let  $r \in I$ , and take  $q_j \in C_j$  with  $\langle q_j, u_r \rangle = h_{C_j}(r)$ . As the  $q_j$  lie in  $[-R_0, R_0] \times [0, 1]$ , a subsequence of  $(q_j)$  converges to a point  $q$ ; passing to the limit in  $\langle q_j, u_s \rangle \leq h_{C_j}(s)$ ,  $s \in I$ , gives  $q \in K_\infty$ , and  $\langle q, u_r \rangle = h_\infty(r)$ . So  $K_\infty$  is nonempty and  $h_{K_\infty}(r) = h_\infty(r)$ . Hence  $K_\infty$  is a convex body, with the support values 1, 1, 0, 0 at  $\omega, \pi/2, \omega + \pi, 3\pi/2$ , and an intersection of half-planes with normal angles in  $I$ ; that is,  $K_\infty \in \mathcal{K}_k^c$ . Also  $h_{C_j}(s) \rightarrow h_{K_\infty}(s)$  for every  $s \in I$ .

*$K_\infty$  is a penalized maximizer.* The sampled normals lie in  $\Theta_k^\diamond$ , so  $\Pi_k(C_j) \rightarrow \Pi_k(K_\infty)$ . Let  $\delta > 0$ . For large  $j$ ,  $h_{C_j}(s) < h_\infty(s) + \delta$  at the finitely many  $s \in \Theta_k^\diamond$ , and  $h_{C_j}(s) = h_\infty(s) = 0$  for  $s = \omega + \pi, 3\pi/2$ ; so  $C_j \subset K_\infty^\delta$ . As  $\delta \downarrow 0$  the bounded sets  $K_\infty^\delta$  decrease to  $K_\infty$ , so  $|K_\infty^\delta| \rightarrow |K_\infty|$ , and therefore  $\limsup_j |C_j| \leq |K_\infty|$ . Let  $p \in \mathcal{N}_{\Theta_k}(K_\infty)$ :  $p \in F_\omega$ , and  $\langle p, u_t \rangle < h_\infty(t) - 1$  and  $\langle p, v_t \rangle < h_\infty(t + \pi/2) - 1$  for some  $t \in \Theta_k$ . These strict inequalities hold with  $h_{C_j}$  in place of  $h_\infty$  for large  $j$ , so  $p \in \mathcal{N}_{\Theta_k}(C_j)$  for all large  $j$ . As  $h_{C_j} \leq R_0 + 1$ , all the niches  $\mathcal{N}_{\Theta_k}(C_j)$  lie in the set of points  $p \in F_\omega$  with  $\langle p, u_t \rangle < R_0$  and  $\langle p, v_t \rangle < R_0$  for some  $t \in \Theta_k$ , which is bounded:  $p_y \geq 0$  gives  $p_x < R_0/\cos t$  and  $-p_x < R_0/\sin t$ , and  $p_y = \sin t \langle p, u_t \rangle + \cos t \langle p, v_t \rangle < 2R_0$ . So their areas are finite, and Fatou's lemma gives  $|\mathcal{N}_{\Theta_k}(K_\infty)| \leq \liminf_j |\mathcal{N}_{\Theta_k}(C_j)|$ . Since  $\Psi_k = |\cdot| - |\mathcal{N}_{\Theta_k}(\cdot)| - \Pi_k$  on  $\mathcal{K}_k^c$  (Fact 2.8),

$$M_k = \lim_j \Psi_k(C_j) \leq \limsup_j |C_j| - \liminf_j |\mathcal{N}_{\Theta_k}(C_j)| - \lim_j \Pi_k(C_j) \leq \Psi_k(K_\infty).$$

As  $\Psi_k \leq M_k$  on  $\mathcal{K}_k^c$ ,  $K_\infty$  is a penalized maximizer at stage  $k$ .

*The lower bound and the box.* A penalized maximizer  $K$  at stage  $k$  has  $\Psi_k(K) \geq \Psi_k(K_k^r) \geq \mathcal{A}_\omega(K_*)$  by Lemma 4.2. If  $K^{(k)}$  is a penalized maximizer at stage  $k$  for every  $k$ , then  $\Psi_k(K^{(k)}) \geq \mathcal{A}_\omega(K_*) > 0$ , and the first step puts every  $K^{(k)}$  in  $[-R_0, R_0] \times [0, 1]$ .  $\square$

#### 4.2. Convergence to the given cap.

**Lemma 4.6** (limits of polygon caps with nested angle sets). *Let  $k_1 < k_2 < \dots$  and  $R \geq 0$ , and let  $C_j \in \mathcal{K}_{k_j}^c$  lie in  $[-R, R] \times [0, 1]$  and converge in  $d_H$  to a convex body  $K_\infty$ , whose niche  $\mathcal{N}(K_\infty)$  and sofa area  $\mathcal{A}_\omega(K_\infty)$  are given by the formulas of Section 2.2. Then for every  $\varepsilon > 0$ ,  $\mathcal{A}_{\Theta_{k_j}}(C_j) \leq \mathcal{A}_\omega(K_\infty) + \varepsilon$  for all large  $j$ .*

*Proof.* Let  $\varepsilon > 0$ . By the continuity of the area (Fact 2.1(c)),  $|C_j| \leq |K_\infty| + \varepsilon/2$  for large  $j$ . Let  $p \in \mathcal{N}(K_\infty)$ :  $p \in F_\omega \cap Q_{K_\infty}^-(t)$  for some  $t \in (0, \omega)$ . The two strict inequalities that define  $Q_{K_\infty}^-(t)$  hold at  $p$  with a margin  $\delta > 0$  at some dyadic angle  $t'$  near  $t$ , since  $h_{K_\infty}$  is continuous. For large  $j$ ,  $t' \in \Theta_{k_j}$  and  $\sup |h_{C_j} - h_{K_\infty}| < \delta/2$ , so  $p \in F_\omega \cap Q_{C_j}^-(t') \subset \mathcal{N}_{\Theta_{k_j}}(C_j)$ . The sets  $\mathcal{N}_{\Theta_{k_j}}(C_j)$  lie in  $[-R, R] \times [0, 2R]$  by Lemma 4.4 (so their areas are finite), and Fatou's lemma gives  $|\mathcal{N}(K_\infty)| \leq \liminf_j |\mathcal{N}_{\Theta_{k_j}}(C_j)|$ ; so  $|\mathcal{N}(K_\infty)| \leq |\mathcal{N}_{\Theta_{k_j}}(C_j)| + \varepsilon/2$  for large  $j$ . As  $\mathcal{A}_{\Theta_{k_j}}(C_j) = |C_j| - |\mathcal{N}_{\Theta_{k_j}}(C_j)|$  (Fact 2.8), for large  $j$

$$\mathcal{A}_{\Theta_{k_j}}(C_j) \leq |K_\infty| - |\mathcal{N}(K_\infty)| + \varepsilon = \mathcal{A}_\omega(K_\infty) + \varepsilon.$$

□

**Lemma 4.7** (dyadic supports determine a cap). *If  $K, K' \in \mathcal{K}_\omega^c$  and  $h_K(s) = h_{K'}(s)$  for all  $s \in \bigcup_m (\Theta_m \cup (\Theta_m + \pi/2))$ , then  $K = K'$ .*

*Proof.* By continuity and density,  $h_K = h_{K'}$  on  $J_\omega$ , and  $h_K = h_{K'} = 0$  at  $\omega + \pi$  and  $3\pi/2$ . For a cap  $K$ ,  $F_\omega = H_K(\omega + \pi) \cap H_K(3\pi/2)$ , so by (2.2)  $K$  is the intersection of its supporting half-planes  $H_K(s)$ ,  $s \in J_\omega \cup \{\omega + \pi, 3\pi/2\}$ ; the same holds for  $K'$ , and these half-planes are those of  $K$ . □

**Proposition 4.8** (selection). *If  $K_*$  maximizes  $\mathcal{A}_\omega$  on  $\mathcal{K}_\omega^c$ , there are  $k_1 < k_2 < \dots$ , penalized maximizers  $K_j$  at stage  $k_j$ , and  $R_0 \geq 0$  such that  $K_j \subset [-R_0, R_0] \times [0, 1]$  for every  $j$  and  $K_j \rightarrow K_*$  in  $d_H$ .*

*Proof.* By Proposition 4.5, for every  $k$  there is a penalized maximizer  $K^{(k)}$  at stage  $k$ , and there is  $R_0 \geq 0$  with  $K^{(k)} \subset [-R_0, R_0] \times [0, 1]$  for every  $k$ . By Blaschke's theorem (Fact 2.1(c)) there are  $k_1 < k_2 < \dots$  with  $K_j = K^{(k_j)} \rightarrow K_\infty$  for a convex body  $K_\infty$ ; the  $K_j$  lie in  $[-R_0, R_0] \times [0, 1]$ .

$K_\infty$  is a cap with rotation angle  $\omega$ . The support values at  $\omega, \pi/2, \omega + \pi, 3\pi/2$  pass to the limit. Let  $(a, b)$  be a gap of  $J_\omega \cup \{\omega + \pi, 3\pi/2\}$  in the circle; it has length at most  $\pi/2$ . A polygon cap  $K_j$  has no normal angle strictly between  $a$  and  $b$ , so the supporting lines  $l_{K_j}(a)$  and  $l_{K_j}(b)$  meet at a vertex  $q$  of  $K_j$  and  $h_{K_j}(r) = \langle q, u_r \rangle$  for  $a \leq r \leq b$ ; with  $\sin(b-a)u_r = \sin(b-r)u_a + \sin(r-a)u_b$  this gives  $\sin(b-a)h_{K_j}(r) = \sin(b-r)h_{K_j}(a) + \sin(r-a)h_{K_j}(b)$ . The identity passes to  $h_{K_\infty}$ , and its coefficients are nonnegative, so  $H_{K_\infty}(a) \cap H_{K_\infty}(b) \subset H_{K_\infty}(r)$ . Hence  $K_\infty = \bigcap_r H_{K_\infty}(r)$  is the intersection of the  $H_{K_\infty}(r)$  with  $r \in J_\omega \cup \{\omega + \pi, 3\pi/2\}$ .

The penalties tend to 0. By Proposition 4.5,  $0 \leq \Pi_{k_j}(K_j) \leq \mathcal{A}_{\Theta_{k_j}}(K_j) - \mathcal{A}_\omega(K_*)$ . Let  $\varepsilon > 0$ . By Lemma 4.6 with  $R = R_0$ , and the maximality of  $K_*$ ,

$\mathcal{A}_{\Theta_{k_j}}(K_j) \leq \mathcal{A}_\omega(K_\infty) + \varepsilon \leq \mathcal{A}_\omega(K_*) + \varepsilon$  for all large  $j$ . So  $\Pi_{k_j}(K_j) \leq \varepsilon$  for all large  $j$ , and  $\Pi_{k_j}(K_j) \rightarrow 0$ .

$K_\infty = K_*$ . Let  $t \in \Theta_m$ . For  $k_j \geq m$ , (4.2) gives  $\nu_m(h_{K_j}(s) - h_*(s))^2 \leq \Pi_{k_j}(K_j)$  for  $s = t, t + \pi/2$ . As  $j \rightarrow \infty$ ,  $\Pi_{k_j}(K_j) \rightarrow 0$  and  $h_{K_j}(s) \rightarrow h_{K_\infty}(s)$ , and  $\nu_m > 0$ ; so  $h_{K_\infty}(s) = h_*(s)$ . Hence  $h_{K_\infty} = h_*$  at all dyadic normals, and  $K_\infty = K_*$  by Lemma 4.7.  $\square$

For a selection as in Proposition 4.8, put  $\eta_j = d_H(K_j, K_*) \rightarrow 0$ ; at every sampled normal,  $|h_{K_j}(s) - h_*(s)| \leq \eta_j$ .

## 5. PENALIZED MAXIMIZERS ARE ALMOST BALANCED

A maximum polygon cap is balanced [1, Thm. 3.4.9]:  $\sigma_K(t) = \tau_K(t)$  for every  $t \in \Theta^\circ$ , since otherwise pushing one side would increase  $\mathcal{A}_\Theta$ . A penalized maximizer need not be balanced, but pushing a side changes the penalty by a controlled amount, and the argument gives a one-sided bound on the defect  $\zeta_K(t) = \sigma_K(t) - \tau_K(t)$ . In this section we fix  $k \geq 0$ ,  $\Theta = \Theta_k$ , a penalized maximizer  $K \in \mathcal{K}_k^c$  at stage  $k$ , and  $\eta \geq 0$  with  $|h_K(s) - h_*(s)| \leq \eta$  at every sampled normal  $s$ . Let  $\bar{\nu} = \sum_s \nu_k(s)$  be the total weight;  $\bar{\nu} \leq 1$  by (4.1).

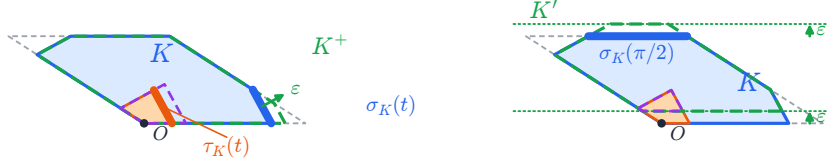


FIGURE 4. Pushing one side of a polygon cap  $K$  with angle set  $\{\omega/2\}$ ,  $\omega = 1$ ; the grey dashed parallelogram is  $P_\omega$ , and the niche of  $K$  is orange. Left: raising the height at the floating normal  $t = \omega/2$  by  $\varepsilon$  gives the cap  $K^+$  (green, dashed) and its niche (purple, dashed); the cap gains a strip of area  $\sigma_K(t)\varepsilon + O(\varepsilon^2)$  along its edge with normal  $t$ , and the niche a strip of area  $\tau_K(t)\varepsilon + O(\varepsilon^2)$  along the side on the inner wall  $b_K(t)$ . Right: raising the pinned height at  $\pi/2$  by  $\varepsilon$  moves both lines of the horizontal strip up by  $\varepsilon$  (green, dotted) and gives  $K'$  (green, dashed), with its niche (purple, dashed); the cap gains a strip along its top edge, of length  $\sigma_K(\pi/2)$ , and loses one along its bottom edge, and  $K'$  is a translate of a polygon cap.

### 5.1. Floating normals.

**Lemma 5.1** (raising a floating height). *Let  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \Theta^\circ \setminus \{\omega, \pi/2\}$ ,  $\varepsilon \geq 0$ , and  $K^+ = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$ . Then  $K^+ \in \mathcal{K}_\Theta^c$ ,  $K \subset K^+$ ,  $h_{K^+}(s) = h_K(s)$  for  $s \in \Theta^\circ \setminus \{t\}$ , and  $h_K(t) \leq h_{K^+}(t) \leq h_K(t) + \varepsilon$ .*

*Proof.* Since  $h_K(\omega) = h_K(\pi/2) = 1$ ,  $P_{h_K^{t,\varepsilon}} = P_\omega$ , so

$$K^+ = P_\omega \cap \bigcap_s H_-(s, h_K^{t,\varepsilon}(s)),$$

the intersection over the floating normals  $s$ . As  $h_K \leq h_K^{t,\varepsilon}$ ,  $K = \mathcal{C}_\Theta(h_K) \subset K^+ \subset P_\omega$ . So  $K^+$  is a convex body (it is bounded, since the half-planes with normals  $t'$  and  $t' + \pi/2$ ,  $t' \in \Theta$ , bound it also when  $\omega = \pi/2$ ), it has the support values 1, 1, 0, 0 of  $P_\omega$  at  $\omega, \pi/2, \omega + \pi, 3\pi/2$ , and it is an intersection of half-planes with the normal angles of a polygon cap. So  $K^+ \in \mathcal{K}_\Theta^c$ . Next,  $h_{K^+} \geq h_K$  since  $K \subset K^+$ , and  $h_{K^+}(s) \leq h_K^{t,\varepsilon}(s)$  since  $K^+ \subset H_-(s, h_K^{t,\varepsilon}(s))$ .  $\square$

**Proposition 5.2** (floating defects). *For every floating normal  $t \in \Theta^\diamond \setminus \{\omega, \pi/2\}$ ,*

$$\sigma_K(t) - \tau_K(t) \leq 2\eta \nu_k(t).$$

*Proof.* Let  $K_\varepsilon^+ = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$ . By Lemma 5.1,  $K_\varepsilon^+$  is a polygon cap that has the support values of  $K$  on  $\Theta^\diamond \setminus \{t\}$ , and differs from  $K$  at  $t$  by at most  $\varepsilon$ ; so by (4.4),  $\Pi_k(K_\varepsilon^+) - \Pi_k(K) \leq \nu_k(t)(2\eta\varepsilon + \varepsilon^2)$ . Penalized maximality gives  $\mathcal{A}_\Theta(K_\varepsilon^+) - \mathcal{A}_\Theta(K) \leq \Pi_k(K_\varepsilon^+) - \Pi_k(K)$ . By Fact 2.11(c) (with  $C = K_\varepsilon^+$ ,  $v = 0$ ) and (a), for small  $\varepsilon$

$$\mathcal{A}_\Theta(K_\varepsilon^+) \geq \mathcal{A}_\Theta(h_K^{t,\varepsilon}) \geq \mathcal{A}_\Theta(K) + (\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2.$$

Hence  $(\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2 \leq \nu_k(t)(2\eta\varepsilon + \varepsilon^2)$ . Divide by  $\varepsilon$  and let  $\varepsilon \rightarrow 0$ .  $\square$

**5.2. Pinned normals.** In this subsection  $\omega < \pi/2$ , so that the two pinned normals  $\omega$  and  $\pi/2$  are distinct and every cap contains  $O$  and  $o_\omega$  (Lemma 2.5); in particular  $h_K \geq 0$ . Moving a pinned normal moves both lines of a strip, and the result is in general only a translate of a polygon cap.

**Lemma 5.3** (the pinned sandwich). *Let  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \{\omega, \pi/2\}$ ,  $\varepsilon \in [0, 1]$ , and  $K' = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$ . Then*

$$(1 - \varepsilon)K + \varepsilon o_\omega \subset K' \subset (1 + \varepsilon)K.$$

*If  $R \geq 0$  and  $|h_K(s)| \leq R$  for all  $s$ , then  $|h_{K'}(s) - h_K(s)| \leq 2R\varepsilon$  for every  $s$ .*

*Proof.* Let  $t'$  be the other pinned normal. Then  $K' = \{p : \varepsilon \leq \langle p, u_t \rangle \leq 1 + \varepsilon, 0 \leq \langle p, u_{t'} \rangle \leq 1, \langle p, u_s \rangle \leq h_K(s) \text{ for floating } s\}$ . Let  $p \in K$  and  $q = (1 - \varepsilon)p + \varepsilon o_\omega \in K'$ . Then  $q$  satisfies every inequality of  $K'$  except possibly the one for  $t$ , and  $\langle q, u_t \rangle = (1 - \varepsilon)\langle p, u_t \rangle + \varepsilon \in [\varepsilon, 1]$  because  $\langle o_\omega, u_t \rangle = 1$  and  $0 \leq \langle p, u_t \rangle \leq 1$ . So  $q \in K'$ . Conversely, let  $p \in K'$ . For floating  $s$ ,  $\langle p/(1 + \varepsilon), u_s \rangle \leq h_K(s)/(1 + \varepsilon) \leq h_K(s)$  as  $h_K(s) \geq 0$ ;  $\langle p, u_t \rangle/(1 + \varepsilon) \in [0, 1]$  and  $\langle p, u_{t'} \rangle/(1 + \varepsilon) \in [0, 1]$ . So  $p/(1 + \varepsilon) \in K$ . Taking support functions,  $(1 - \varepsilon)h_K(s) + \varepsilon \langle o_\omega, u_s \rangle \leq h_{K'}(s) \leq (1 + \varepsilon)h_K(s)$ . As  $o_\omega \in K$ ,  $-\langle o_\omega, u_s \rangle = \langle o_\omega, u_{s+\pi} \rangle \leq h_K(s + \pi) \leq R$ ; with  $|h_K(s)| \leq R$  this gives  $-2R\varepsilon \leq h_{K'}(s) - h_K(s) \leq R\varepsilon$ .  $\square$

**Lemma 5.4** (moving back to standard position). *Let  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \{\omega, \pi/2\}$ ,  $\varepsilon \geq 0$ , and suppose that  $K' = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$  is a translate  $C + v$  of a polygon cap  $C \in \mathcal{K}_\Theta^c$ ,  $v \in \mathbb{R}^2$ . Then  $|\langle v, u_s \rangle| \leq (2/\cos \omega + 1)\varepsilon$  for all  $s$ . If moreover  $\varepsilon \leq 1$ ,  $R \geq 0$  and  $|h_K(s)| \leq R$  for all  $s$ , then*

$$|h_C(s) - h_K(s)| \leq \gamma(R)\varepsilon \quad \text{for every } s, \quad \gamma(R) = 2R + \frac{2}{\cos \omega} + 1.$$

*Proof.* Let  $t'$  be the other pinned normal. For  $s \in \{\omega, \pi/2\}$ ,  $K'$  lies in the strip  $a_s \leq \langle p, u_s \rangle \leq a_s + 1$ , where  $a_t = \varepsilon$  and  $a_{t'} = 0$ . As  $C$  is a cap, it has width exactly 1 in the direction  $u_s$  (its support values at  $s$  and  $s + \pi$  are 1 and 0), and so has  $C + v$ ; hence  $\langle v, u_s \rangle = a_s$ . With  $s = \pi/2$  and  $s = \omega$  this gives  $v_y \in [0, \varepsilon]$  and  $v_x \cos \omega + v_y \sin \omega \in [0, \varepsilon]$ , so  $|v_x| \leq 2\varepsilon/\cos \omega$  and  $|\langle v, u_s \rangle| \leq |v_x| + |v_y| \leq (2/\cos \omega + 1)\varepsilon$ . Finally  $h_C(s) = h_{K'}(s) - \langle v, u_s \rangle$ , and Lemma 5.3 gives  $|h_{K'}(s) - h_K(s)| \leq 2R\varepsilon$ ; so  $|h_C(s) - h_K(s)| \leq 2R\varepsilon + (2/\cos \omega + 1)\varepsilon = \gamma(R)\varepsilon$ .  $\square$

**Proposition 5.5** (pinned defects). *Let  $\omega < \pi/2$ ,  $R \geq 0$  with  $|h_K(s)| \leq R$  for all  $s$ , and  $t \in \{\omega, \pi/2\}$ . Then*

$$\sigma_K(t) - \tau_K(t) \leq 2\bar{\nu}\eta\gamma(R), \quad \gamma(R) = 2R + \frac{2}{\cos \omega} + 1.$$

*Proof.* If  $\sigma_K(t) = 0$ , the left side is  $-\tau_K(t) \leq 0$ . Let  $\sigma_K(t) > 0$ . By Fact 2.11(b), for small  $\varepsilon \in (0, 1]$  the set  $\mathcal{C}_\Theta(h_K^{t,\varepsilon})$  is a translate  $C_\varepsilon + v_\varepsilon$  of a polygon cap  $C_\varepsilon$ . By Lemma 5.4,  $|h_{C_\varepsilon}(s) - h_K(s)| \leq \gamma(R)\varepsilon$  at every  $s$ , so (4.3) gives  $\Pi_k(C_\varepsilon) - \Pi_k(K) \leq \bar{\nu}(2\eta\gamma(R)\varepsilon + \gamma(R)^2\varepsilon^2)$ . Penalized maximality compares  $K$  with  $C_\varepsilon$ , and Fact 2.11(c), (a) give

$$\mathcal{A}_\Theta(C_\varepsilon) \geq \mathcal{A}_\Theta(h_K^{t,\varepsilon}) \geq \mathcal{A}_\Theta(K) + (\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2.$$

So  $(\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2 \leq \bar{\nu}(2\eta\gamma(R)\varepsilon + \gamma(R)^2\varepsilon^2)$ ; divide by  $\varepsilon$  and let  $\varepsilon \rightarrow 0$ .  $\square$

### 5.3. A two-sided bound.

**Corollary 5.6** (two-sided defect bound). *Let  $\omega < \pi/2$ , and  $R \geq 0$  with  $|h_K(s)| \leq R$  for all  $s$ . For every  $t \in \Theta^\circ$ ,*

$$|\sigma_K(t) - \tau_K(t)| \leq \frac{2\bar{\nu}\eta + 4\bar{\nu}\eta\gamma(R)}{\sin t}, \quad \gamma(R) = 2R + \frac{2}{\cos \omega} + 1.$$

*Proof.* Put  $\zeta(s) = \sigma_K(s) - \tau_K(s)$ , and  $e(s) = 2\eta\nu_k(s)$  for floating  $s$ ,  $e(s) = 2\bar{\nu}\eta\gamma(R)$  for the two pinned  $s$ . By Propositions 5.2 and 5.5,  $\zeta \leq e$  on  $\Theta^\circ$ , and  $e \geq 0$  with  $\sum_s e(s) \leq 2\bar{\nu}\eta + 4\bar{\nu}\eta\gamma(R) =: E$ . By Fact 2.10(c), the sums  $\sum_s \sigma_K(s) \sin s$  and  $\sum_s \tau_K(s) \sin s$  both equal  $A_K^-(0)_x - C_K^+(\omega)_x$ , so  $\sum_s \sin s \zeta(s) = 0$ ; and  $0 < \sin s \leq 1$  on  $\Theta^\circ \subset (0, \pi)$ . So

$$\sin t \zeta(t) = - \sum_{s \neq t} \sin s \zeta(s) \geq - \sum_{s \neq t} \sin s e(s) \geq -E,$$

$$\sin t \zeta(t) \leq \sin t e(t) \leq E.$$

$\square$

## 6. THE PINNED BOUNDS AND THE RIGHT ANGLE

Baek shows that a balanced maximum sofa  $S_\omega$  with  $|S_\omega| \geq 11/5$  and rotation angle  $\omega \in [\omega_0, \pi/2]$  has a rotated copy that turns through a right angle [1, Thm. 1.5.2]. The proof uses the balance only through two inequalities,  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $z_K^\circ \leq \sigma_K(\omega)$ , which bound the wedge gaps on the two floors  $y = 0$  and  $l(\omega, 0)$  by the lengths of the sides of the cap on the opposite walls  $y = 1$  and  $l(\omega, 1)$  [1, Thm. 4.1.4]. We prove these inequalities for  $K_*$  from the defect bounds of Section 5, and then follow Baek's proof with them as hypotheses. In Sections 6.1 and 6.2 we have  $\omega < \pi/2$ , except in Lemma 6.6(c), and  $K_*$  is as in Section 4; Proposition 6.9 also covers  $\omega = \pi/2$ .

**6.1. The pinned bounds.** For a cap  $K \in \mathcal{K}_\omega^c$  and  $t \in (0, \omega)$  let

$$w_K(t) = h_K(0) - \frac{h_K(t) - 1}{\cos t}, \quad z_K(t) = h_K(\omega + \frac{\pi}{2}) - \frac{h_K(t + \pi/2) - 1}{\cos(\omega - t)},$$

the *wedge gaps*. With  $A = (h_K(0), 0) = A_K^-(0)$  and  $C = h_K(\omega + \pi/2)v_\omega = C_K^+(\omega)$  the ends of the upper boundary of  $K$  (see (2.3)),  $w_K(t) = \langle A - W_K(t), u_0 \rangle$ , where  $W_K(t) = ((h_K(t) - 1)/\cos t, 0)$  is the point where the inner wall  $b_K(t)$  meets the  $x$ -axis, and  $z_K(t) = \langle C - Z_K(t), v_\omega \rangle$ , where  $Z_K(t)$  is the point where  $d_K(t)$  meets  $l(\omega, 0)$  [1, Def. 2.5.5]. Let  $w_K^\circ = \inf_{t \in (0, \omega)} w_K(t)$  and  $z_K^\circ = \inf_{t \in (0, \omega)} z_K(t)$  [1, Def. 4.1.1]. As  $t \rightarrow \omega$ ,  $w_K(t) \rightarrow h_K(0)$ , so  $w_K^\circ \leq h_K(0)$ ; as  $t \rightarrow 0$ ,  $z_K(t) \rightarrow h_K(\omega + \pi/2)$ , so  $z_K^\circ \leq h_K(\omega + \pi/2)$ .

**Lemma 6.1** (wedge gaps and the polyline). *Every polygon cap  $K \in \mathcal{K}_\Theta^c$  satisfies  $w_K^\circ \leq \tau_K(\pi/2)$  and  $z_K^\circ \leq \tau_K(\omega)$ .*

This is the first half of the proof of [1, Thm. 4.1.2].

*Proof.* Choose  $t_0 \in \Theta$  with  $w_K(t_0) = \min_{t \in \Theta} w_K(t)$ . A point  $(x, 0)$  of  $\mathcal{N}_\Theta(K)$  lies in  $F_\omega$ , so  $x \geq 0$ , and in some  $Q_K^-(t)$ ,  $t \in \Theta$ , so  $x \cos t < h_K(t) - 1$ , that is  $x < h_K(0) - w_K(t) \leq h_K(0) - w_K(t_0)$ . Hence  $\mathcal{N}_\Theta(K) \cap l(\pi/2, 0) \subset [0, h_K(0) - w_K(t_0)) \times \{0\}$ , and by Fact 2.10(b)

$$\sigma_K(3\pi/2) - \tau_K(\pi/2) = \mathcal{H}^1(\mathcal{N}_\Theta(K) \cap l(\pi/2, 0)) \leq \max(h_K(0) - w_K(t_0), 0).$$

The bottom edge  $e_K(3\pi/2)$  contains  $O$  and  $A$  (Lemmas 2.5 and 2.6), so  $\sigma_K(3\pi/2) \geq h_K(0)$ . If  $w_K(t_0) \geq h_K(0)$ , then  $\tau_K(\pi/2) \geq \sigma_K(3\pi/2) \geq h_K(0) \geq w_K^\circ$ . Otherwise  $\tau_K(\pi/2) \geq \sigma_K(3\pi/2) - h_K(0) + w_K(t_0) \geq w_K(t_0) \geq w_K^\circ$ , as  $t_0 \in \Theta \subset (0, \omega)$ . The second inequality is the same argument on the line  $l(\omega, 0)$ , with  $C$ ,  $z_K$  and  $h_K(\omega + \pi/2)$  in place of  $A$ ,  $w_K$  and  $h_K(0)$ : a point  $\mu v_\omega$  of  $\mathcal{N}_\Theta(K)$  has  $\mu \geq 0$ , as it lies in  $F_\omega$ , and  $\mu \cos(\omega - t) = \langle \mu v_\omega, v_t \rangle < h_K(t + \pi/2) - 1$  for some  $t \in \Theta$ , that is  $\mu < h_K(\omega + \pi/2) - z_K(t)$ ; the edge  $e_K(\omega + \pi)$  contains  $O$  and  $C$ , so  $\sigma_K(\omega + \pi) \geq h_K(\omega + \pi/2)$ ; and Fact 2.10(b) is used with  $t = \omega$ .  $\square$

**Lemma 6.2** (continuity). *For caps  $K, K' \in \mathcal{K}_\omega^c$  with  $d_H(K, K') = \varepsilon$ ,  $|w_K^\circ - w_{K'}^\circ| \leq (1 + \sec \omega)\varepsilon$  and  $|z_K^\circ - z_{K'}^\circ| \leq (1 + \sec \omega)\varepsilon$ .*

This is [1, Lemma 4.1.1], with the analogous bound for  $z^\circ$ .

*Proof.* For  $t \in (0, \omega)$ ,  $|w_K(t) - w_{K'}(t)| \leq \varepsilon + \varepsilon \sec t \leq (1 + \sec \omega)\varepsilon$ ; similarly for  $z$ , since  $\cos(\omega - t) \geq \cos \omega$ . The infima differ by at most as much.  $\square$

Baek obtains [1, Thm. 4.1.4] from the weak convergence of surface area measures [1, Thm. 4.1.3] and the Portmanteau theorem. Instead we use the following bound on edge lengths.

**Lemma 6.3** (upper semicontinuity of edge lengths). *Let convex bodies  $K_j$  converge to a convex body  $K$  in  $d_H$ , let  $t \in \mathbb{R}$ , and let real numbers  $a_j \rightarrow a$  satisfy  $a_j \leq \sigma_{K_j}(t)$  for every  $j$ . Then  $a \leq \sigma_K(t)$ .*

*Proof.* For  $0 < \delta < \pi$ ,

$$\begin{aligned} h_K(t + \delta) &\geq \langle v_K^+(t), u_{t+\delta} \rangle = h_K(t) \cos \delta + \langle v_K^+(t), v_t \rangle \sin \delta, \\ h_K(t - \delta) &\geq \langle v_K^-(t), u_{t-\delta} \rangle = h_K(t) \cos \delta - \langle v_K^-(t), v_t \rangle \sin \delta. \end{aligned}$$

Adding, and using  $\sigma_K(t) = \langle v_K^+(t) - v_K^-(t), v_t \rangle$ ,

$$\sigma_K(t) \leq \hat{\sigma}_K(\delta) = \frac{h_K(t + \delta) + h_K(t - \delta) - 2h_K(t) \cos \delta}{\sin \delta}.$$

The same holds for each  $K_j$ . Since  $h_{K_j} \rightarrow h_K$  pointwise,  $\hat{\sigma}_{K_j}(\delta) \rightarrow \hat{\sigma}_K(\delta)$ , and  $a_j \leq \sigma_{K_j}(t) \leq \hat{\sigma}_{K_j}(\delta)$  gives  $a \leq \hat{\sigma}_K(\delta)$  for every  $\delta \in (0, \pi)$ . As  $\delta \downarrow 0$ ,

$$\frac{h_K(t + \delta) - h_K(t) \cos \delta}{\sin \delta} \rightarrow h'_K(t^+), \quad \frac{h_K(t) \cos \delta - h_K(t - \delta)}{\sin \delta} \rightarrow h'_K(t^-),$$

because  $(1 - \cos \delta)/\delta \rightarrow 0$  and  $\delta/\sin \delta \rightarrow 1$ . By Fact 2.1(a),  $\hat{\sigma}_K(\delta) \rightarrow h'_K(t^+) - h'_K(t^-) = \langle v_K^+(t) - v_K^-(t), v_t \rangle = \sigma_K(t)$ , so  $a \leq \sigma_K(t)$ .  $\square$

**Proposition 6.4** (pinned bounds). *Let  $\omega \in (0, \pi/2)$  and let  $K_* \in \mathcal{K}_\omega^c$  be a maximizing cap with  $\mathcal{A}_\omega(K_*) > 0$ . Then  $w_{K_*}^\circ \leq \sigma_{K_*}(\pi/2)$  and  $z_{K_*}^\circ \leq \sigma_{K_*}(\omega)$ .*

*Proof.* Take the selection  $K_j \rightarrow K_*$  of Proposition 4.8, with  $K_j \subset [-R_0, R_0] \times [0, 1]$ , and  $\eta_j = d_H(K_j, K_*) \rightarrow 0$ . Every point  $p$  of this box has  $|\langle p, u_s \rangle| \leq R_0 + 1$ , so  $|h_{K_j}(s)| \leq R_0 + 1 = R$  for all  $s$ , where  $R \geq 0$ ; at every sampled normal  $|h_{K_j} - h_*| \leq \eta_j$ , and the total weight is  $\bar{\nu} \leq 1$  by (4.1). By Corollary 5.6 at  $t = \pi/2$ , with  $e_j = (2 + 4\gamma(R))\eta_j$ ,  $|\sigma_{K_j}(\pi/2) - \tau_{K_j}(\pi/2)| \leq e_j$ ; so by Lemma 6.1

$$w_{K_j}^\circ - e_j \leq \tau_{K_j}(\pi/2) - e_j \leq \sigma_{K_j}(\pi/2).$$

By Lemma 6.2,  $|w_{K_j}^\circ - w_{K_*}^\circ| \leq (1 + \sec \omega)\eta_j$ , so  $w_{K_j}^\circ - e_j \rightarrow w_{K_*}^\circ$ , and Lemma 6.3 with  $t = \pi/2$  and  $a_j = w_{K_j}^\circ - e_j$  gives the first inequality. For the second use  $t = \omega$ , where Corollary 5.6 divides by  $\sin \omega > 0$ : by Lemma 6.1,  $z_{K_j}^\circ - e_j/\sin \omega \leq \sigma_{K_j}(\omega)$ , and Lemmas 6.2 and 6.3 conclude as before.  $\square$

**6.2. The triangle in the niche.** Except in Lemma 6.6(c), let  $\omega \in [\omega_0, \pi/2)$ , so that  $\omega > \pi/4$ . Put  $c = c_\omega = \sec \omega - \tan \omega > 0$ , so that  $o_\omega - v_0 = cu_0$  and  $o_\omega - u_\omega = cv_\omega$  [1, Prop. 4.2.1], and let  $\Delta_\omega$  be the triangle with vertices  $O$ ,  $cu_0$ ,  $cv_\omega$ , which cuts off the corner  $O$  of  $P_\omega$ . Let  $d_{\min} = 5/4$  if  $\omega < \tan^{-1}(11/5)$  and  $d_{\min} = 11/10$  otherwise, and for  $d \in [0, \tan \omega]$  let

$R_{\omega,d} = P_{\omega} \cap H_{-}(0, d+c) \cap H_{-}(\omega + \pi/2, d+c)$  be  $P_{\omega}$  without its two acute corners.

**Fact 6.5** (numerical inequalities at the corner). *Let  $\omega \in [\omega_0, \pi/2)$ .*

- (a)  $|R_{\omega,d_{\min}}| < 11/5$ . [1, Lemma 4.2.2]
- (b) *If  $d \in [d_{\min}, \tan \omega]$  and  $r_y = 1 - d \cot \omega \geq 0$ , then the points  $q_0 = o_{\omega} - v_0 + d u_0$  and  $q_1 = o_{\omega} - g u_0$ ,  $g = \sqrt{1 - r_y^2}$ , satisfy*

$$\langle q_0 - c u_0, u_{\pi/2-\omega} \rangle > 1 \quad \text{and} \quad \langle q_1 - c v_{\omega}, v_{\pi/2-\omega} \rangle > 1.$$

[1, Lemma 4.2.4, in the range  $[\sec^{-1}(2.2), \pi/2)$  of its proof and of its use (E9)]

- Lemma 6.6** (the triangle lies in the niche). (a) *Let  $\omega \in [\omega_0, \pi/2)$ , let  $K$  be a compact set, and let  $q_0, q_1 \in K$  satisfy  $\langle q_0 - c u_0, u_{\pi/2-\omega} \rangle > 1$  and  $\langle q_1 - c v_{\omega}, v_{\pi/2-\omega} \rangle > 1$ . Then  $O$ ,  $c u_0$  and  $c v_{\omega}$  lie in  $Q_K^{-}(\pi/2 - \omega)$ .*
- (b) *Let  $\omega \in [\omega_0, \pi/2)$  and let  $K \in \mathcal{K}_{\omega}^c$  satisfy  $w_K^{\circ} \leq \sigma_K(\pi/2)$  and  $h_K(0) \geq d_{\min} + c$ . Then  $O$ ,  $c u_0$  and  $c v_{\omega}$  lie in  $Q_K^{-}(\pi/2 - \omega)$ .*
- (c) *For every  $\omega \in \mathbb{R}$ , every compact set  $K$  and every  $t \in \mathbb{R}$ , the reflection  $M_{\omega}$  maps  $Q_{M_{\omega}(K)}^{-}(t)$  into  $Q_K^{-}(\omega - t)$ .*
- (d) *Let  $\omega \in [\omega_0, \pi/2)$  and let  $K \in \mathcal{K}_{\omega}^c$  satisfy  $\mathcal{A}_{\omega}(K) \geq 11/5$ ,  $w_K^{\circ} \leq \sigma_K(\pi/2)$  and  $z_K^{\circ} \leq \sigma_K(\omega)$ . Then for some  $t_{\sharp} \in (0, \omega)$  the three vertices  $O$ ,  $c u_0$ ,  $c v_{\omega}$  of  $\Delta_{\omega}$  lie in the closure of  $Q_K^{-}(t_{\sharp})$ .*

Part (d) is [1, Thm. 4.2.5] for a cap that satisfies the pinned bounds, in place of a balanced maximum cap; the proof there uses the balance only through them. Baek's theorem puts the vertices in the closure of the quarter-plane, while its proof gives the open quarter-plane (E10). Parts (a) and (b) are steps of that proof, and (c) is a consequence of [1, Prop. 2.5.4].

*Proof.* (a) Put  $t = \pi/2 - \omega$ . As  $q_0, q_1 \in K$ ,  $h_K(t) \geq \langle q_0, u_t \rangle$  and  $h_K(t + \pi/2) \geq \langle q_1, v_t \rangle$ . So every point  $p$  with  $\langle p, u_t \rangle \leq \langle c u_0, u_t \rangle$  and  $\langle p, v_t \rangle \leq \langle c v_{\omega}, v_t \rangle$  lies in  $Q_K^{-}(t)$ , because the hypotheses give

$$\begin{aligned} \langle p, u_t \rangle &\leq \langle c u_0, u_t \rangle < \langle q_0, u_t \rangle - 1 \leq h_K(t) - 1, \\ \langle p, v_t \rangle &\leq \langle c v_{\omega}, v_t \rangle < \langle q_1, v_t \rangle - 1 \leq h_K(t + \pi/2) - 1. \end{aligned}$$

Now  $\langle c u_0, u_t \rangle = c \sin \omega > 0$  and  $\langle c v_{\omega}, v_t \rangle = c \cos(\omega - t) = c \sin 2\omega > 0$ , while  $\langle c u_0, v_t \rangle = -c \sin t = -c \cos \omega < 0$  and  $\langle c v_{\omega}, u_t \rangle = c \sin(t - \omega) = c \cos 2\omega \leq 0$ , since  $\pi/4 < \omega < \pi/2$ . So  $O$ ,  $c u_0$  and  $c v_{\omega}$  satisfy both conditions.

(b) Put  $d = h_K(0) - c \geq d_{\min}$  and  $q_0 = (h_K(0), 0)$ , which is  $A_K^{-}(0) \in K$  by Lemma 2.6; then  $q_0 = (c+d)u_0 = o_{\omega} - v_0 + d u_0$ . As  $q_0 \in P_{\omega}$ ,  $h_K(0) \cos \omega \leq 1$ , so  $d \cos \omega \leq 1 - c \cos \omega = \sin \omega$ . Hence  $d \leq \tan \omega$ , and  $r_y = 1 - d \cot \omega$  lies in  $[0, 1]$ . The point  $r = (h_K(0), r_y)$  lies on  $l_K(0)$  and on  $l(\omega, 1)$ , since  $\langle r, u_{\omega} \rangle = (c+d) \cos \omega + \sin \omega - d \cos \omega = 1$ . Let  $g = \sqrt{1 - r_y^2}$  and  $s = q_0 - g u_0$ : the triangle  $s, q_0, r$  has a right angle at  $q_0$ , and  $r - s = (g, r_y)$  is a unit vector.

For  $t \in (0, \omega)$ ,  $\sin \omega u_t = \sin(\omega - t) u_0 + \sin t u_\omega$  with nonnegative coefficients, and every  $p \in K$  has  $\langle p, u_0 \rangle \leq h_K(0) = \langle r, u_0 \rangle$  and  $\langle p, u_\omega \rangle \leq 1 = \langle r, u_\omega \rangle$ , as  $K \subset P_\omega$ . So  $h_K(t) \leq \langle r, u_t \rangle$ , and since  $\langle r - s, u_t \rangle \leq \|r - s\| = 1$ ,

$$h_K(t) - 1 \leq \langle r, u_t \rangle - \langle r - s, u_t \rangle = \langle s, u_t \rangle = (h_K(0) - g) \cos t.$$

So  $W_K(t)$  lies on or to the left of  $s$ , that is  $w_K(t) \geq g$ . Hence  $0 \leq g \leq w_K^\circ \leq \sigma_K(\pi/2)$ .

The top edge  $e_K(\pi/2)$  lies on the line  $l(\pi/2, 1)$ , whose points in  $P_\omega$  have abscissa at most  $c$ . Its end  $v_K^-(\pi/2)$  is such a point, and its other end  $v_K^+(\pi/2) = v_K^-(\pi/2) - \sigma_K(\pi/2)u_0$  (Fact 2.1(b), with  $v_{\pi/2} = -u_0$ ) has abscissa at most  $c - \sigma_K(\pi/2) \leq c - g$ . So  $q_1 = o_\omega - g u_0 = (c - g, 1)$  lies on the segment from  $v_K^+(\pi/2)$  to  $o_\omega$ , which lies in  $K$ , since  $o_\omega \in K$  (Lemma 2.5) and  $K$  is convex.

Finally  $d \in [d_{\min}, \tan \omega]$  and  $r_y \geq 0$ , so by Fact 6.5(b) the points  $q_0, q_1 \in K$  satisfy the hypotheses of (a), and (a) gives the claim.

(c) The reflection  $M_\omega$  is a symmetric linear map with  $M_\omega u_s = u_{\omega+\pi/2-s}$ , so  $\langle M_\omega p, u_s \rangle = \langle p, u_{\omega+\pi/2-s} \rangle$  for all  $p$  and  $s$ ; in particular  $h_{M_\omega(K)}(s) = h_K(\omega + \pi/2 - s)$ . Let  $p \in Q_{M_\omega(K)}^-(t)$ , that is  $\langle p, u_t \rangle < h_K(\omega + \pi/2 - t) - 1$  and  $\langle p, u_{t+\pi/2} \rangle < h_K(\omega - t) - 1$ . Then

$$\begin{aligned} \langle M_\omega p, u_{\omega-t} \rangle &= \langle p, u_{t+\pi/2} \rangle < h_K(\omega - t) - 1, \\ \langle M_\omega p, u_{\omega-t+\pi/2} \rangle &= \langle p, u_t \rangle < h_K(\omega - t + \pi/2) - 1, \end{aligned}$$

so  $M_\omega p \in Q_K^-(\omega - t)$ .

(d) If  $h_K(0) < d_{\min} + c$  and  $h_K(\omega + \pi/2) < d_{\min} + c$ , then  $K \subset R_{\omega, d_{\min}}$ , as  $K \subset P_\omega$ , and  $|K| \leq |R_{\omega, d_{\min}}| < 11/5$  by Fact 6.5(a), whereas  $|K| \geq \mathcal{A}_\omega(K) \geq 11/5$ . So  $h_K(0) \geq d_{\min} + c$  or  $h_K(\omega + \pi/2) \geq d_{\min} + c$ . In the first case (b) puts  $O$ ,  $cu_0$  and  $cv_\omega$  in  $Q_K^-(\pi/2 - \omega)$ , hence in its closure, and  $t_\sharp = \pi/2 - \omega$  lies in  $(0, \omega)$ , as  $\omega > \pi/4$ .

In the second case we use the mirror image  $K^m = M_\omega(K)$ , a cap with  $h_{K^m}(s) = h_K(\omega + \pi/2 - s)$  and  $\sigma_{K^m}(\pi/2) = \sigma_K(\omega)$  (Fact 2.4(d)). Then  $h_{K^m}(0) = h_K(\omega + \pi/2) \geq d_{\min} + c$ , and the formulas for the wedge gaps give  $w_{K^m}(t) = z_K(\omega - t)$  for  $t \in (0, \omega)$ , so  $w_{K^m}^\circ = z_K^\circ \leq \sigma_K(\omega) = \sigma_{K^m}(\pi/2)$ . By (b),  $O$ ,  $cu_0$  and  $cv_\omega$  lie in  $Q_{K^m}^-(\pi/2 - \omega)$ . As  $M_\omega$  fixes  $O$  and exchanges  $u_0$  and  $v_\omega$ , (c) puts  $O$ ,  $cv_\omega$  and  $cu_0$  in  $Q_K^-(\omega - (\pi/2 - \omega))$ , hence in its closure, and  $t_\sharp = 2\omega - \pi/2$  lies in  $(0, \omega)$ , as  $\pi/4 < \omega < \pi/2$ .  $\square$

### 6.3. A rotated copy turns through the right angle.

**Lemma 6.7** (the width of the sofa). *Let  $\omega \in (0, \pi/2)$ ,  $c = c_\omega$  and  $K \in \mathcal{K}_\omega^c$ , and suppose that  $O$ ,  $cu_0$  and  $cv_\omega$  lie in the closure of  $Q_K^-(t_\sharp)$  for some  $t_\sharp \in (0, \omega)$ . Then*

- (a) *every point  $\lambda u_0 + \mu v_\omega$  with  $\lambda, \mu \geq 0$  and  $\lambda + \mu < c$  lies in  $\mathcal{N}(K)$ ;*
- (b)  *$\langle p - q, u_t \rangle \leq 1$  for all  $p \in K$ ,  $q \in K \setminus \mathcal{N}(K)$  and  $t \in [\omega, \pi/2]$ .*

Baek asserts the width bound (b) without proof, for the pentagon  $P_\omega \setminus \Delta_\omega$ , in the proof of [1, Thm. 1.5.2] (E10).

*Proof.* (a) The closure of  $Q_K^-(t_\sharp)$  lies in the closed quarter-plane of the points  $p$  with  $\langle p, u_{t_\sharp} \rangle \leq \alpha$  and  $\langle p, v_{t_\sharp} \rangle \leq \beta$ , where  $\alpha = h_K(t_\sharp) - 1$  and  $\beta = h_K(t_\sharp + \pi/2) - 1$ . It contains  $cu_0$  and  $cv_\omega$ , and  $\langle cu_0, u_{t_\sharp} \rangle = c \cos t_\sharp > 0$ ,  $\langle cv_\omega, v_{t_\sharp} \rangle = c \cos(\omega - t_\sharp) > 0$ , so  $\alpha > 0$  and  $\beta > 0$ . The point  $p = \lambda u_0 + \mu v_\omega$  is  $(\lambda/c)cu_0 + (\mu/c)cv_\omega$ , with  $\lambda/c, \mu/c \geq 0$  and  $(\lambda + \mu)/c < 1$ . So  $\langle p, u_{t_\sharp} \rangle \leq (\lambda + \mu)\alpha/c < \alpha$  and  $\langle p, v_{t_\sharp} \rangle \leq (\lambda + \mu)\beta/c < \beta$ , and  $p \in Q_K^-(t_\sharp)$ . Also  $p \in F_\omega$ , since  $\langle p, u_\omega \rangle = \lambda \cos \omega \geq 0$  and  $p_y = \mu \cos \omega \geq 0$ ; as  $t_\sharp \in (0, \omega)$ ,  $p \in \mathcal{N}(K)$ .

(b) Write  $q = \lambda u_0 + \mu v_\omega$ . As  $q \in P_\omega$ ,  $\lambda = \langle q, u_\omega \rangle / \cos \omega \geq 0$  and  $\mu = q_y / \cos \omega \geq 0$ , and  $\lambda + \mu \geq c$  by (a). For  $t \in [\omega, \pi/2]$  both  $\cos t$  and  $\sin(t - \omega)$  are nonnegative, so  $\langle q, u_t \rangle = \lambda \cos t + \mu \sin(t - \omega) \geq c \min(\cos t, \sin(t - \omega))$ . Also  $\cos \omega u_t = \cos t u_\omega + \sin(t - \omega) u_{\pi/2}$ , and  $\langle p, u_\omega \rangle \leq 1$ ,  $p_y \leq 1$  as  $p \in P_\omega$ , so  $\cos \omega \langle p, u_t \rangle \leq \cos t + \sin(t - \omega)$ , that is  $\langle p, u_t \rangle \leq c \cos t + \sin t = \langle o_\omega, u_t \rangle$ , as  $c \cos \omega = 1 - \sin \omega$ . If  $\cos t \leq \sin(t - \omega)$ , then  $\langle p - q, u_t \rangle \leq \sin t \leq 1$ . Otherwise, using  $c(1 + \sin \omega) = \cos \omega$  and  $c \cos \omega = 1 - \sin \omega$ ,

$$\langle p - q, u_t \rangle \leq c \cos t + \sin t - c \sin(t - \omega) = \cos(t - \omega) \leq 1. \quad \square$$

**Lemma 6.8** (turning inside the horizontal side). *Let  $S$  be a moving sofa with rotation angle  $\omega < \pi/2$  such that  $\langle p - q, u_t \rangle \leq 1$  for all  $p, q \in S$  and  $t \in [\omega, \pi/2]$ . Then  $R_\psi S$ ,  $\psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ .*

*Proof.* Let  $(\theta, \mathbf{c})$  be a movement of  $S$  with rotation angle  $\omega$ , as in Definition 2.2; its translation  $\mathbf{c}$  is set in bold, to distinguish it from  $c_\omega$ . As  $S$  is compact, there is  $R$  with  $\|p\| \leq R$  for all  $p \in S$ . For  $\rho \in [0, \psi]$  let  $b(\rho) = (1 - 2R, h_S(3\pi/2 - \rho))$ , which depends continuously on  $\rho$ , as  $h_S$  is continuous. Then  $R_\rho S + b(\rho) \subset H_L$ : for  $p \in S$  the abscissa of  $R_\rho p + b(\rho)$  is  $p_x \cos \rho - p_y \sin \rho + 1 - 2R \leq |p_x| + |p_y| + 1 - 2R \leq 1$ ; and, as  $(R_\rho p)_y = \langle p, u_{\pi/2-\rho} \rangle$  and  $u_{3\pi/2-\rho} = -u_{\pi/2-\rho}$ , its ordinate is

$$\langle p, u_{\pi/2-\rho} \rangle + h_S(3\pi/2 - \rho) = \langle p, u_{\pi/2-\rho} \rangle - \min_{q \in S} \langle q, u_{\pi/2-\rho} \rangle,$$

which lies in  $[0, 1]$  by the hypothesis with  $t = \pi/2 - \rho \in [\omega, \pi/2]$ . Define  $\theta'$  and  $\mathbf{c}'$  on  $[0, 1]$  by

$$\begin{aligned} \theta'(s) &= -3\psi s, & \mathbf{c}'(s) &= b(\psi - 3\psi s) & \text{on } [0, \tfrac{1}{3}], \\ \theta'(s) &= -\psi, & \mathbf{c}'(s) &= (2 - 3s)b(0) + (3s - 1)\mathbf{c}(0) & \text{on } [\tfrac{1}{3}, \tfrac{2}{3}], \\ \theta'(s) &= \theta(3s - 2) - \psi, & \mathbf{c}'(s) &= \mathbf{c}(3s - 2) & \text{on } [\tfrac{2}{3}, 1]. \end{aligned}$$

The definitions agree at  $s = 1/3$  and at  $s = 2/3$ , so  $\theta'$  and  $\mathbf{c}'$  are continuous;  $\theta'(0) = 0$  and  $\theta'(1) = \theta(1) - \psi = -\pi/2$ . The image of  $R_\psi S$  under  $p \mapsto R_{\theta'(s)}p + \mathbf{c}'(s)$  is  $R_{\theta'(s)+\psi}S + \mathbf{c}'(s)$ . On  $[0, 1/3]$  it is  $R_\rho S + b(\rho) \subset H_L$  with  $\rho = \psi(1 - 3s)$ , so the sofa turns clockwise by  $\psi$  inside  $H_L$ . On  $[1/3, 2/3]$  it is  $S + (2 - 3s)b(0) + (3s - 1)\mathbf{c}(0)$ , which lies in  $H_L$ , since  $H_L$  is convex and contains  $S + b(0)$  and the starting position  $S + \mathbf{c}(0)$  of the movement of  $S$ . On  $[2/3, 1]$  it is the position of  $S$  at the time  $3s - 2$  of its movement, which lies in  $L$ , and in  $V_L$  at  $s = 1$ . So  $(\theta', \mathbf{c}')$  is a movement of  $R_\psi S$  with rotation angle  $\pi/2$ , and  $R_\psi S$  is closed and connected.  $\square$

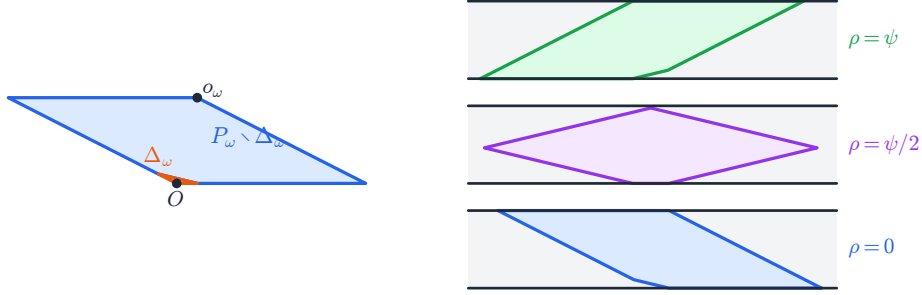


FIGURE 5. Lemmas 6.7 and 6.8, for  $\omega = 1.1$ . Left: the parallelogram  $P_\omega$  (dashed), its corner triangle  $\Delta_\omega$  (orange) and the pentagon  $P_\omega \setminus \Delta_\omega$  (blue); the sofa  $K \setminus \mathcal{N}(K)$  lies in the pentagon, up to the far side of  $\Delta_\omega$ . Right: the pentagon turned by  $\rho = \psi$ ,  $\psi/2$  and  $0$ ,  $\psi = \pi/2 - \omega$ , in a horizontal strip of width one: it fits for every  $\rho \in [0, \psi]$ , since its width in the direction  $u_t$  is at most 1 for  $t \in [\omega, \pi/2]$ .

**Proposition 6.9** (a right-angle monotone sofa). *Let  $\omega \in [\omega_0, \pi/2]$  and let  $T$  be a monotone sofa with rotation angle  $\omega$  and  $|T| = |G|$  (such as the sofa  $T$  of Proposition 3.2). Then  $R_\psi T$ ,  $\psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ . Consequently there are  $w_1 \in \mathbb{R}^2$  and a monotone sofa  $U$  with rotation angle  $\pi/2$  such that*

$$R_\psi T + w_1 \subset U, \quad |U| = |G|,$$

and the cap  $\mathcal{C}(U) \in \mathcal{K}_{\pi/2}^c$  is maximizing, with  $\mathcal{A}(\mathcal{C}(U)) = |G|$ .

*Proof.* If  $\omega = \pi/2$ , then  $\psi = 0$ , and  $T$  is a moving sofa with rotation angle  $\pi/2$  by Fact 2.4(a). Let  $\omega < \pi/2$  and  $K = \mathcal{C}(T)$ . By Fact 2.4(a), (b) and (c),  $T$  is a moving sofa with rotation angle  $\omega$  in standard position,  $K$  is a cap,  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |T| = |G|$ ; so  $K$  is maximizing by Fact 2.7, and  $\mathcal{A}_\omega(K) > 11/5 > 0$ . Proposition 6.4, applied to  $K_* = K$ , gives the pinned bounds  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $z_K^\circ \leq \sigma_K(\omega)$ . With them and  $\mathcal{A}_\omega(K) \geq 11/5$ , Lemma 6.6(d) gives  $t_\sharp \in (0, \omega)$  such that  $O$ ,  $cu_0$  and  $cv_\omega$  lie in the closure of  $Q_K^-(t_\sharp)$ . As  $T = K \setminus \mathcal{N}(K)$ , Lemma 6.7(b) gives  $\langle p - q, u_t \rangle \leq 1$  for  $p, q \in T$  and  $t \in [\omega, \pi/2]$ , and Lemma 6.8 shows that  $R_\psi T$  is a moving sofa with rotation angle  $\pi/2$ . Also  $|R_\psi T| = |T| = |G|$ , and the second part is Proposition 3.2 for the moving sofa  $S' = R_\psi T$ .  $\square$

By Propositions 3.2 and 6.9, there are  $w_0, w_1 \in \mathbb{R}^2$  such that, with  $g_1(p) = R_\psi(p + w_0) + w_1$ ,

$$g_1(S) \subset U, \quad |U| = |G|, \quad (6.1)$$

where  $U$  is a monotone sofa with rotation angle  $\pi/2$  whose cap is maximizing.

## 7. CURVATURE BOUNDS AND THE INJECTIVITY CONDITION

Baek proves the injectivity condition for a balanced maximum cap  $K$  with rotation angle  $\pi/2$  from the inequality  $\sigma_K \leq k_0(g_K^+(t))dt$  on  $[0, \pi/2)$  [1, Thm. 6.4.3]. This inequality is the limit of  $\sigma_{K_n}(t) \leq k_0(g_{K_n}^+(t))\delta + O(\delta^2)$  for maximum polygon caps  $K_n$  with step size  $\delta$  [1, Thm. 6.3.3], whose proof uses the balance  $\sigma = \tau$ . For the polygon caps of Section 4 the defect bound of Proposition 5.2 replaces the balance, and its errors add up to at most  $2\eta$  by (4.1). In this section we use Section 4, Proposition 5.2 and (4.1) with  $\omega = \pi/2$  and  $\mathcal{A} = \mathcal{A}_{\pi/2}$ :  $K_* \in \mathcal{K}_{\pi/2}^c$  is a maximizing cap with  $\mathcal{A}(K_*) > 0$ ,  $\Theta_k = \{j\delta : 0 < j < n\}$  with  $n = 2^{k+1}$  and step size  $\delta = \delta_k = \pi/(2n) \leq \pi/4$ . All the normals of  $\Theta_k$  are floating. A polygon cap  $K \in \mathcal{K}_k^c$  has no edge with normal angle 0 or  $\pi$  and none with normal in  $(-\pi/2, 0)$ ; its edge normals are among  $\delta, 2\delta, \dots, (2n-1)\delta$  and  $3\pi/2$ . For an angle  $u$  let  $H_K^b(u)$  and  $H_K^d(u)$  be as in Section 2.5, so that  $Q_K^-(u)$  is the complement of  $H_K^b(u) \cup H_K^d(u)$ .

## 7.1. One normal of a polygon cap.

**Lemma 7.1** (the inner wall near the corner). *Let  $K \in \mathcal{K}_k^c$ ,  $t \in \Theta_k$ ,  $q = \tan(\delta/2)$ , and  $\mathbf{x} = \mathbf{x}_K(t)$ . Let  $\vec{b} = \vec{b}_K(t) = \{\mathbf{x} - sv_t : s \geq 0\}$ . Then the set of  $s \geq 0$  with  $\mathbf{x} - sv_t \in H_K^d(t - \delta)$  is  $[0, \alpha_-]$  and the set of  $s \geq 0$  with  $\mathbf{x} - sv_t \in H_K^d(t + \delta)$  is  $[0, \alpha_+]$ , where*

$$\alpha_- = \tan \delta (g_K^-(t) - 1 + q), \quad \alpha_+ = \tan \delta (1 - g_K^+(t) + q)$$

(an interval is empty if its right end is negative); and the set of  $s \in \mathbb{R}$  with  $\mathbf{x} - sv_t \in H_K^b(t - \delta) \cap H_K^b(t + \delta)$  is an interval, empty or of length  $2q - \sigma_K(t)$ .

This is [1, Lemma 6.3.2], together with the length  $2 \tan(\delta/2) - \sigma_K(t)$  that Baek asserts in the proof of [1, Thm. 6.3.3]. They are stated there for maximum polygon caps, but the proofs use only that no normal angle of  $K$  lies in  $(t - \delta, t) \cup (t, t + \delta) \cup (t + \frac{\pi}{2} - \delta, t + \frac{\pi}{2}) \cup (t + \frac{\pi}{2}, t + \frac{\pi}{2} + \delta)$ , which holds for every  $K \in \mathcal{K}_k^c$ .

*Proof.* Since no normal angle of  $K$  lies between  $t + \pi/2$  and  $t + \delta + \pi/2$ , the supporting lines  $l_K(t + \pi/2)$  and  $l_K(t + \delta + \pi/2)$  meet at a vertex  $r$  of  $K$ , namely  $r = C_K^+(t) = \mathbf{y}_K(t) - g_K^+(t)u_t$ , and  $h_K(t + \delta + \pi/2) = \langle r, v_{t+\delta} \rangle$ . As  $\mathbf{x} - r = (g_K^+(t) - 1)u_t - v_t$ ,

$$\begin{aligned} \langle \mathbf{x}, v_{t+\delta} \rangle - h_K(t + \delta + \frac{\pi}{2}) + 1 &= \langle \mathbf{x} - r, v_{t+\delta} \rangle + 1 \\ &= -(g_K^+(t) - 1) \sin \delta - \cos \delta + 1. \end{aligned}$$

Now  $\langle \mathbf{x} - sv_t, v_{t+\delta} \rangle = \langle \mathbf{x}, v_{t+\delta} \rangle - s \cos \delta$ , so  $\mathbf{x} - sv_t \in H_K^d(t + \delta)$  iff  $s \leq \alpha_+ = [(1 - g_K^+(t)) \sin \delta + 1 - \cos \delta] / \cos \delta$ , and  $(1 - \cos \delta) / \cos \delta = \tan \delta \tan(\delta/2)$ . For  $t - \delta$  use  $r' = C_K^-(t) = \mathbf{y}_K(t) - g_K^-(t)u_t$  in the same way. Next,  $\langle \mathbf{x} - sv_t, u_{t \mp \delta} \rangle = \langle \mathbf{x}, u_{t \mp \delta} \rangle \pm s \sin \delta$ , so the conditions  $\langle p, u_{t-\delta} \rangle \geq h_K(t - \delta) - 1$

and  $\langle p, u_{t+\delta} \rangle \geq h_K(t+\delta) - 1$  on  $p = \mathbf{x} - sv_t$  read  $s \geq s_-$  and  $s \leq s_+$  with

$$\begin{aligned} s_+ - s_- &= \frac{2 \cos \delta (h_K(t) - 1) - h_K(t+\delta) - h_K(t-\delta) + 2}{\sin \delta} \\ &= \frac{2(1 - \cos \delta) - \sigma_K(t) \sin \delta}{\sin \delta} = 2q - \sigma_K(t), \end{aligned}$$

where we used  $u_{t+\delta} + u_{t-\delta} = 2 \cos \delta u_t$ ,  $\langle \mathbf{x}, u_t \rangle = h_K(t) - 1$  and  $h_K(t+\delta) + h_K(t-\delta) - 2 \cos \delta h_K(t) = \sigma_K(t) \sin \delta$ . The last identity holds because no normal angle of  $K$  lies in  $(t-\delta, t) \cup (t, t+\delta)$ : the vertices  $v_K^-(t) = v_K(t-\delta, t)$  and  $v_K^+(t) = v_K(t, t+\delta)$  lie on  $l_K(t-\delta)$  and on  $l_K(t+\delta)$ , so that  $h_K(t \pm \delta) = h_K(t) \cos \delta \pm \langle v_K^\pm(t), v_t \rangle \sin \delta$ , while  $\sigma_K(t) = \langle v_K^+(t) - v_K^-(t), v_t \rangle$  by Fact 2.1(b).  $\square$

**Lemma 7.2** (the polygon bound at one normal). *For  $K \in \mathcal{K}_k^c$  and  $t \in \Theta_k$ ,*

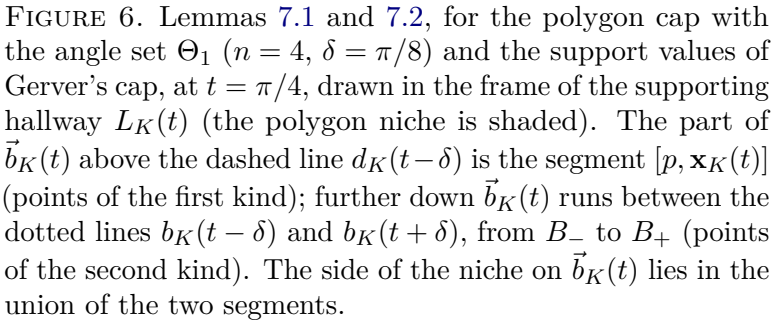
$$\begin{aligned} \tau_K(t) &\leq \max(\tan \delta (g_K^-(t) - 1 + q), \tan \delta (1 - g_K^+(t) + q), 0) \\ &\quad + \max(2q - \sigma_K(t), 0), \quad q = \tan \frac{\delta}{2}. \end{aligned}$$

*Proof.* Let  $X = \partial \mathcal{N}_{\Theta_k}(K) \cap \vec{b}_K(t)$ ; by Fact 2.10(a),  $\tau_K(t) = \mathcal{H}^1(X)$ . Let  $I = \{t - \delta, t, t + \delta\}$ . The points of  $X$  that lie in  $H_K^d(u)$  for some  $u \in I$  are of the first kind. On  $\vec{b}_K(t)$  the half-plane  $H_K^d(t)$  contains only the corner  $\mathbf{x}_K(t)$ , so by Lemma 7.1 the points of the first kind form a set of length at most  $\max(0, \alpha_-, \alpha_+)$ . Let  $p \in X$  not be of the first kind. As  $X \subset \text{cl} \mathcal{N}_{\Theta_k}(K) \subset \{y \geq 0\}$ ,  $p_y \geq 0$ , and  $p_y = 0$  for at most one point of  $X$ , since the line  $b_K(t)$  is not horizontal. Let  $p_y > 0$ . For  $u \in \Theta_k$ ,  $p \notin Q_K^-(u)$ : otherwise the open set  $\{y > 0\} \cap Q_K^-(u)$ , which lies in  $\mathcal{N}_{\Theta_k}(K)$ , would be a neighborhood of  $p$  contained in the niche, and  $p$  would not be a boundary point. Also  $Q_K^-(0)$  and  $Q_K^-(\pi/2)$  lie in  $\{y < 0\}$ , since one of the two inequalities that define each of them is  $y < h_K(\pi/2) - 1 = 0$ . As  $t - \delta, t + \delta \in \Theta_k \cup \{0, \pi/2\}$  and  $Q_K^-(u)$  is the complement of  $H_K^b(u) \cup H_K^d(u)$ , we get  $p \in H_K^b(t - \delta) \cap H_K^b(t + \delta)$ . These are the points of the second kind, and by Lemma 7.1 they form a set of length at most  $\max(0, 2q - \sigma_K(t))$ . So  $X$  consists of points of the two kinds and of at most one point on the  $x$ -axis, and  $\mathcal{H}^1(X)$  is at most the sum of the two lengths.  $\square$

**Lemma 7.3** (with a defect). *Let  $K \in \mathcal{K}_k^c$ ,  $t \in \Theta_k$ ,  $e \geq 0$ ,  $D \geq 0$  with  $g_K^+(t) \leq D$  and  $\sigma_K(t) \leq \tau_K(t) + e$ . Then*

$$\sigma_K(t) \leq k_0(g_K^+(t)) \delta + (D + 4)\delta^2 + e.$$

*Proof.* Let  $M = |g_K^+(t) - 1|$ . As  $C_K^+(t) \in K$ ,  $g_K^+(t) = h_K(t) - \langle C_K^+(t), u_t \rangle \geq 0$ ; with  $g_K^+(t) \leq D$  this gives  $M \leq D + 1$ . By Fact 2.1(b) and  $v_{t+\pi/2} = -u_t$ ,  $C_K^+(t) = C_K^-(t) - \sigma_K(t + \pi/2)u_t$ , so  $g_K^+(t) - g_K^-(t) = \sigma_K(t + \pi/2) \geq 0$ , and each of  $g_K^-(t) - 1 + q$  and  $1 - g_K^+(t) + q$  is at most  $M + q$ . So the first maximum in Lemma 7.2 is at most  $\tan \delta (M + q) \leq \delta M + (D + 3)\delta^2$ , using  $\tan \delta \leq \delta + \delta^2$ ,  $q \leq \delta$  and  $\delta \leq 1$ . If  $\sigma_K(t) \geq 2q$ , the second maximum



## 7.2. From normals to intervals.

$$g_K^+(s) = g_K^-(s), \quad g_K^+(s) \leq g_K^+(t), \quad g_K^-(t + \delta) \leq g_K^-(s),$$

This is [1, Lemma 6.4.1], with  $D$  in place of the bound 5 for maximum polygon caps.

*Proof.* Since there is no normal angle of  $K$  strictly between  $t$  and  $t + \delta$ , nor between  $t + \pi/2$  and  $t + \delta + \pi/2$ , the supporting lines  $l_K(s)$ ,  $s \in [t, t + \delta]$ , pass

through the vertex  $A = v_K(t, t + \delta)$  of  $K$ , and the lines  $l_K(s + \pi/2)$  through the vertex  $C = v_K(t + \pi/2, t + \delta + \pi/2)$ ; moreover  $C_K^+(t) = C$ ,  $C_K^-(t + \delta) = C$ , and  $C_K^\pm(s) = C$  for  $s \in (t, t + \delta)$ . As  $\langle \mathbf{y}_K(s), u_s \rangle = h_K(s) = \langle A, u_s \rangle$ , the function  $G(s) = \langle A - C, u_s \rangle$  equals  $g_K^+$  at  $t$ ,  $g_K^-$  at  $t + \delta$ , and both  $g_K^+$  and  $g_K^-$  on  $(t, t + \delta)$ . Its derivative  $G'(s) = \langle A - C, v_s \rangle$  is at most 0 on  $[t, t + \delta]$ , because  $\langle A, v_s \rangle \leq h_K(s + \pi/2) = \langle C, v_s \rangle$ , and at least  $-\|A - C\| \geq -D$ , as  $A, C \in K$ . So  $G$  is nonincreasing on  $[t, t + \delta]$ , which gives the inequalities for  $s \in (t, t + \delta)$ , and  $G(s) + Ds$  is nondecreasing there, which gives  $G(t) - G(t + \delta) \leq D\delta$ .  $\square$

**Lemma 7.5** (summing the steps). *Let  $B, C, D \geq 0$ , let  $K \in \mathcal{K}_k^c$  with  $\|p - q\| \leq D$  for all  $p, q \in K$  and  $k_0(g_K^+) \leq B$  on  $[0, \pi/2]$ , and suppose that for every  $t \in \{0\} \cup \Theta_k$*

$$\sigma_K(t) \leq k_0(g_K^+(t))\delta + C\delta^2 + e(t), \quad e(t) \geq 0.$$

Then for  $0 \leq a \leq b \leq \pi/2$

$$\sigma_K([a, b]) \leq \int_a^b k_0(g_K^+(u))du + \left(2B + \frac{\pi}{2}(C + D)\right)\delta + \sum_{j=0}^{n-1} e(j\delta),$$

and for  $a \geq -\pi/2$  with  $a' = \max(a, 0) \leq b \leq \pi/2$

$$\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+(u))du + \left(2B + \frac{\pi}{2}(C + D)\right)\delta + \sum_{j=0}^{n-1} e(j\delta).$$

*Proof.* Let  $t \in \{0\} \cup \Theta_k$  and  $s \in (t, t + \delta)$ . By Lemma 7.4,  $g_K^+(s) \leq g_K^+(t)$  and  $g_K^+(t) - g_K^+(s) = g_K^+(t) - g_K^-(s) \leq g_K^+(t) - g_K^-(t + \delta) \leq D\delta$ ; as  $k_0$  is 1-Lipschitz,  $k_0(g_K^+(s)) \geq k_0(g_K^+(t)) - D\delta$ . Integrating over  $(t, t + \delta)$  gives  $k_0(g_K^+(t))\delta \leq \int_t^{t+\delta} k_0(g_K^+(u))du + D\delta^2$ , hence

$$\sigma_K(t) \leq \int_t^{t+\delta} k_0(g_K^+(u))du + (C + D)\delta^2 + e(t).$$

For  $0 \leq j < n$ ,  $K$  has no normal angle in  $(j\delta, j\delta + \delta)$ , so  $v_K^+$  is constant on  $[j\delta, j\delta + \delta)$ , equal to a vertex  $w$ . There  $h_K(s) = \langle w, u_s \rangle$ , and the function  $G_K(s) = \langle w, v_s \rangle + \int_0^s h_K$  of Fact 2.1(b) has the derivative  $-\langle w, u_s \rangle + h_K(s) = 0$ ; so  $\sigma_K((j\delta, j\delta + \delta)) = 0$  and  $\sigma_K([j\delta, j\delta + \delta)) = \sigma_K(j\delta)$ . Adding the bounds at  $t = j\delta$  for  $i \leq j < m$  gives, for integers  $0 \leq i \leq m \leq n$ ,

$$\sigma_K([i\delta, m\delta]) \leq \int_{i\delta}^{m\delta} k_0(g_K^+(u))du + (m - i)(C + D)\delta^2 + \sum_{j=i}^{m-1} e(j\delta).$$

Now let  $0 \leq a \leq b \leq \pi/2$ ,  $i = \lfloor a/\delta \rfloor$  and  $m = \lceil b/\delta \rceil$ . Then  $i\delta \leq a < i\delta + \delta$ ,  $b \leq m\delta < b + \delta$  and  $i \leq m \leq n$ , as  $b \leq \pi/2 = n\delta$ ; so  $[a, b] \subset [i\delta, m\delta]$ . Hence  $\sigma_K([a, b]) \leq \sigma_K([i\delta, m\delta])$ , and the first bound follows: the integrals of  $k_0(g_K^+)$  over  $[i\delta, a]$  and over  $[b, m\delta]$ , intervals of length less than  $\delta$  in  $[0, \pi/2]$ , are at most  $B\delta$  each;  $(m - i)(C + D)\delta^2 \leq n(C + D)\delta^2 = \frac{\pi}{2}(C + D)\delta$ ; and the errors  $e(j\delta)$  with  $i \leq j < m$  are among the nonnegative terms of  $\sum_{j=0}^{n-1} e(j\delta)$ . For the second bound,  $K$  has no normal angle in  $(-\pi/2, 0)$ , so  $\sigma_K((-\pi/2, 0)) = 0$

in the same way. As  $(a, b) \subset (-\pi/2, 0) \cup [a', b)$ ,  $\sigma_K((a, b)) \leq \sigma_K([a', b))$ , and the first bound applies to  $[a', b)$ .  $\square$

**Lemma 7.6** (the bound on an open interval in the limit). *Let  $K_j$  be polygon caps (with any angle sets) that converge in  $d_H$  to  $K \in \mathcal{K}_{\pi/2}^c$ , and let  $\varepsilon_j \rightarrow 0$  be such that, for every  $j$  and all  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ ,*

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j.$$

*Then  $\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+)$  for all such  $a$  and  $b$ .*

With Lemma 7.7, this is the passage to the limit in the proof of [1, Thm. 6.4.3]. Baek obtains the lower semicontinuity of the mass of an open interval from the weak convergence of  $\sigma_{K_n}$  and the Portmanteau theorem; the proof here uses support functions instead.

*Proof.* Let  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ . By Fact 2.14(a),  $\int_0^{\pi/2} |g_{K_j}^+ - g_K^+| \rightarrow 0$ ; as  $k_0$  is 1-Lipschitz and  $[a', b] \subset [0, \pi/2]$ , the right sides  $c_j = \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j$  converge to  $c = \int_{a'}^b k_0(g_K^+)$ . For  $0 < \varepsilon < \pi$  let  $\Phi_K(\varepsilon) = \langle v_K(b - \varepsilon, b), v_b \rangle - \langle v_K(a, a + \varepsilon), v_a \rangle + \int_a^b h_K$ , and define  $\Phi_{K_j}(\varepsilon)$  in the same way. A point  $p = h_K(b)u_b + mv_b$  of  $e_K(b)$  lies in  $H_K(b - \varepsilon)$ , that is,  $h_K(b) \cos \varepsilon - m \sin \varepsilon \leq h_K(b - \varepsilon)$ , so  $\langle v_K^-(b), v_b \rangle \geq \langle v_K(b - \varepsilon, b), v_b \rangle$ ; likewise  $\langle v_K^+(a), v_a \rangle \leq \langle v_K(a, a + \varepsilon), v_a \rangle$ . By (2.1),  $\Phi_K(\varepsilon) \leq \sigma_K((a, b))$ , and in the same way  $\Phi_{K_j}(\varepsilon) \leq \sigma_{K_j}((a, b)) \leq c_j$ . Fix  $\varepsilon$ . The points  $v_{K_j}(b - \varepsilon, b)$  and  $v_{K_j}(a, a + \varepsilon)$  depend continuously on the support values of  $K_j$  at  $a, a + \varepsilon, b - \varepsilon$  and  $b$ , and  $h_{K_j} \rightarrow h_K$  uniformly; so  $\Phi_{K_j}(\varepsilon) \rightarrow \Phi_K(\varepsilon)$ , and therefore  $\Phi_K(\varepsilon) \leq c$ . As  $\varepsilon \downarrow 0$ ,  $\Phi_K(\varepsilon) \rightarrow \sigma_K((a, b))$  by Fact 2.1(a) and (2.1), hence  $\sigma_K((a, b)) \leq c$ .  $\square$

**Lemma 7.7** (passing to the limit). *Let  $K_j$  be polygon caps (with any angle sets) that converge in  $d_H$  to  $K \in \mathcal{K}_{\pi/2}^c$ , and let  $\varepsilon_j \rightarrow 0$  be such that, for all  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ ,*

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j.$$

*Then  $\sigma_K(E) \leq \int_E k_0(g_K^+(t))dt$  for every Borel set  $E \subset [0, \pi/2)$ .*

*Proof.* By Lemma 7.6,  $\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+)$  for all  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ . Let  $\mu$  and  $\nu$  be the restrictions of  $\sigma_K$  and of  $k_0(g_K^+(t))dt$  to  $[0, \pi/2)$ ; the claim is  $\mu \leq \nu$ . Both are finite measures on  $\mathbb{R}$ :  $\sigma_K([0, \pi/2))$  is finite, and  $k_0(g_K^+) \leq g_K^+ + 1$  is bounded, since  $g_K^+(t) = h_K(t) - \langle C_K^+(t), u_t \rangle$  lies between 0 and  $h_K(t) + h_K(t + \pi)$ , as  $C_K^+(t) \in K$ . Let  $a < b$  be real,  $a' = \max(a, 0)$  and  $b' = \min(b, \pi/2)$ . If  $b' \leq a'$ , then  $(a, b) \cap [0, \pi/2)$  is empty. Otherwise  $(a, b) \cap [0, \pi/2)$  contains  $(a', b')$  and lies

in  $(\max(a, -\pi/2), b')$ , an interval to which the bound above applies, so

$$\mu((a, b)) \leq \sigma_K((\max(a, -\pi/2), b')) \leq \int_{a'}^{b'} k_0(g_K^+) \leq \nu((a, b)).$$

As  $\mu$  and  $\nu$  are finite and  $(a, b]$  is the intersection of the decreasing intervals  $(a, b+1/i)$ , also  $\mu((a, b]) \leq \nu((a, b])$  for all  $a < b$ . Hence  $F(x) = \nu((-\infty, x]) - \mu((-\infty, x])$  is nondecreasing; it is also right-continuous, and its Lebesgue–Stieltjes measure  $dF$  satisfies  $\nu = \mu + dF$ , since both sides give every interval  $(a, b]$  the mass  $\nu((a, b])$ ; hence  $\mu \leq \mu + dF = \nu$ .  $\square$

### 7.3. The curvature bounds.

**Definition 7.8** (curvature bounds). A cap  $K \in \mathcal{K}_{\pi/2}^c$  satisfies the *curvature bounds* if, as measures,

$$\sigma_K \leq k_0(g_K^+(t))dt \text{ on } [0, \pi/2), \quad \sigma_K \leq k_0(f_K^-(t - \pi/2))dt \text{ on } (\pi/2, \pi].$$

**Proposition 7.9** (maximizing caps satisfy the curvature bounds). *Every maximizing cap  $K_* \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K_*) > 0$  satisfies the curvature bounds.*

*Proof.* Take the selection  $K_j \rightarrow K_*$  of Proposition 4.8, with stages  $k_j$ , step sizes  $\delta_j \rightarrow 0$ , in  $[-R_0, R_0] \times [0, 1]$ , and  $\eta_j = d_H(K_j, K_*) \rightarrow 0$ . Let  $D = 2R_0 + 2$ . Two points of the box differ by at most  $2R_0$  in  $x$  and by at most 1 in  $y$ , so  $\|p - q\| \leq D$  for all  $p, q \in K_j$ . Also  $|h_{K_j}| \leq R_0 + 1$ , and as  $C_{K_j}^+(t) \in K_j$ ,

$$0 \leq g_{K_j}^+(t) = h_{K_j}(t) - \langle C_{K_j}^+(t), u_t \rangle \leq h_{K_j}(t) + h_{K_j}(t + \pi) \leq D;$$

since  $k_0(x) \leq |x| + 1$ ,  $k_0(g_{K_j}^+) \leq B = D + 1$ . Let  $e_j(t) = 2\eta_j \nu_{k_j}(t) \geq 0$ . For  $t \in \Theta_{k_j}$ , Proposition 5.2 gives  $\sigma_{K_j}(t) \leq \tau_{K_j}(t) + e_j(t)$ , so by Lemma 7.3

$$\sigma_{K_j}(t) \leq k_0(g_{K_j}^+(t))\delta_j + (D + 4)\delta_j^2 + e_j(t).$$

This holds also at  $t = 0$ , where the right side is nonnegative and  $\sigma_{K_j}(0) = 0$ , since  $K_j$  has no normal angle in  $(-\pi/2, \delta_j)$  and so  $v_{K_j}^+(0) = v_{K_j}^-(0)$ . The grid points  $i\delta_j$ ,  $0 \leq i < 2^{k_j+1}$ , are distinct, and the weights  $\nu_{k_j}(s)$  add up to at most 1 by (4.1), so the errors  $e_j(i\delta_j)$  add up to at most  $2\eta_j$ . By Lemma 7.5 with  $C = D + 4$ , for  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' \leq b$ ,

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + (2B + \pi(D + 2))\delta_j + 2\eta_j.$$

By Lemma 7.7,  $K_*$  satisfies the first curvature bound. The reflection  $K_*^m$  is a cap with the same sofa area (Fact 2.4(d)), hence maximizing with positive sofa area, and the argument above, applied to  $K_*^m$  in place of  $K_*$ , shows that it satisfies the first curvature bound. Let  $E \subset (\pi/2, \pi]$  be a Borel set, so that  $\pi - E \subset [0, \pi/2)$ . By Fact 2.4(d),  $\sigma_{K_*}(E) = \sigma_{K_*^m}(\pi - E)$ , and by Fact 2.13(c),  $g_{K_*^m}^+(s) = f_{K_*}^-(\pi/2 - s)$ ; hence

$$\sigma_{K_*}(E) \leq \int_{\pi-E} k_0(f_{K_*}^-(\frac{\pi}{2} - s))ds = \int_E k_0(f_{K_*}^-(t - \frac{\pi}{2}))dt,$$

which is the second curvature bound for  $K_*$ .  $\square$

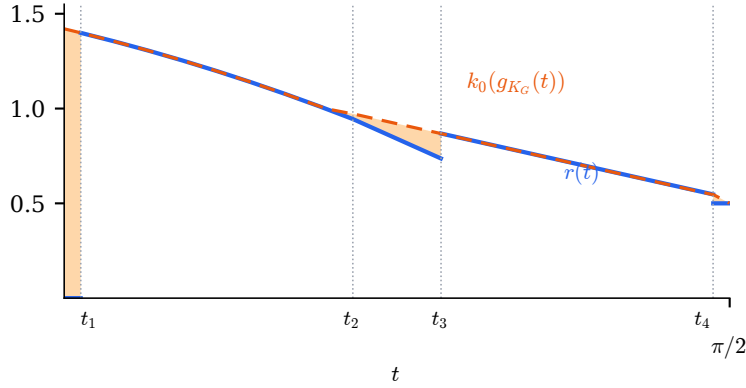


FIGURE 7. The first curvature bound for Gerver's cap, which maximizes  $\mathcal{A}$ , as  $\mathcal{A}(K_G) = |G|$  (Fact 2.16(b), Fact 2.4(c), Fact 2.7). The density  $r(t)$  of  $\sigma_{K_G}$  on  $[0, \pi/2)$  (blue) and  $k_0(g_{K_G}(t))$  (orange, dashed) are computed from Romik's rotation path. In the figure the blue curve lies below the orange one, and the two agree on  $[t_1, t_*]$  and on  $[t_3, t_4]$ , where  $t_* = 0.616\dots$  is the angle at which  $g_{K_G}$  equals 2; the gap between them is shaded.

#### 7.4. The injectivity condition.

**Lemma 7.10** (integral inequalities). *Let  $K \in \mathcal{K}_{\pi/2}^c$  satisfy the curvature bounds. Then  $K$  satisfies condition (1) of Definition 2.12, and*

$$\begin{aligned} f_K(t) - 1 &\geq \int_0^t m_0(g_K(u)) du \quad (0 \leq t < \frac{\pi}{2}), \\ g_K(t) - 1 &\geq \int_t^{\pi/2} m_0(f_K(u)) du \quad (0 < t \leq \frac{\pi}{2}). \end{aligned}$$

*Proof.* Each restriction of  $\sigma_K$  is a finite measure, dominated by a measure with a density, hence absolutely continuous with respect to Lebesgue measure; so it has a density (Radon–Nikodym), and condition (1) holds. By Fact 2.13(d),  $f_K, g_K$  are defined and continuous on  $[0, \pi/2]$ ,  $f_K^+ = f_K$  on  $[0, \pi/2)$ , and  $f_K(0) = g_K(\pi/2) = 1$ . For  $t \in [0, \pi/2)$ ,  $(0, t] \subset [0, \pi/2)$ , so the first curvature bound and Fact 2.13(b) give

$$\begin{aligned} f_K(t) - 1 &= f_K^+(t) - f_K^+(0) = \int_0^t g_K - \sigma_K((0, t]) \\ &\geq \int_0^t (g_K - k_0(g_K)) = \int_0^t m_0(g_K). \end{aligned}$$

For  $a \leq b$ ,

$$g_K^+(b) - g_K^+(a) = \sigma_K((a + \frac{\pi}{2}, b + \frac{\pi}{2}]) - \int_a^b f_K^+.$$

As  $\langle \mathbf{y}_K(s), u_s \rangle = h_K(s)$ ,  $\langle \mathbf{y}_K(s), v_s \rangle = h_K(s + \pi/2)$ ,  $A_K^+(s) = v_K^+(s)$ ,  $C_K^+(s) = v_K^+(s + \pi/2)$  and  $u_s = -v_{s+\pi/2}$ , we have  $g_K^+(s) = h_K(s) + \langle v_K^+(s + \pi/2), v_{s+\pi/2} \rangle$  and  $f_K^+(s) = h_K(s + \pi/2) - \langle v_K^+(s), v_s \rangle$ . By Fact 2.1(b),  $\langle v_K^+(d), v_d \rangle - \langle v_K^+(c), v_c \rangle = \sigma_K((c, d]) - \int_c^d h_K$  for  $c \leq d$ . The product rule [1, Lemma 5.1.3] for  $h_K(s) = \langle v_K^+(s), u_s \rangle$ , with  $v_K^+(d) - v_K^+(c) = \int_{(c, d]} v_s d\sigma_K(s)$  (Fact 2.1(b)) and  $\frac{d}{ds} u_s = v_s$ , gives  $h_K(b) - h_K(a) = \int_{(a, b]} \langle v_s, u_s \rangle d\sigma_K(s) + \int_a^b \langle v_K^+(s), v_s \rangle ds = \int_a^b \langle v_K^+(s), v_s \rangle ds$ . Hence

$$g_K^+(b) - g_K^+(a) = h_K(b) - h_K(a) + \sigma_K((a + \frac{\pi}{2}, b + \frac{\pi}{2}]) - \int_{a+\pi/2}^{b+\pi/2} h_K,$$

$$\int_a^b f_K^+ = \int_{a+\pi/2}^{b+\pi/2} h_K - (h_K(b) - h_K(a)),$$

and the identity follows. Let  $t \in (0, \pi/2]$ , and take  $a = t$ ,  $b = \pi/2$ . As  $g_K = g_K^+$ ,  $g_K(\pi/2) = 1$ ,  $f_K^+ = f_K$  on  $[t, \pi/2)$  and  $(t + \pi/2, \pi] \subset (\pi/2, \pi]$ , the second curvature bound gives

$$g_K(t) - 1 = \int_t^{\pi/2} f_K - \sigma_K((t + \frac{\pi}{2}, \pi])$$

$$\geq \int_t^{\pi/2} f_K - \int_{t+\pi/2}^{\pi} k_0(f_K^-(u - \frac{\pi}{2})) du = \int_t^{\pi/2} m_0(f_K). \quad \square$$

**Lemma 7.11** (the lower sequence). *Let  $f, g : [0, \pi/2] \rightarrow [0, \infty)$  be continuous with  $f(t) - 1 \geq \int_0^t m_0(g)$  for  $t \in [0, \pi/2)$  and  $g(t) - 1 \geq \int_t^{\pi/2} m_0(f)$  for  $t \in (0, \pi/2]$ . Then for every  $j \geq 0$ ,  $f_j(t) \leq f(t)$  on  $[0, \pi/2)$  and  $f_j(\pi/2 - t) \leq g(t)$  on  $(0, \pi/2]$ .*

*Proof.* Induction on  $j$ , the case  $j = 0$  being  $f_0 = 0 \leq f, g$ . If it holds for  $j$ , then for  $t \in [0, \pi/2)$ , since  $m_0$  is nondecreasing and  $f_j(\pi/2 - u) \leq g(u)$  for  $u \in (0, t]$ ,  $\mathcal{F}f_j(t) = 1 + \int_0^t m_0(f_j(\pi/2 - u)) du \leq 1 + \int_0^t m_0(g) \leq f(t)$ , and with  $f_j \leq f$  we get  $f_{j+1} = \max(f_j, \mathcal{F}f_j) \leq f$  on  $[0, \pi/2)$ . For  $t \in (0, \pi/2]$ ,  $\mathcal{F}f_j(\pi/2 - t) = 1 + \int_t^{\pi/2} m_0(f_j(s)) ds \leq 1 + \int_t^{\pi/2} m_0(f) \leq g(t)$ , and with  $f_j(\pi/2 - t) \leq g(t)$  this gives  $f_{j+1}(\pi/2 - t) \leq g(t)$ .  $\square$

**Proposition 7.12** (injectivity). *Every  $K \in \mathcal{K}_{\pi/2}^c$  that satisfies the curvature bounds satisfies the injectivity condition, and  $f_K, g_K > 1$  on  $(0, \pi/2)$ .*

*Proof.* By Lemma 7.10,  $K$  satisfies (1) and the integral inequalities; by Fact 2.13(d),  $f_K$  and  $g_K$  are continuous on  $[0, \pi/2]$ , and they are nonnegative, since  $f_K(t) = h_K(t + \pi/2) - \langle A_K(t), v_t \rangle$  and  $g_K(t) = h_K(t) - \langle C_K(t), u_t \rangle$  with  $A_K(t), C_K(t) \in K$ . By Lemma 7.11 with  $j = 11$  and Fact 2.14(b), for  $t \in (0, \pi/2)$ ,  $1 < f_{11}(t) \leq f_K(t)$  and  $1 < f_{11}(\pi/2 - t) \leq g_K(t)$ . By Fact 2.13(d),  $\mathbf{x}_K$  is continuously differentiable on  $[0, \pi/2]$  with  $\mathbf{x}'_K = -(f_K - 1)u_t + (g_K - 1)v_t$ , which gives (2) and (3).  $\square$

**Corollary 7.13** (maximizing right-angle caps are in  $\mathcal{K}^i$ ). *Every  $K \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K) = |G|$  lies in  $\mathcal{K}^i$ .*

*Proof.*  $K$  is maximizing (Fact 2.7). Gerver's cap  $K_G \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(K_G) = |G|$  (Fact 2.16(b), Fact 2.4(c)), so  $\mathcal{A}(K) \geq \mathcal{A}(K_G) = |G| > 0$ . Hence  $K$  satisfies the curvature bounds (Proposition 7.9) and the injectivity condition (Proposition 7.12); and  $|K| = \mathcal{A}(K) + |\mathcal{N}(K)| \geq |G| \geq 11/5$ .  $\square$

## 8. EQUALITY IN BAEK'S UPPER BOUND

By Corollary 7.13, every cap  $K \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K) = |G|$  lies in  $\mathcal{K}^i$ . For a cap  $K \in \mathcal{K}^i$ , write  $K_0 = K_G$  for Gerver's cap and  $K_1 = K$ , let  $\xi_0 = \xi_G$  and  $\xi_1 = \xi_K$  be their canonical triples (Section 2.5), and for  $\lambda \in [0, 1]$  let  $K_\lambda$ ,  $B_\lambda$ ,  $D_\lambda$  be the components of  $\xi_\lambda = (1 - \lambda)\xi_0 + \lambda\xi_1$ . Thus  $K_\lambda = (1 - \lambda)K_0 + \lambda K_1$ ,  $B_\lambda = (1 - \lambda)B_{K_0} + \lambda B_{K_1}$  and  $D_\lambda = (1 - \lambda)D_{K_0} + \lambda D_{K_1}$ . Write  $h_i = h_{K_i}$  for  $i = 0, 1$ , and  $\Delta = h_1 - h_0$ .

### 8.1. Equality in the concavity of $\mathcal{Q}$ .

**Lemma 8.1** (equality in the Mamikon terms). *Let  $K \in \mathcal{K}^i$  with  $\mathcal{A}(K) = |G|$ . Then for every  $\lambda \in [0, 1]$*

$$\begin{aligned}\mathcal{S}_{K_\lambda} &= (1 - \lambda)\mathcal{S}_{K_0} + \lambda\mathcal{S}_{K_1}, \\ \mathcal{R}_{B_\lambda} &= (1 - \lambda)\mathcal{R}_{B_0} + \lambda\mathcal{R}_{B_1}, \\ \mathcal{L}_{D_\lambda} &= (1 - \lambda)\mathcal{L}_{D_0} + \lambda\mathcal{L}_{D_1}.\end{aligned}$$

*Proof.* By Fact 2.15(a), (c),  $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_1) \leq \mathcal{Q}(\xi_0) = |G|$ , so  $\mathcal{Q}(\xi_1) = \mathcal{Q}(\xi_0)$ , and  $\mathcal{Q} \leq \mathcal{Q}(\xi_0)$  on  $\mathcal{T}$ . Let  $\lambda \in [0, 1]$ . The set  $\mathcal{T}$  is closed under Minkowski combinations, so  $\xi_\lambda \in \mathcal{T}$ . By Fact 2.15(b),  $\mathcal{Q} = \Lambda - \mathcal{S} - \mathcal{R} - \mathcal{L}$  with  $\Lambda$  convex-linear and  $\mathcal{S}$ ,  $\mathcal{R}$ ,  $\mathcal{L}$  convex, so  $\mathcal{Q}$  is concave on  $\mathcal{T}$ , and

$$\mathcal{Q}(\xi_\lambda) \geq (1 - \lambda)\mathcal{Q}(\xi_0) + \lambda\mathcal{Q}(\xi_1) = \mathcal{Q}(\xi_0) \geq \mathcal{Q}(\xi_\lambda).$$

So  $\mathcal{Q}(\xi_\lambda) = (1 - \lambda)\mathcal{Q}(\xi_0) + \lambda\mathcal{Q}(\xi_1)$ , and since  $\Lambda$  is convex-linear this says

$$\begin{aligned}[(1 - \lambda)\mathcal{S}_{K_0} + \lambda\mathcal{S}_{K_1} - \mathcal{S}_{K_\lambda}] &+ [(1 - \lambda)\mathcal{R}_{B_0} + \lambda\mathcal{R}_{B_1} - \mathcal{R}_{B_\lambda}] \\ &+ [(1 - \lambda)\mathcal{L}_{D_0} + \lambda\mathcal{L}_{D_1} - \mathcal{L}_{D_\lambda}] = 0.\end{aligned}$$

Each bracket is  $\geq 0$  by the convexity of  $\mathcal{S}$ ,  $\mathcal{R}$  and  $\mathcal{L}$ , so each is 0.  $\square$

**8.2. Equal tangent displacements.** In this subsection,  $K_0$  and  $K_1$  are any two convex bodies,  $\lambda \in (0, 1)$ ,  $K_\lambda = (1 - \lambda)K_0 + \lambda K_1$ ,  $h_i = h_{K_i}$  and  $\Delta = h_1 - h_0$ . Let  $a < b < a + \pi$ , and let  $\mathbf{z}_K$  be curves on  $[a, b]$  as in Section 2.5, one for every convex body  $K$ , that are convex-linear in  $K$ :  $\mathbf{z}_{K_\lambda} = (1 - \lambda)\mathbf{z}_{K_0} + \lambda\mathbf{z}_{K_1}$ . By Fact 2.15(b), the curves of the four terms of  $\mathcal{S}_K$  are of this kind. For  $i \in \{0, \lambda, 1\}$  write  $\varrho_i = \varrho_{K_i, \mathbf{z}_{K_i}}$  and  $\mathcal{M}_i = \mathcal{M}_{K_i}(a, b; \mathbf{z}_{K_i})$ . As  $v_{K_\lambda}^+(t) = (1 - \lambda)v_{K_0}^+(t) + \lambda v_{K_1}^+(t)$  for every  $t$  [1, Thm. 7.1.2],  $\varrho_\lambda = (1 - \lambda)\varrho_0 + \lambda\varrho_1$  on  $[a, b]$ . By Mamikon's theorem [1, Thm. 7.4.1],  $\varrho_{K, \mathbf{z}}$  is almost everywhere measurable and bounded on  $[a, b]$  for

every convex body  $K$  and every curve  $\mathbf{z}$  as in Section 2.5; so  $\varrho_0^2$ ,  $\varrho_1^2$  and  $\varrho_0\varrho_1$  are integrable on  $[a, b]$ . Since  $(1-\lambda)x^2 + \lambda y^2 - ((1-\lambda)x + \lambda y)^2 = \lambda(1-\lambda)(x-y)^2$ ,

$$(1-\lambda)\mathcal{M}_0 + \lambda\mathcal{M}_1 - \mathcal{M}_\lambda = \frac{\lambda(1-\lambda)}{2} \int_a^b (\varrho_0(t) - \varrho_1(t))^2 dt. \quad (8.1)$$

As  $\lambda(1-\lambda) > 0$ , the equality  $\mathcal{M}_\lambda = (1-\lambda)\mathcal{M}_0 + \lambda\mathcal{M}_1$  implies that  $\varrho_0 = \varrho_1$  almost everywhere on  $(a, b)$ . If moreover  $\varrho_0$  and  $\varrho_1$  are continuous on  $(a, b)$ , then  $\varrho_0 = \varrho_1$  at every point of  $(a, b)$ , because the set where they differ is open and has measure 0.

Now let  $K_0, K_1 \in \mathcal{K}_{\pi/2}^c$  satisfy condition (1) of Definition 2.12, and let  $[a, b] \subset [0, \pi/2]$  or  $[a, b] \subset [\pi/2, \pi]$ , so that  $(a, b) \subset (0, \pi/2)$  or  $(a, b) \subset (\pi/2, \pi)$ . By Fact 2.13(d),  $v_{K_i}^+(t) = v_{K_i}^-(t)$  for  $t \in [0, \pi/2) \cup (\pi/2, \pi]$ ; so, by the one-sided derivatives of Fact 2.1(a),  $h_i$  is differentiable at every such  $t$ , with  $h'_i(t) = \langle v_{K_i}^+(t), v_t \rangle$ . Also by Fact 2.13(d),  $v_{K_i}^+ = A_{K_i}$  on  $(0, \pi/2)$  and  $v_{K_i}^+(t) = C_{K_i}(t - \pi/2)$  for  $t \in (\pi/2, \pi)$ , where  $A_{K_i}$  and  $C_{K_i}$  are continuous on  $[0, \pi/2]$ . So, for  $i = 0, 1$ ,  $v_{K_i}^+$  is continuous on  $(a, b)$ , and so is  $\varrho_i(t) = \langle \mathbf{z}_{K_i}(t) - v_{K_i}^+(t), v_t \rangle$ .

**Lemma 8.2** (the tangent equations). *Let  $K_0, K_1 \in \mathcal{K}_{\pi/2}^c$  satisfy condition (1) of Definition 2.12, let  $\lambda \in (0, 1)$ , and let  $a < b$  with  $[a, b] \subset [0, \pi/2]$  or  $[a, b] \subset [\pi/2, \pi]$ .*

- (a) *If  $T - \pi < a$ ,  $b \leq T$  and  $\mathcal{M}_{K_\lambda}(a, b; \mathbf{l}_{K_\lambda}^T) = (1-\lambda)\mathcal{M}_{K_0}(a, b; \mathbf{l}_{K_0}^T) + \lambda\mathcal{M}_{K_1}(a, b; \mathbf{l}_{K_1}^T)$ , then there are  $p, q \in \mathbb{R}$  with  $\Delta(t) = p \cos t + q \sin t$  for  $t \in [a, b]$ , and  $\Delta(T) = p \cos T + q \sin T$ .*
- (b) *If  $\mathcal{M}_{K_\lambda}(a, b; \mathbf{y}_{K_\lambda}) = (1-\lambda)\mathcal{M}_{K_0}(a, b; \mathbf{y}_{K_0}) + \lambda\mathcal{M}_{K_1}(a, b; \mathbf{y}_{K_1})$ , then  $\Delta(t) = \Delta(b) - \int_t^b \Delta(u + \pi/2) du$  for  $t \in [a, b]$ .*

*Proof.* Both  $\Delta$  and  $t \mapsto \Delta(t + \pi/2)$  are continuous, as support functions are.

(a) Here  $b \leq T < a + \pi$  and  $[a, b] \subset (T - \pi, T]$ . For  $t \in [a, b]$ ,  $\mathbf{l}_K^T(t)$  lies on  $l_K(t)$ : it is the point  $v_K(t, T)$  of  $l_K(t)$  if  $t < T$ , and  $v_K^-(T) \in l_K(T)$  if  $t = T$ . By [1, Thm. 8.3.1],  $\mathbf{l}_K^T$  is continuous and of bounded variation on  $[a, b]$ , and by [1, Thm. 8.3.2],  $\mathbf{l}_{K_\lambda}^T = (1-\lambda)\mathbf{l}_{K_0}^T + \lambda\mathbf{l}_{K_1}^T$  on  $[a, b]$ . So these curves are of the kind considered above, and  $\varrho_0 = \varrho_1$  on  $(a, b)$ . Let  $t \in (a, b)$ . Then  $0 < T - t < \pi$ ,  $\mathbf{l}_{K_i}^T(t) = v_{K_i}(t, T)$  and  $\langle v_{K_i}(t, T), v_t \rangle = (h_i(T) - h_i(t) \cos(T - t)) / \sin(T - t)$ , so

$$\varrho_i(t) = \frac{h_i(T) - h_i(t) \cos(T - t)}{\sin(T - t)} - h'_i(t).$$

Multiplied by  $\sin(T - t)$ , the equality  $\varrho_0(t) = \varrho_1(t)$  is

$$\sin(T - t)\Delta'(t) + \cos(T - t)\Delta(t) = \Delta(T).$$

Then  $\chi(t) = (\Delta(t) - \Delta(T) \cos(T - t)) / \sin(T - t)$  has  $\chi' = [\Delta' \sin(T - t) + \Delta \cos(T - t) - \Delta(T)] / \sin^2(T - t) = 0$  on  $(a, b)$ , so  $\chi$  is a constant  $\kappa$  there, and  $\Delta(t) = \Delta(T) \cos(T - t) + \kappa \sin(T - t)$  for  $t \in (a, b)$ . Both sides are continuous, so this holds on  $[a, b]$ . It reads  $\Delta(t) = p \cos t + q \sin t$

with  $p = \Delta(T) \cos T + \kappa \sin T$  and  $q = \Delta(T) \sin T - \kappa \cos T$ , and then  $p \cos T + q \sin T = \Delta(T)$ .

(b) Here  $b - a \leq \pi/2 < \pi$ . By the formula  $\mathbf{y}_K(t) = h_K(t)u_t + h_K(t + \pi/2)v_t$  for the outer corner (Section 2.2),  $\langle \mathbf{y}_K(t), u_t \rangle = h_K(t)$ , that is,  $\mathbf{y}_K(t) \in l_K(t)$ ; and  $\mathbf{y}_K$  is convex-linear in  $K$ , as  $h_K$  is. As  $K$  is bounded,  $h_K$  is Lipschitz, and so is  $\mathbf{y}_K$  on  $[a, b]$ , which makes  $\mathbf{y}_K$  continuous and of bounded variation there. These curves are of the kind considered above, and  $\varrho_0 = \varrho_1$  on  $(a, b)$ . As  $\langle \mathbf{y}_{K_i}(t), v_t \rangle = h_i(t + \pi/2)$ , we have  $\varrho_i(t) = h_i(t + \pi/2) - h'_i(t)$ , and  $\varrho_0 = \varrho_1$  says  $\Delta'(t) = \Delta(t + \pi/2)$  for  $t \in (a, b)$ . For  $t \in [a, b]$ ,  $\Delta$  is continuous on  $[t, b]$  and has the continuous derivative  $u \mapsto \Delta(u + \pi/2)$  on  $(t, b)$ , so  $\Delta(b) - \Delta(t) = \int_t^b \Delta(u + \pi/2) du$ .  $\square$

### 8.3. The cap is a horizontal translate.

**Proposition 8.3** (the difference of support functions). *Let  $K \in \mathcal{K}^i$  with  $\mathcal{A}(K) = |G|$ . Then there is  $s \in \mathbb{R}$  with  $h_K(t) - h_{K_G}(t) = s \cos t$  for  $t \in [0, \pi]$ .*

*Proof.* We use the notation of the beginning of this section, with  $K_0 = K_G$ ,  $K_1 = K$  and  $\Delta = h_K - h_{K_G}$ . By Lemma 8.1 with  $\lambda = 1/2$ ,  $\mathcal{S}_{K_{1/2}} = \frac{1}{2}\mathcal{S}_{K_0} + \frac{1}{2}\mathcal{S}_{K_1}$ . Each of the four terms of  $\mathcal{S}$  is convex (Fact 2.15(b)), so its value at  $K_{1/2}$  is at most the mean of its values at  $K_0$  and  $K_1$ ; as the sums are equal, each term takes at  $K_{1/2}$  the mean of its values at  $K_0$  and  $K_1$ . The cap  $K_0 = K_G$  lies in  $\mathcal{K}^i$  by Fact 2.16(b), and so does  $K_1 = K$ ; so both satisfy condition (1). Also  $0 < \varphi < \pi/4$  (Appendix A). So Lemma 8.2 applies with  $\lambda = 1/2$ : part (a) to the first, third and fourth terms, with  $(a, b, T)$  equal to  $(0, \varphi, \pi/2)$ ,  $(\pi/2 - \varphi, \pi/2, \pi - \varphi)$  and  $(\pi/2, \pi, \pi)$ , and part (b) to the second, with  $(a, b) = (\varphi, \pi/2 - \varphi)$ . It gives  $p_j, q_j \in \mathbb{R}$  such that

- (i)  $\Delta(t) = p_1 \cos t + q_1 \sin t$  for  $t \in [0, \varphi]$ , and  $\Delta(\pi/2) = q_1$ ;
- (ii)  $\Delta(t) = \Delta(\pi/2 - \varphi) - \int_t^{\pi/2 - \varphi} \Delta(u + \pi/2) du$  for  $t \in [\varphi, \pi/2 - \varphi]$ ;
- (iii)  $\Delta(t) = p_3 \cos t + q_3 \sin t$  for  $t \in [\pi/2 - \varphi, \pi/2]$ , and  $\Delta(\pi - \varphi) = p_3 \cos(\pi - \varphi) + q_3 \sin(\pi - \varphi)$ ;
- (iv)  $\Delta(t) = p_4 \cos t + q_4 \sin t$  for  $t \in [\pi/2, \pi]$ , and  $\Delta(\pi) = -p_4$ .

Moreover  $\Delta(\pi/2) = 1 - 1 = 0$ , since every cap with rotation angle  $\pi/2$  has the support value 1 at  $\pi/2$ . Let  $s = -\Delta(\pi)$ .

On  $[\pi/2, \pi]$ : by (iv) at  $t = \pi/2$ ,  $q_4 = \Delta(\pi/2) = 0$ ; so  $\Delta(t) = p_4 \cos t$ , and  $p_4 = -\Delta(\pi) = s$ .

On  $[\pi/2 - \varphi, \pi/2]$ : by (iii) at  $t = \pi/2$ ,  $q_3 = \Delta(\pi/2) = 0$ , so  $\Delta(\pi - \varphi) = p_3 \cos(\pi - \varphi)$ . As  $\pi - \varphi \in [\pi/2, \pi]$ , the previous step gives  $\Delta(\pi - \varphi) = s \cos(\pi - \varphi)$ , and  $\cos(\pi - \varphi) = -\cos \varphi \neq 0$ ; so  $p_3 = s$ .

On  $[\varphi, \pi/2 - \varphi]$ : let  $b = \pi/2 - \varphi$ . For  $u \in [\varphi, b]$  we have  $u + \pi/2 \in [\pi/2, \pi]$ , so  $\Delta(u + \pi/2) = s \cos(u + \pi/2) = -s \sin u$  by the first step, and  $\Delta(b) = s \cos b$  by the second. By (ii),  $\Delta(t) = s \cos b + s \int_t^b \sin u du = s \cos t$ .

On  $[0, \varphi]$ : by (i),  $q_1 = \Delta(\pi/2) = 0$ , so  $\Delta(\varphi) = p_1 \cos \varphi$ , while  $\Delta(\varphi) = s \cos \varphi$  by the third step. As  $\cos \varphi \neq 0$ ,  $p_1 = s$ . So  $\Delta(t) = s \cos t$  on each of the four intervals, which cover  $[0, \pi]$ .  $\square$

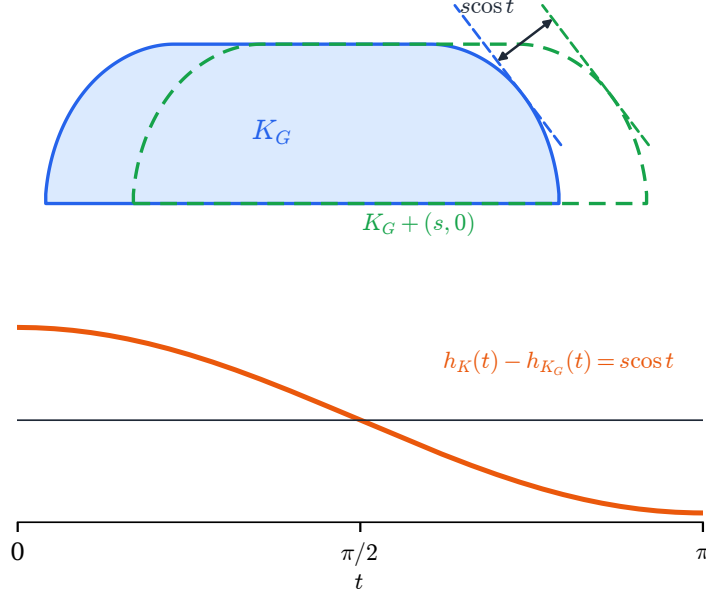


FIGURE 8. Proposition 8.3 and Lemma 8.4. Top: Gerver's cap  $K_G$  and its horizontal translate by  $(s, 0)$ ; their supporting lines at the normal  $t$  lie  $s \cos t$  apart. Bottom: the difference  $h_K - h_{K_G} = s \cos t$  of the support functions on  $[0, \pi]$ .

**Lemma 8.4** (caps with translated supports). *Let  $C \in \mathcal{K}_{\pi/2}^c$  and  $s \in \mathbb{R}$ .*

- (a) *Let  $K$  be a cap in  $\mathcal{K}_{\pi/2}^c$  with  $h_K(t) - h_C(t) = s \cos t$  for  $t \in [0, \pi]$ . Then  $K = C + (s, 0)$ ,  $\mathcal{N}(K) = \mathcal{N}(C) + (s, 0)$  and  $K \setminus \mathcal{N}(K) = (C \setminus \mathcal{N}(C)) + (s, 0)$ .*
- (b) *The translate  $C + (s, 0)$  lies in  $\mathcal{K}_{\pi/2}^c$ ,  $h_{C+(s,0)}(t) - h_C(t) = s \cos t$  for every  $t$ , and  $\mathcal{A}(C + (s, 0)) = \mathcal{A}(C)$ .*

*Proof.* (a) A point  $p$  lies in  $K$  if and only if  $p_y \geq 0$  and  $\langle p, u_t \rangle \leq h_K(t)$  for  $t \in [0, \pi]$ , and likewise for  $C$ . Indeed,  $K$  lies in  $P_{\pi/2} = \mathbb{R} \times [0, 1]$ , and  $\langle p, u_t \rangle \leq h_K(t)$  for  $p \in K$ . Conversely,  $K$  is an intersection of closed half-planes with normal angles in  $J_{\pi/2} \cup \{3\pi/2\} = [0, \pi] \cup \{3\pi/2\}$ ; each of them contains  $K$ , hence contains the supporting half-plane  $H_K(t)$  at its normal angle  $t$ , and  $H_K(3\pi/2) = \{p : p_y \geq 0\}$  because  $h_K(3\pi/2) = 0$ . For  $t \in [0, \pi]$ ,  $\langle p - (s, 0), u_t \rangle = \langle p, u_t \rangle - s \cos t$  and  $h_C(t) = h_K(t) - s \cos t$ , and  $p - (s, 0)$  has the ordinate of  $p$ ; so  $p \in K$  if and only if  $p - (s, 0) \in C$ , that is,  $K = C + (s, 0)$ .

By the definition of the niche,  $p \in \mathcal{N}(K)$  if and only if  $p_y \geq 0$  and there is  $t \in (0, \pi/2)$  with  $\langle p, u_t \rangle < h_K(t) - 1$  and  $\langle p, v_t \rangle < h_K(t + \pi/2) - 1$ , and likewise for  $C$ . For  $t \in (0, \pi/2)$  we have  $h_K(t) = h_C(t) + s \cos t$  and  $h_K(t + \pi/2) = h_C(t + \pi/2) - s \sin t$ , while  $\langle p - (s, 0), u_t \rangle = \langle p, u_t \rangle - s \cos t$  and  $\langle p - (s, 0), v_t \rangle = \langle p, v_t \rangle + s \sin t$ . So  $p \in \mathcal{N}(K)$  if and only if  $p - (s, 0) \in \mathcal{N}(C)$ ,

that is,  $\mathcal{N}(K) = \mathcal{N}(C) + (s, 0)$ . Together with  $K = C + (s, 0)$ , this gives  $K \setminus \mathcal{N}(K) = (C \setminus \mathcal{N}(C)) + (s, 0)$ .

(b) As  $h_{C+w}(t) = h_C(t) + \langle w, u_t \rangle$ ,  $h_{C+(s,0)}(t) - h_C(t) = s \cos t$ . This vanishes at  $t = \pi/2$  and  $t = 3\pi/2$ , so  $h_{C+(s,0)}(\pi/2) = 1$  and  $h_{C+(s,0)}(3\pi/2) = 0$ . The convex body  $C + (s, 0)$  is the intersection of the translates by  $(s, 0)$  of the closed half-planes whose intersection is  $C$ , and these have the same normal angles; so  $C + (s, 0) \in \mathcal{K}_{\pi/2}^c$ . By (a) with  $K = C + (s, 0)$ ,  $\mathcal{N}(C + (s, 0)) = \mathcal{N}(C) + (s, 0)$ , and as translations preserve area,  $\mathcal{A}(C + (s, 0)) = |C| - |\mathcal{N}(C)| = \mathcal{A}(C)$ .  $\square$

**Theorem 8.5** (maximizing right-angle caps). *Let  $K$  be a cap in  $\mathcal{K}_{\pi/2}^c$ . Then  $\mathcal{A}(K) = |G|$  if and only if  $K = K_G + (s, 0)$  for some  $s \in \mathbb{R}$ , and in this case  $K \setminus \mathcal{N}(K) = G + (s, 0)$ . The maximizing caps in  $\mathcal{K}_{\pi/2}^c$  (Definition 3.1) are the horizontal translates  $K_G + (s, 0)$  of Gerver's cap, and every monotone sofa  $U$  with rotation angle  $\pi/2$  and  $|U| = |G|$  is a horizontal translate  $G + (s, 0)$  of  $G$ .*

*Proof.* If  $\mathcal{A}(K) = |G|$ , then  $K \in \mathcal{K}^i$  by Corollary 7.13, Proposition 8.3 gives  $s \in \mathbb{R}$  with  $h_K(t) - h_{K_G}(t) = s \cos t$  for  $t \in [0, \pi]$ , and  $K = K_G + (s, 0)$  by Lemma 8.4(a) with  $C = K_G$ . Conversely, if  $K = K_G + (s, 0)$ , then  $\mathcal{A}(K) = \mathcal{A}(K_G)$  by Lemma 8.4(b), and  $\mathcal{A}(K_G) = |G|$  by Fact 2.4(c), as  $G$  is a monotone sofa with cap  $K_G$  (Fact 2.16(b)). In this case Lemma 8.4(b) and then (a), applied to  $K_G + (s, 0)$  and  $K_G$ , give  $K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ , since  $G = K_G \setminus \mathcal{N}(K_G)$  (Fact 2.16(b)).

By Fact 2.7,  $\mathcal{A} \leq |G|$  on  $\mathcal{K}_{\pi/2}^c$ , and  $K_G \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(K_G) = |G|$ . So a cap  $K \in \mathcal{K}_{\pi/2}^c$  is maximizing if and only if  $\mathcal{A}(K) = |G|$ , that is, if and only if  $K = K_G + (s, 0)$  for some  $s$ ; and every such translate lies in  $\mathcal{K}_{\pi/2}^c$  by Lemma 8.4(b). Finally, let  $U$  be a monotone sofa with rotation angle  $\pi/2$  and  $|U| = |G|$ . Its cap  $K = \mathcal{C}(U)$  lies in  $\mathcal{K}_{\pi/2}^c$ , with  $U = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}(K) = |U| = |G|$  (Fact 2.4(b), (c)). So  $K \in \mathcal{K}^i$  by Corollary 7.13, and Proposition 8.3 and Lemma 8.4(a) with  $C = K_G$  give  $s \in \mathbb{R}$  with  $U = K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ .  $\square$

## 9. RECOVERING THE SOFA; PROOF OF THEOREM 1.1

By (6.1) and Theorem 8.5, a rigid image of  $S$  lies in a horizontal translate of  $G$  and has the same area. A closed subset of a set of finite area, with the same area, is the whole set if that set is the closure of its interior (Lemma 9.1). So it remains to show that  $G$  is the closure of its interior.

**Lemma 9.1** (recovering a set from its area). *Let  $X \subset Y \subset \mathbb{R}^2$  with  $X$  closed,  $Y = \text{cl}(\text{int } Y)$ ,  $|Y| < \infty$  and  $|X| = |Y|$ . Then  $X = Y$ .*

*Proof.* As  $X$  is closed, hence measurable, and  $|Y| < \infty$ ,  $|Y \setminus X| = |Y| - |X| = 0$ . The set  $\text{int } Y \setminus X$  is open and lies in  $Y \setminus X$ , so it has area 0; as a nonempty open set has positive area, it is empty. Hence  $\text{int } Y \subset X$ , and since  $X$  is closed,  $Y = \text{cl}(\text{int } Y) \subset X \subset Y$ .  $\square$

**Lemma 9.2** (removing the region under an envelope). *Let  $K \subset \mathbb{R}^2$  with  $K = \text{cl}(\text{int } K)$  and  $[a, b] \times [0, 1] \subset K$ . Let  $\Gamma$  be a compact set of points with abscissa in  $[a, b]$  and height in  $[0, 1]$ , and let  $\Gamma_{<}$  be the set of points  $q$  with  $q_y \geq 0$  that lie strictly below a point  $\gamma$  of  $\Gamma$  with  $\gamma_x = q_x$ . If  $X = K \setminus \Gamma_{<}$  is closed, then  $X = \text{cl}(\text{int } X)$ .*

*Proof.* Let  $\bar{\Gamma}$  be the set of points  $q$  with  $q_y \geq 0$  that lie on or below a point of  $\Gamma$  with the same abscissa. As the heights of  $\Gamma$  are nonnegative,  $\bar{\Gamma}$  is the image of the compact set  $\Gamma \times [0, 1]$  under  $(\gamma, \lambda) \mapsto (\gamma_x, \lambda\gamma_y)$ , so it is compact; and  $\Gamma_{<} \subset \bar{\Gamma}$ .

First, every point  $p \in K \setminus \bar{\Gamma}$  lies in  $\text{cl}(\text{int } X)$ . Indeed,  $p$  has a neighborhood outside the closed set  $\bar{\Gamma}$ , and since  $K = \text{cl}(\text{int } K)$ ,  $p$  is a limit of interior points of  $K$  outside  $\bar{\Gamma}$ . These points form an open subset of  $K \setminus \bar{\Gamma} \subset K \setminus \Gamma_{<} = X$ , so they are interior points of  $X$ .

Now let  $p \in X$ . If  $p \notin \bar{\Gamma}$ , then  $p \in \text{cl}(\text{int } X)$  by the first step. If  $p \in \bar{\Gamma}$ , there is  $\gamma \in \Gamma$  with  $\gamma_x = p_x$  and  $p_y \leq \gamma_y$ , so  $p_x \in [a, b]$  and  $p_y < 1$ . For  $0 < s < 1 - p_y$ , the point  $(p_x, p_y + s)$  lies in the rectangle  $[a, b] \times [0, 1] \subset K$ , and not in  $\bar{\Gamma}$ : a point  $\delta \in \Gamma$  with  $\delta_x = p_x$  and  $\delta_y \geq p_y + s > p_y$  would put  $p$  in  $\Gamma_{<}$ . By the first step these points lie in  $\text{cl}(\text{int } X)$ , and so does their limit  $p$  as  $s \rightarrow 0$ .

So  $X \subset \text{cl}(\text{int } X)$ , and  $\text{cl}(\text{int } X) \subset X$  since  $X$  is closed.  $\square$

**Proposition 9.3** (Gerver's sofa is regular closed).  $G = \text{cl}(\text{int } G)$ .

*Proof.* Let  $\Gamma = \mathbf{D}([t_0, t_2]) \cup \mathbf{x}([t_1, t_4]) \cup \mathbf{B}([t_3, t_5])$ ,  $a = \mathbf{D}(0)_x$  and  $b = \mathbf{B}(\pi/2)_x$  (Figure 9). The set  $\Gamma$  is compact, as the union of the images of three closed intervals under the continuous curves  $\mathbf{D}$ ,  $\mathbf{x}$  and  $\mathbf{B}$ . By Fact 2.16(c), the abscissa increases along  $\mathbf{D}$  and  $\mathbf{B}$  and, as  $t$  grows, decreases along  $\mathbf{x}$ , and the endpoints are joined, so

$$a = \mathbf{D}(0)_x < \mathbf{D}(t_2)_x = \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x = \mathbf{B}(t_3)_x < \mathbf{B}(t_5)_x = b,$$

and the abscissa of every point of  $\Gamma$  lies in  $[a, b]$ . The ordinate of  $\mathbf{D}$  is nondecreasing, since  $\mathbf{D}'$  is a positive multiple of  $u_t$  and  $\sin t \geq 0$ ; it starts at 0 and ends at  $\mathbf{x}(t_4)_y < 1$ . The ordinate of  $\mathbf{B}$  is nonincreasing, since  $\mathbf{B}'$  is a negative multiple of  $v_t$  and  $\cos t \geq 0$ ; it ends at 0 and starts at  $\mathbf{x}(t_1)_y < 1$ . The ordinate of  $\mathbf{x}$  on  $[t_1, t_4]$  lies in  $(0, 1)$  by Fact 2.17. So the heights of  $\Gamma$  lie in  $[0, 1]$ .

The points  $(a, 1) = \mathbf{D}(0) + v_0 = \mathbf{C}(0)$  and  $(b, 1) = \mathbf{B}(\pi/2) + u_{\pi/2} = \mathbf{A}(\pi/2)$  lie in  $K_G$ , since  $\mathbf{A} = A_{K_G}$  and  $\mathbf{C} = C_{K_G}$  on  $[0, \pi/2]$  (Fact 2.16(b)). As  $K_G$  is convex, it contains the segment from  $(a, 1)$  to  $(b, 1)$ . A cap  $K$  with rotation angle  $\pi/2$  is closed downward above the floor: if  $p \in K$  and  $0 \leq y \leq p_y$ , then  $(p_x, y) \in K$ . Indeed,  $K$  is an intersection of closed half-planes with normal angles in  $[0, \pi] \cup \{3\pi/2\}$ . Lowering  $p$  does not increase  $\langle p, u_s \rangle$  for  $s \in [0, \pi]$ , as  $\sin s \geq 0$ , and a closed half-plane with normal angle  $3\pi/2$  that contains  $K$  contains  $\{y \geq 0\}$ , as  $h_K(3\pi/2) = 0$ . Hence  $K_G \supset [a, b] \times [0, 1]$ . As  $a < b$ ,  $K_G$  is a closed convex set with nonempty interior, so  $K_G = \text{cl}(\text{int } K_G)$ .

By Fact 2.16(b) and (c),  $G = K_G \setminus \mathcal{N}(K_G) = K_G \setminus \Gamma_{<}$ , and  $G$  is closed, being a moving sofa (Fact 2.16(b), Fact 2.4(a)). So Lemma 9.2 applies with  $K = K_G$ .  $\square$

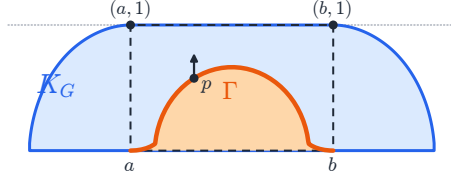


FIGURE 9. Proposition 9.3. Gerver's cap  $K_G$  with its niche (orange) below the curve  $\Gamma$  formed by  $\mathbf{D}$ ,  $\mathbf{x}$  and  $\mathbf{B}$ ; the dashed rectangle  $[a, b] \times [0, 1]$  lies in  $K_G$ , and  $\Gamma$  stays below height one (Fact 2.17). Every point  $p$  of  $\Gamma$  is a limit of points of  $G$  above it.

*Proof of Theorem 1.1.* Let  $S$  be a moving sofa with  $|S| = |G|$ . By Proposition 3.2 and Proposition 6.9, there are  $\omega \in [\omega_0, \pi/2]$  and  $w_0, w_1 \in \mathbb{R}^2$  such that, with  $\psi = \pi/2 - \omega$  and  $g_1(p) = R_\psi(p + w_0) + w_1$ , (6.1) holds:  $g_1(S) \subset U$  for a monotone sofa  $U$  with rotation angle  $\pi/2$  and  $|U| = |G|$ . By Theorem 8.5, there is  $s \in \mathbb{R}$  with

$$g_1(S) \subset U = G + (s, 0).$$

Let  $g(p) = g_1(p) - (s, 0) = R_\psi p + v$  with  $v = R_\psi w_0 + w_1 - (s, 0)$ . Then  $g(S) \subset G$ , and  $\psi \in [0, \pi/2 - \omega_0]$ . The set  $g(S)$  is closed, since  $S$  is closed and  $g$  is a homeomorphism, and  $|g(S)| = |S| = |G|$ , since rotations and translations preserve area; and  $|G| < \infty$  by Fact 2.16(a). By Proposition 9.3,  $G = \text{cl}(\text{int } G)$ , so  $g(S) = G$  by Lemma 9.1, that is,  $R_\psi S + v = G$ . This is the assertion with  $\theta = \psi$ .  $\square$

**Corollary 9.4** (the maximal sofas). *A moving sofa  $S$  has maximal area, that is,  $|S'| \leq |S|$  for every moving sofa  $S'$ , if and only if  $R_\theta S + v = G$  for some  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R}^2$ .*

*Proof.* By Fact 2.7,  $|S'| \leq |G|$  for every moving sofa  $S'$ , and  $G$  is a moving sofa (Fact 2.16(b), Fact 2.4(a)). If  $S$  has maximal area, then  $|G| \leq |S| \leq |G|$ , and Theorem 1.1 gives  $\theta$  and  $v$ . Conversely, if  $R_\theta S + v = G$ , then  $|S| = |G| \geq |S'|$  for every moving sofa  $S'$ , since rotations and translations preserve area.  $\square$

**Lemma 9.5** (the widths of Gerver's sofa). (a) *There are  $a, b, x_- \in \mathbb{R}$  with  $a < 0$  and  $b - x_- > 1$  such that the points  $(a, 1)$ ,  $(b, 1)$ ,  $(x_-, 0)$  and  $(1, 0)$  lie in  $G$ .*

(b) *For every  $r \in [0, \pi]$  with  $r \neq \pi/2$  there are  $p, q \in G$  with  $\langle p - q, u_r \rangle > 1$ .*

(c) *If  $w \in \mathbb{R}^2$  is a unit vector and  $\langle p - q, w \rangle \leq 1$  for all  $p, q \in G$ , then  $w_x = 0$ .*

*Proof.* (a) Let  $a = \mathbf{D}(0)_x$ ,  $b = \mathbf{B}(\pi/2)_x$  and  $x_- = \mathbf{x}(\pi/2)_x - 1$ , and let  $\Gamma$  be as in the proof of Proposition 9.3. A point  $q \in \Gamma_<$  lies strictly below a point  $\gamma \in \Gamma$  with  $\gamma_x = q_x$ , so  $\gamma_y > q_y \geq 0$ . Write  $\gamma = \mathbf{D}(t)$  with  $t \in [t_0, t_2]$ ,  $\gamma = \mathbf{x}(t)$  with  $t \in [t_1, t_4]$ , or  $\gamma = \mathbf{B}(t)$  with  $t \in [t_3, t_5]$ . As  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$  lie on the  $x$ -axis,  $t \neq t_0$  in the first case and  $t \neq t_5$  in the third. By the strict monotonicity of the abscissa along  $\mathbf{D}$  on  $[t_0, t_2]$ , along  $\mathbf{x}$  on  $[t_1, t_4]$  and along  $\mathbf{B}$  on  $[t_3, t_5]$  (Fact 2.16(c)), and the chain  $a = \mathbf{D}(0)_x < \mathbf{D}(t_2)_x = \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x = \mathbf{B}(t_3)_x < b$  of the proof of Proposition 9.3,  $q_x = \gamma_x \in (a, b)$ . As  $G = K_G \setminus \mathcal{N}(K_G) = K_G \setminus \Gamma_<$  (Fact 2.16(b), (c)), every point of  $K_G$  with abscissa outside  $(a, b)$  lies in  $G$ .

The points  $(a, 1) = \mathbf{C}(0)$  and  $(b, 1) = \mathbf{A}(\pi/2)$  lie in  $K_G$ , as in the proof of Proposition 9.3. So do  $\mathbf{A}(0) = (1, 0)$  (Appendix A, condition (2)) and  $\mathbf{C}(\pi/2) = (x_-, 0)$  (Fact 2.16(d)), as  $\mathbf{A} = A_{K_G}$  and  $\mathbf{C} = C_{K_G}$  on  $[0, \pi/2]$  (Fact 2.16(b)). By the formula for the inner corner (Section 2.2), applied to  $\mathbf{x} = \mathbf{x}_{K_G}$  (Fact 2.16(b)) at  $t = 0$  and  $t = \pi/2$ ,  $h_{K_G}(0) = \langle \mathbf{x}(0), u_0 \rangle + 1 = 1$ , as  $\mathbf{x}(0) = O$  (Appendix A, condition (2)), and  $h_{K_G}(\pi) = \langle \mathbf{x}(\pi/2), v_{\pi/2} \rangle + 1 = -x_-$ . So every point  $p$  of  $K_G$  has  $p_x = \langle p, u_0 \rangle \leq h_{K_G}(0) = 1$  and  $-p_x = \langle p, u_\pi \rangle \leq h_{K_G}(\pi) = -x_-$ ; in particular  $x_- \leq a$  and  $b \leq 1$ . Hence none of  $a, b, x_-$  and 1 lies in  $(a, b)$ , and the four points lie in  $G$ .

By condition (3) of the injectivity condition, which  $K_G$  satisfies with  $\mathbf{x}_{K_G} = \mathbf{x}$  (Fact 2.16(b)),  $\alpha < 0 < \beta$  on  $(0, \pi/2)$ . So the abscissa of  $\mathbf{x}$ , whose derivative is  $\alpha \cos t - \beta \sin t$ , is decreasing on  $[0, \pi/2]$ . With the chain above,  $a < \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x \leq \mathbf{x}(0)_x = 0$  and  $b > \mathbf{x}(t_1)_x > \mathbf{x}(t_4)_x \geq \mathbf{x}(\pi/2)_x = x_- + 1$ .

(b) Let  $a, b$  and  $x_-$  be as in (a). Since  $0 \leq \sin r \leq 1$ ,  $\sin r \geq \sin^2 r$ , and likewise  $|\cos r| \geq \cos^2 r$ ; so  $|\cos r| + \sin r \geq 1$ . For  $r \in [0, \pi/2)$ ,  $\cos r > 0$  and  $\langle (b, 1) - (x_-, 0), u_r \rangle = (b - x_-) \cos r + \sin r > \cos r + \sin r \geq 1$ . For  $r \in (\pi/2, \pi]$ ,  $\cos r < 0$  and  $\langle (a, 1) - (1, 0), u_r \rangle = (1 - a)|\cos r| + \sin r > |\cos r| + \sin r \geq 1$ . These four points lie in  $G$  by (a).

(c) Suppose first that  $w_y \geq 0$ . Then  $w = u_r$  with  $r = \arccos w_x \in [0, \pi]$ , and if  $w_x \neq 0$ , then  $r \neq \pi/2$ , and (b) contradicts the hypothesis. If  $w_y < 0$ , the same applies to  $-w$ , since  $\langle p - q, -w \rangle = \langle q - p, w \rangle \leq 1$  for  $p, q \in G$ .  $\square$

**Corollary 9.6** (no rotation is needed). (a) If  $S$  is a moving sofa with  $|S| = |G|$ , then  $S + v = G$  for some  $v \in \mathbb{R}^2$ , so that  $S$  is a translate of  $G$ ; in particular,  $S$  is a moving sofa with rotation angle  $\pi/2$ .

(b) If  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R}^2$  satisfy  $R_\theta G + v \subset H_L$ , then  $\sin \theta = 0$ . So  $R_\theta G$  is a moving sofa only if  $\theta$  is a multiple of  $\pi$ ; in particular,  $R_{\pi/2} G$  is not a moving sofa.

*Proof.* (a) By the proof of Theorem 1.1, there are  $\psi \in [0, \pi/2 - \omega_0]$  and  $v \in \mathbb{R}^2$  such that  $g(p) = R_\psi p + v$  maps  $S$  onto  $G$ . A moving sofa lies in a horizontal strip of height one: the start position  $\Phi_0(S)$  of a movement of  $S$  is a translate of  $S$  and lies in  $H_L$  (Definition 2.2), so  $p_y - q_y \leq 1$  for  $p, q \in S$ . For  $p, q \in S$ ,

$$\langle g(p) - g(q), v_\psi \rangle = \langle R_\psi(p - q), R_\psi v_0 \rangle = \langle p - q, v_0 \rangle = p_y - q_y \leq 1,$$

so  $\langle x - y, u_{\psi+\pi/2} \rangle \leq 1$  for all  $x, y \in G = g(S)$ . If  $\psi > 0$ , then  $\psi + \pi/2 \in (\pi/2, \pi)$ , as  $\psi \leq \pi/2 - \omega_0 < \pi/2$ , and Lemma 9.5(b) contradicts this. So  $\psi = 0$ , and  $S + v = g(S) = G$ . As  $G$  is a moving sofa with rotation angle  $\pi/2$  (Fact 2.16(b), Fact 2.4(a)), so is its translate  $S = G - v$ .

(b) For  $p, q \in G$ , the points  $R_\theta p + v$  and  $R_\theta q + v$  of  $H_L$  have ordinates in  $[0, 1]$ , so the ordinate of  $R_\theta(p - q)$ , which is  $\langle p - q, w \rangle$  with the unit vector  $w = (\sin \theta, \cos \theta)$ , is at most 1. By Lemma 9.5(c),  $\sin \theta = w_x = 0$ . If  $R_\theta G$  is a moving sofa, the start position of a movement of it is a translate of  $R_\theta G$  that lies in  $H_L$  (Definition 2.2), so  $\sin \theta = 0$  by the first statement. As  $\sin(\pi/2) = 1$ ,  $R_{\pi/2}G$  is not a moving sofa.  $\square$

## 10. FORMALIZATION, THE USE OF AI, AND OPEN QUESTIONS

**10.1. The formalization.** The repository [14] (Apache-2.0) contains three Lean 4 libraries on top of Mathlib (Lean and Mathlib v4.35.0-rc3):

- **MovingSofaOptimality** (51 files, about 40,000 lines): Baek's paper [1], organized by its chapters, with the results that it cites (Schneider's area formula for planar convex bodies; the existence and uniqueness of the solution of Romik's system in the box of Fact 2.16) and the structure of Gerver's sofa (Fact 2.16(b),(c)), which [1] states without proof (Thm. 8.4.1) or proves invalidly (Thm. 6.1.2);
- **MovingSofaUniqueness** (8 files, about 5,500 lines): Sections 4 to 9, one module per step of the argument;
- **MovingSofaBridge** (4 files, about 2,600 lines): the connection with the definitions of formal-conjectures [4], which are different (a motion there is a path of isometries that starts at the identity, and Gerver's sofa is defined from Gerver's four constants); it shows that a set is a moving sofa for formal-conjectures if and only if it lies in the horizontal side of the hallway and is a moving sofa in the sense of Appendix C, and that the optimal areas and the Gerver's sofas of the two formulations agree, so that formal-conjectures' statements follow from Baek's theorem and from Theorem 1.1, with the existence and uniqueness of Gerver's four constants, which the bridge proves; the uniqueness statement, which it lists as open, follows from Theorem 1.1.

The file `Challenge.lean` states twelve theorems, among them Baek's theorem (`Baek.gerver_sofa_optimal`) and Theorem 1.1 (`Baek.gerver_sofa_unique`). The five theorems of the namespace `Baek` use only Mathlib's vocabulary and the definitions of Appendices A and C; the other seven use the definitions of formal-conjectures, restated verbatim in a namespace of their own (an anonymous topology instance is given a name), and the three theorems of the namespace `Bridge` also use Baek's definitions. `Solution.lean`, with the modules that it imports, proves them: the Challenge is a file of statements, so its twelve theorems end in `sorry` by design, and `Comparator` checks that `Solution.lean` proves exactly the statements

of `Challenge.lean`. `lake build` succeeds, and an audit checks that every declaration of the three libraries depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The project is registered in the Palomar registry as entry PALOMAR-2026-10-02-000008, whose version 4 registers an earlier commit with the same Challenge. The continuous integration and Palomar’s preflight both passed on the commit cited in [14]. To rebuild, run `lake exe cache get && lake build`, then the commands listed in the repository’s `README.md`. The Challenge states Theorem 1.1; the other results of this paper are proved in the libraries, and Appendix C lists the declarations.

The machine check shows that Theorem 1.1, as the Lean statement formulates it, follows from Lean’s axioms and Mathlib. The formal proof contains Baek’s argument and the argument of this paper. The machine check does not check that the definitions say what they should. Appendix C reproduces the Lean definitions (moving sofa, hallway, the rotation path and Gerver’s sofa) in mathematical notation, so that they can be read and compared with the intended meaning. Every proof of a result of [1] in the library follows Baek’s argument, except at sixteen steps, each forced by an error or gap of the paper, by mathematics that Mathlib lacks (the Jordan curve theorem, Green’s theorem, the Brunn–Minkowski inequality, mixed volumes), or by the representations chosen in the formalization (the surface area measure as a Lebesgue–Stieltjes measure, the area functional of a convex arc as  $\frac{1}{2} \int h d\sigma$ , no orientations of curves). A route check in continuous integration compares the results used by each Lean proof with those cited by Baek’s proof; the other details were compared with Baek’s TeX source by AI agents, not by a person.

**10.2. The use of AI.** ChatGPT Pro 6 (OpenAI) wrote the first version of the proof of Theorem 1.1 on 2 October 2026, at the author’s request, together with drafts of its formalization and of the connection with formal-conjectures; neither Lean draft had ever been compiled. Claude Opus 5.5 (Anthropic), in Claude Code with sub-agents, formalized [1] and completed the two drafts, following the author’s `formalize-math-paper` procedure, in rounds between 1 and 3 October 2026; 54 proofs of the first draft and 78 proofs of the second did not compile and were replaced and proved again. The repository [14] records who did what, with times and sizes. Claude Sonnet 5.5 (Anthropic), in Claude Code, wrote the first version of this manuscript on 4 October 2026 from the Lean library, the illustrated text of the proofs in [14] and the text of [1]. Eight of the nine figures were computed from the definitions by the script `docs/paper/figures/make_figures.py` of the repository, which was added after the commit cited in [14]; the ninth is drawn in TikZ. The author asked for each of these steps and decided its scope. None of the three systems is an author of this paper, since none of them can take responsibility for it.

The text of this paper is not machine-checked; the dictionary of Appendix C gives its correspondence with the Lean statements. On 4 October

2026, six runs of Claude Opus 5.5, each assigned some sections, compared the statements and proofs of the first version of this paper with the Lean statements and with the TeX source of [1]; a run of Claude Sonnet 5.5 checked the claims about the formalization, the dictionary and the references; and a run of Claude Opus 5.5 read the whole paper without access to either. The findings were corrected, and a further run of Claude Opus 5.5 checked the passages rewritten in response. Two more runs of Claude Opus 5.5 checked a rewrite of the prose for the patterns of machine-written text and for changes of claim. The same day, the Lean library was extended with the results of this paper that it did not yet prove, and Sections 2 and 4 to 9 were rewritten, by sub-agents running Claude Opus 5.5, to state what the Lean states and to follow its proofs. Five runs of Claude Opus 5.5, each assigned some sections and appendices, then compared the result with the Lean, and a second round of the same runs compared the corrected text again. The changes that followed the second round were not checked again. At the time of writing, the argument has not been refereed, and no person has checked it independently of the formalization.

**10.3. Open questions.** *Stability.* Is there a quantitative version of Theorem 1.1: a bound for the distance of a moving sofa  $S$  to the nearest copy of  $G$  in terms of  $|G| - |S|$ ? The obstacle is that Baek’s bound  $\mathcal{A} \leq \mathcal{Q}$  is available only for caps with the injectivity condition, which we establish for sofas of maximal area, through limits that carry no rate.

*Other corners.* For a hallway whose corner has an angle  $\alpha \neq \pi/2$  one can ask for the largest area as a function of  $\alpha$  and for the extremal shapes. Each step of Baek’s method has an analogue to prove, and its numerical inequalities, such as those of Fact 6.5, are specific to the right angle.

*The ambidextrous sofa.* Romik [10] found a sofa of area 1.6449... that can turn both left and right around a right-angled corner. As far as we know, it is not known whether it is optimal. Baek’s reduction to monotone sofas uses one hallway; for a sofa that must turn both ways, the cap, the niche and the bound  $\mathcal{Q}$  all need two-sided analogues.

**Acknowledgments.** This paper builds on the results of [1]. The audit of [1] owes twelve findings to the notes of another formalization of that paper [9], and was also compared with the blueprint of a third [12].

## APPENDIX A. GERVER’S SOFA

We recall Romik’s description of Gerver’s sofa [10, Section 4], in the form used by [1] and by the formalization. Let  $t_0 = 0 < t_1 = \varphi < t_2 = \theta < t_3 = \pi/2 - \theta < t_4 = \pi/2 - \varphi < t_5 = \pi/2$ . A *parameter tuple* consists of the angles  $\varphi, \theta$ , numbers  $a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2, e_1, e_2$  and points  $\kappa_1, \dots, \kappa_5 \in \mathbb{R}^2$  (22 real parameters). The *rotation path*  $\mathbf{x} : [0, \pi/2] \rightarrow \mathbb{R}^2$  is  $\mathbf{x} = \mathbf{x}_i$  on the  $i$ th

phase, where the phases are  $[0, t_1]$ ,  $[t_1, t_2]$ ,  $[t_2, t_3]$ ,  $(t_3, t_4]$  and  $(t_4, t_5]$ , and

$$\begin{aligned}\mathbf{x}_1(t) &= R_t(a_1 \cos t + a_2 \sin t - 1, -a_2 \cos t + a_1 \sin t - \tfrac{1}{2}) + \kappa_1, \\ \mathbf{x}_2(t) &= R_t(-\tfrac{t^2}{4} + b_1 t + b_2, \tfrac{t}{2} - b_1 - 1) + \kappa_2, \\ \mathbf{x}_3(t) &= R_t(c_1 - t, c_2 + t) + \kappa_3, \\ \mathbf{x}_4(t) &= R_t(-\tfrac{t}{2} + d_1 - 1, -\tfrac{t^2}{4} + d_1 t + d_2) + \kappa_4, \\ \mathbf{x}_5(t) &= R_t(e_1 \cos t + e_2 \sin t - \tfrac{1}{2}, -e_2 \cos t + e_1 \sin t - 1) + \kappa_5.\end{aligned}$$

The pieces of a solution agree at the ends of the phases, by condition (3) below, so the conventions at the ends matter only for other tuples. For a differentiable curve  $\mathbf{x}$  let  $\alpha = \langle \mathbf{x}', u_t \rangle$ ,  $\beta = \langle \mathbf{x}', v_t \rangle$ , and let  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$  be as in Section 2.6. The tuple *solves Romik's equations* if  $0 < \varphi < \theta < \pi/4$  and:

- (1) (left–right symmetry)  $e_1 = a_1$ ,  $e_2 = -a_2$ ,  $d_1 = \frac{\pi}{4} - b_1$ ,  $d_2 = b_2 + \frac{\pi}{4}(2b_1 - \frac{\pi}{4})$ ,  $c_2 = c_1 - \frac{\pi}{2}$ ;
- (2) (the start)  $\kappa_1 = (1 - a_1, \frac{1}{4})$  and  $a_2 = -\frac{1}{4}$ , so that  $\mathbf{x}(0) = O$  and  $\mathbf{A}(0) = (1, 0)$ ;
- (3) (continuous differentiability)  $\mathbf{x}_i(t_i) = \mathbf{x}_{i+1}(t_i)$  and  $\mathbf{x}'_i(t_i) = \mathbf{x}'_{i+1}(t_i)$  for  $i = 1, 2, 3, 4$ ;
- (4) (the contact conditions)  $\mathbf{x}_1(t_1) = \mathbf{B}_4(t_3)$  and  $\mathbf{x}_5(t_4) = \mathbf{D}_2(t_2)$ , where  $\mathbf{B}_4 = \mathbf{x}_4 + \alpha_4 v_t$  and  $\mathbf{D}_2 = \mathbf{x}_2 - \beta_2 u_t$  are the contact curves of  $\mathbf{x}_4$  and  $\mathbf{x}_2$ .

These are Romik's equations (27)–(44). *Gerver's sofa* is the shape of the rotation path of the solution with  $\varphi \in [0.039, 0.04]$  and  $\theta \in [0.68, 0.69]$ , which is unique in this box (Fact 2.16(a)). Its numerical values, to ten decimal places, are

$$\begin{aligned}\varphi &= 0.0391773648, & \theta &= 0.6813015094, \\ a_1 &= e_1 = 1.2103224221, & a_2 &= -e_2 = -0.25, \\ b_1 &= -0.5276245980, & b_2 &= 0.9202583852, \\ c_1 &= 0.6260455228, & c_2 &= -0.9447508039, \\ d_1 &= 1.3130227614, & d_2 &= -0.5253826704,\end{aligned}$$

$$\begin{aligned}\kappa_1 &= (-0.2103224221, 0.25), & \kappa_5 &= (-1.0172040368, 0.25), \\ \kappa_2 &= (-0.9191792928, 0.4724066198), \\ \kappa_3 &= (-0.6137632294, 0.8896264790), \\ \kappa_4 &= (-0.3083471661, 0.4724066198),\end{aligned}$$

and  $\mathbf{x}(\pi/2) = (-1.2275264589, 0)$ ,  $\max_t \mathbf{x}(t)_y = \mathbf{x}(\pi/4)_y = 0.6642678135$ ,  $\mathbf{x}(t_1)_y = \mathbf{x}(t_4)_y = 0.0551867605$ . The area is  $|G| = 2.21953166887\dots$  (Gerver, Romik), and  $2.2192 \leq |G| \leq 2.2199$  is proved in [14]. In Gerver's own parametrization the constants  $A = 0.0944265608$  and  $B = 1.3992037273$  together with  $\varphi$  and  $\theta$  solve a system of four equations [3], from which the 22 parameters above are computed.

The contact curves  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$  are the points where the sides of the hallway touch the sofa. The sofa touches the walls  $a(t)$  and  $c(t)$  at  $\mathbf{A}(t)$  and  $\mathbf{C}(t)$  for all  $t$ ; the wall  $d(t)$  at  $\mathbf{D}(t)$  for  $t \in [t_0, t_2]$ ; the wall  $b(t)$  at  $\mathbf{B}(t)$  for  $t \in [t_3, t_5]$ ;

and the inner corner touches it at  $\mathbf{x}(t)$  for  $t \in [t_1, t_4]$ . As in [3, 10], the boundary of  $G$  consists of 18 pieces: **A** on four phases, the top edge, **C** on four phases, an edge on the  $x$ -axis, **D** on two phases, **x** on three phases, **B** on two phases, and a second edge on the  $x$ -axis.

#### APPENDIX B. CORRECTIONS TO BAEK'S PAPER THAT ARE USED HERE

The audit that accompanies the formalization compared every result of [1] with its TeX source and with the formal statement [14]. Every result of the paper holds with the intended statement. Table 1 lists the corrections that concern the results used in this paper, with the labels  $En$  of the audit. The complete list of the audit has 27 items, E22 only pointing to E21, and two missing hypotheses: [1, Thm. 2.1.1] (Schneider's Theorem 4.2.3:  $\sigma_K(X)$  is the length of  $\bigcup_{t \in X} e_K(t)$ ) needs  $K$  to have interior points, and [1, Thm. 3.1.2] needs a bounded Nef polygon; every use satisfies them (the text itself uses  $\sigma_K$  only as defined in Fact 2.1(b), which does not depend on Thm. 2.1.1). Twelve of the findings in the audit come from the notes of another formalization of Baek's proof [9].

Table 1: Corrections to [1] that concern the results used here.

| Item | Result of [1]                                           | Correction                                                                                                                                                                                                                                       | Where here                         |
|------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| E1   | Def. 2.3.9                                              | the left and right side of a line are defined for lines not parallel to the $x$ -axis (printed: $y$ -axis)                                                                                                                                       | Fact 2.4(a)                        |
| E2   | Thm. 2.4.1, 2.4.2                                       | the union over $t \in [0, \omega]$ and over $(0, \omega)$ agree, since $F_\omega \cap Q_K^-(0) = F_\omega \cap Q_K^-(\omega) = \emptyset$ ; $\mathcal{C}(S)$ is bounded also for $\omega = \pi/2$                                                | Fact 2.4(b)                        |
| E3   | Prop. 2.5.4                                             | the reflection exchanges the walls ( $a \leftrightarrow c$ , $b \leftrightarrow d$ ) and the ends $W \leftrightarrow Z$ ; $K$ and not $K^m$ stands on the right of the vertex identity                                                           | Fact 2.4(d), Lemma 2.6             |
| E4   | Lemma 2.5.6, Thm. 2.5.8 (and Lemma 3.4.8, Remark 3.4.1) | used without proof, true for every cap: $A_K^-(0) = (h_K(0), 0)$ , $C_K^+(\omega) = h_K(\omega + \pi/2)v_\omega$ , and $O \in K$ if $\omega < \pi/2$ ; the same holds for $o_\omega$ , which Lemma 3.4.8 and Remark 3.4.1 use (not in the audit) | Lemma 2.5, Lemma 2.6, Fact 2.13(d) |

| Item | Result of [1]               | Correction (continued)                                                                                                                                                                                                                                                                                                         | Where here                |
|------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| E5   | Thm. 3.1.2,<br>Lemma 3.4.7  | the theorem needs its simple Nef polygon, a set $\{p : \mathcal{E}(p \in H_1, \dots, p \in H_n)\}$ for closed or open half-planes $H_1, \dots, H_n$ with distinct boundary lines and a monotone Boolean function $\mathcal{E}$ , to be bounded; the two applications of the theorem for a pinned normal change disjoint strips | Fact 2.11(a)              |
| E6   | Def. 3.2.5                  | $\mathcal{N}_\Theta(K) = F_\omega \cap \bigcup_{t \in \Theta} Q_K^-(t)$ (printed: $P_\omega$ in place of $F_\omega$ ); with $P_\omega$ , Prop. 3.3.5 and Lemma 3.4.2 are false                                                                                                                                                 | Section 2.3,<br>Fact 2.9  |
| E7   | Prop. 3.3.1 (2)             | normal angles in $\Theta^\circ \cup \{\omega + \pi, 3\pi/2\}$ (printed: $\Theta^\circ$ )                                                                                                                                                                                                                                       | Section 2.3,<br>Fact 2.11 |
| E8   | Thm. 3.4.3,<br>3.5.2, 3.5.4 | minor gaps: a Hausdorff limit of polygon caps is a polygon cap (Thm. 3.4.3), resp. a cap (Thm. 3.5.2), and each point of the niche of the limit lies in the niches of the polygon caps from some index on (Thm. 3.5.4)                                                                                                         | Fact 2.7                  |
| E9   | Lemma 4.2.4                 | the range is $\omega \in [\sec^{-1}(2.2), \pi/2)$ (printed: $[\tan^{-1}(2.2), \pi/2)$ )                                                                                                                                                                                                                                        | Fact 6.5                  |
| E10  | Thm. 4.2.5,<br>1.5.2        | the width bound for the pentagon $P_\omega \setminus \Delta_\omega$ is asserted without proof, and $\Delta_\omega \subset \mathcal{N}(K)$ is taken from Thm. 4.2.5, which puts the vertices of $\Delta_\omega$ in the closure of a quarter-plane $Q_K^-(t)$ ; its proof puts them in the open quarter-plane                    | Lemma 6.6,<br>Lemma 6.7   |
| E12  | Thm. 6.1.2                  | the proof (Gerver's Thm. 2 makes $G$ a balanced maximum sofa) is invalid; the statement is true and follows from Romik's equations                                                                                                                                                                                             | Fact 2.16(b)              |
| E13  | Prop. 6.2.2                 | $f_{K^m}^\pm(t) = g_K^\mp(\pi/2 - t)$ , and likewise for $g$ (printed: without the change of angle)                                                                                                                                                                                                                            | Fact 2.13(c)              |

| Item | Result of [1]                        | Correction (continued)                                                                                                                                                                                                                                                                                       | Where here                                                                         |
|------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| E14  | Def. 6.2.2,<br>Thm. 6.2.3            | $\partial^+$ is the right derivative<br>(Def. 6.2.2 calls it the left one),<br>and the second display of<br>Thm. 6.2.3 is about $\partial^-\mathbf{x}_K$                                                                                                                                                     | Fact 2.13(a)                                                                       |
| E15  | Lemma 6.3.1                          | cites Thm. 3.5.4 (balanced<br>maximum caps) for<br>$\mathcal{N}(K) \subset K$ of a maximum<br>polygon cap; the applicable<br>result is Thm. 3.4.10, which<br>gives what the lemma needs,<br>and the lemma holds                                                                                              | not used in the<br>proof (the box of<br>Proposition 4.5<br>bounds the<br>diameter) |
| E16  | Def. 6.4.1                           | $f_K$ and $g_K$ map $[0, \pi/2]$ to $\mathbb{R}$<br>(printed: to $\mathbb{R}^2$ )                                                                                                                                                                                                                            | Fact 2.13(d)                                                                       |
| E17  | Lemma 6.5.2–<br>6.5.5,<br>Thm. 6.5.6 | $f_K(0) = 1$ is used without<br>being stated, and so is<br>$g_K(\pi/2) = 1$ (not in the<br>audit); Lemma 6.5.5 is used on<br>$(0, \pi/2]$ (printed: $(0, 1]$ )                                                                                                                                               | Fact 2.13(d),<br>Fact 2.14(b)                                                      |
| E19  | Lemma 8.1.5                          | the proof deduces that the<br>bottom edge $e_K(3\pi/2)$ has<br>length $\sigma_K(3\pi/2) \geq 2.2$ from<br>$ K  \geq 2.2$ without a reason;<br>this holds, since the horizontal<br>width of a cap with rotation<br>angle $\pi/2$ does not increase<br>with the height, so that<br>$ K  \leq \sigma_K(3\pi/2)$ | Fact 2.15(a)                                                                       |
| E20  | Lemma 8.3.5                          | the arc $\mathbf{u}_K^{0,\pi}$ is outside the<br>range $b < a + \pi$ of [1,<br>Thm. 7.3.2]; the area is split at<br>the atom $\pi/2$ instead                                                                                                                                                                 | Fact 2.15(b)                                                                       |
| E21  | Lemma 8.1.7,<br>Thm. 8.2.4           | $\mathcal{N}(K) \subset K$ is used but not<br>available on $\mathcal{K}^1$ ; both<br>statements hold and have<br>proofs that avoid it                                                                                                                                                                        | Fact 2.15(a)                                                                       |
| E23  | Lemma 8.3.6 (3)                      | the tangent line<br>parametrization is evaluated at<br>$\pi/2$ (printed: at $\varphi^L$ , where the<br>statement is false)                                                                                                                                                                                   | Fact 2.15(b)                                                                       |

| Item        | Result of [1]                      | Correction (continued)                                                                                                                                                                                                                                                                                                    | Where here                                             |
|-------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| E24         | Thm. 8.4.1                         | stated without proof; true, and proved from Romik's equations; part (4) holds with one-sided derivatives at the junctions and the ends of the phases. Not in the audit: part (2), read as a Jordan region, omits the open floor segment of $\mathcal{N}(K_G)$ , and the exact description is the identity in Fact 2.16(c) | Fact 2.16                                              |
| E25,<br>E26 | Thm. 8.4.3 (2),<br>Prop. 8.4.4 (4) | $\mathbf{B}(t_3)$ (printed: $\mathbf{D}(t_3)$ ); the measure $\check{\sigma}_D$ lives on $(\pi/2 + t_0, \pi/2 + t_2]$ with a shifted density                                                                                                                                                                              | Fact 2.15(c)                                           |
| E27         | several                            | misprints, among them: the second half-plane of $Q_S^-(t)$ lacks its “−1” (Prop. 2.2.2); the sign of the change of $ \mathcal{N}_\Theta $ in the proof of Lemma 3.4.7; $\mathcal{R}_B$ starts at $\pi + \varphi^R$ , not $\pi/2 + \varphi^R$ (Def. 8.3.3); the statement of Thm. 2.1.3                                    | Section 2.2,<br>Fact 2.1,<br>Fact 2.11,<br>Section 2.5 |

## APPENDIX C. THE LEAN FORMALIZATION: THE STATEMENT AND A DICTIONARY

**C.1. The statement of Theorem 1.1.** The statement of record is `Baek.gerver_sofa_unique` in `Challenge.lean`. It uses Mathlib's vocabulary and the definitions below, which `Challenge.lean` copies verbatim from `ChallengeDefs.lean`; they are those of this paper. In Lean the plane is  $\mathbb{R} \times \mathbb{R}$ , the area of a set is its Lebesgue measure `volume`, and `rot t` is the counterclockwise rotation by  $t$ .

- `horizSide` =  $\{p : p_1 \leq 1, 0 \leq p_2 \leq 1\} = H_L$ , `vertSide` =  $\{p : 0 \leq p_1 \leq 1, p_2 \leq 1\} = V_L$ , `hallway` =  $H_L \cup V_L = L$ .
- `IsMovingSofa S` holds if  $S$  is closed and connected (hence nonempty) and there are  $\theta : \mathbb{R} \rightarrow \mathbb{R}$  and  $c : \mathbb{R} \rightarrow \mathbb{R}^2$ , continuous on  $[0, 1]$ , with  $\theta(0) = 0$ ,  $\text{rot}_{\theta(0)}p + c(0) \in H_L$  for  $p \in S$ ,  $\text{rot}_{\theta(s)}p + c(s) \in L$  for  $p \in S$  and  $s \in [0, 1]$ , and  $\text{rot}_{\theta(1)}p + c(1) \in V_L$  for  $p \in S$  (Definition 2.2, the rotation angle  $-\theta(1)$  being left free).
- `shapeOfPath x` =  $H_L \cap \bigcap_{t \in [0, \pi/2]} (\mathbf{x}(t) + R_t L) \cap (\mathbf{x}(\pi/2) + R_{\pi/2} V_L)$ , written in Lean with images of the hallway sets under  $p \mapsto \mathbf{x}(t) + \text{rot}_t p$ .
- `GerverParams` holds the 22 parameters of Appendix A; `P.path` is the rotation path glued from the five curves of Appendix A; `P.IsSolution`

is the conjunction of  $0 < \varphi < \theta < \pi/4$  and the four groups (1)–(4) of Appendix A (Romik's equations (27)–(44)); `P.InBox` says  $\varphi \in [0.039, 0.04]$ ,  $\theta \in [0.68, 0.69]$ ; and `gerverSofa P = shapeOfPath P.path`.

The theorem states that for every `P : GerverParams` with `P.IsSolution` and `P.InBox`, and every set  $S \subset \mathbb{R} \times \mathbb{R}$  with `IsMovingSofa S` and `volume S = volume (gerverSofa P)`, there are  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R} \times \mathbb{R}$  such that the image  $\{\text{rot}_{\theta} p + v : p \in S\}$  equals `gerverSofa P`. The theorems `Baek.gerver_params_exists` and `Baek.gerver_params_unique` say that the box contains exactly one solution, so that `gerverSofa P` is Gerver's sofa  $G$ , and `Baek.gerver_sofa_area` says that its area lies between 2.2192 and 2.2199.

**C.2. Dictionary.** Table 2 lists the declarations that prove the results of this paper; they are in the repository [14]. Each name is followed, directly or after other names of the same file, by its file in parentheses: a file name without a directory, such as `Main`, is a file of `MovingSofaUniqueness/`, a path with a directory, such as `Basic/SurfaceArea`, is a file of `MovingSofaOptimality/`, and a path ending in `.lean` starts at the root of the repository. The theorems of `Challenge.lean` are stated there with `sorry` and proved in `Solution.lean`. Names are written without the namespace of their library: `image_eq_gerver_of_volume_eq` is `MovingSofaUniqueness.image_eq_gerver_of_volume_eq`. A result numbered  $n_1.n_2.n_3$  in [1] is formalized under the name `theorem $n_1.n_2.n_3$` , or `lemma`, `proposition`, `corollary` or `definition` in place of `theorem`, followed by a suffix, as in `proposition2_5_4_sigma` or `definition8_1_2_exists`, when its parts are separate declarations. Some of the listed declarations, such as `IsCap.zero_mem`, are private to their files.

Table 2: The results of this paper and the Lean declarations that prove them.

| Result           | Lean declarations                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 1.1      | <code>Baek.gerver_sofa_unique</code> (stated in <code>Challenge.lean</code> , proved in <code>Solution.lean</code> ); <code>image_eq_gerver_of_volume_eq</code> , <code>maximizer_contained_in_gerver</code> ( <code>Main</code> )                                                                                                                        |
| Lemma 2.5        | <code>IsCap.zero_mem</code> , <code>IsCap.oPt_mem</code> ( <code>Monotone/CapContainsNiche</code> )                                                                                                                                                                                                                                                       |
| Lemma 2.6, (2.3) | <code>IsCap.corner_mem</code> , <code>IsCap.cPlus_eq_corner</code> , <code>IsCap.aMinus_eq_corner</code> ( <code>Monotone/CapContainsNiche</code> )                                                                                                                                                                                                       |
| Definition 3.1   | <code>IsMaxCap</code> ( <code>Main</code> )                                                                                                                                                                                                                                                                                                               |
| Proposition 3.2  | <code>maximal_envelope</code> , <code>own_cap_isMax</code> , <code>isMaxCap_of_area_eq</code> ( <code>Main</code> ), with <code>theorem2_4_3</code> ( <code>Monotone/CapNiche</code> ) and <code>theorem2_5_10</code> ( <code>Monotone/CapContainsNiche</code> ); the last statement: <code>theorem1_5_1</code> ( <code>Intro/RotationAngleBound</code> ) |

| Result                                           | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (4.1), (4.2), (4.3),<br>(4.4),<br>Definition 4.1 | dyadic_totalWeight, dyadic_totalWeight_le_one;<br>persistent_sample_first, persistent_sample_second;<br>SupportSamples.penalty_change_uniform, with<br>abs_square_increment_le;<br>SupportSamples.penalty_change_one; IsPenalizedMax, with<br>$\Pi_k$ given by dyadicPenalty, dyadicSamples,<br>dyadicLevelWeight (Selection) and $\Theta_k$ by dyadicAngleSet<br>(Balanced/BalancedMaximumSofa) |
| Lemma 4.2                                        | SupportSamples.penalty_recovery_zero,<br>SupportSamples.recovery_objective_ge,<br>dyadic_recovery_ge (Selection)                                                                                                                                                                                                                                                                                 |
| Lemma 4.3                                        | polygon_subset_sampleBox, sampleBoxRadius (Selection)                                                                                                                                                                                                                                                                                                                                            |
| Lemma 4.4                                        | niche_subset_box, polyNiche_subset_box (Selection)                                                                                                                                                                                                                                                                                                                                               |
| Proposition 4.5                                  | existence: exists_dyadic_penalizedMax,<br>exists_penalizedMax, penalty_and_area_le_of_positive<br>(Selection), mpc_exists_limit_polycap,<br>mpc_le_limit_objective (Balanced/MaxPolygonCapExists);<br>the lower bound: selected_objective_ge; the box:<br>selected_sequence_bounded (Selection)                                                                                                  |
| Lemma 4.6                                        | dyadic_objective_limsup (Selection)                                                                                                                                                                                                                                                                                                                                                              |
| Lemma 4.7                                        | caps_eq_of_dyadic_supports, eqOn_of_dyadic_eq,<br>caps_eq_of_upper_supports (Selection), with<br>nef_eq_setOf_supp (Balanced/PolygonCap)                                                                                                                                                                                                                                                         |
| Proposition 4.8                                  | exists_selectedCapSequence, SelectedCapSequence,<br>penalty_tendsto_zero_of_objective (Selection); the limit<br>is a cap: mpc_limit_isCap (Balanced/BalancedMaximumSofa)                                                                                                                                                                                                                         |
| Lemma 5.1                                        | floatingCap, floatingCap_polygon, subset_floatingCap,<br>floatingCap_support_bounds, floatingCap_support_other,<br>floatingCap_support_self (Variation)                                                                                                                                                                                                                                          |
| Proposition 5.2                                  | floating_defect_le, floating_penalty_growth,<br>assigned_comparison_of_actual,<br>polygon_defect_le_penalty_growth (Variation)                                                                                                                                                                                                                                                                   |
| Lemma 5.3                                        | pinned_contract_mem, pinned_div_mem,<br>pinned_raw_support_bound (Variation)                                                                                                                                                                                                                                                                                                                     |
| Lemma 5.4                                        | translated_strip_support, pinned_translation_bound,<br>pinned_normalized_support_bound (Variation)                                                                                                                                                                                                                                                                                               |
| Proposition 5.5                                  | pinned_defect_le, polygon_defect_le_of_zero_facet<br>(Variation)                                                                                                                                                                                                                                                                                                                                 |
| Corollary 5.6                                    | abs_selected_defect_le, polygon_weighted_defect_zero,<br>selectorDefectBound, selectorDefectBound_sum_le,<br>abs_weighted_defect_le (Variation)                                                                                                                                                                                                                                                  |
| Lemma 6.1                                        | ang_wedgeGapWInf_le_tau, ang_wedgeGapZInf_le_tau<br>(Angle/HorizontalSide)                                                                                                                                                                                                                                                                                                                       |
| Lemma 6.2                                        | lemma4_1_1, ang_abs_wedgeGapZInf_sub_le<br>(Angle/HorizontalSide)                                                                                                                                                                                                                                                                                                                                |
| Lemma 6.3                                        | ang_le_sigmaAt_of_tendsto (Angle/HorizontalSide)                                                                                                                                                                                                                                                                                                                                                 |
| Proposition 6.4                                  | pinned_bounds_of_maximal_positive (Variation), with<br>abs_supp_le_box (Basic/ConvexBody)                                                                                                                                                                                                                                                                                                        |

| Result                    | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 6.6                 | (a) <code>ang_three_points_mem</code> , (b) <code>ang_consumed_of_supp_zero</code> ,<br>(c) <code>ang_mirror_mem_qMinus</code> ( <code>Angle/RightAngle</code> ); (d)<br><code>consumed_of_pinned</code> ( <code>AngleExtension</code> ), with<br><code>ang_wedgeGapWInf_mirror</code> , <code>ang_sigmaAt_mirror</code><br>( <code>Angle/HorizontalSide</code> )                                                                                                                                                                                                                                       |
| Lemma 6.7                 | (a) <code>ang_mem_niche_of_small</code> , (b) <code>ang_width_le_one</code><br>( <code>Angle/RightAngle</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Lemma 6.8                 | <code>right_angle_motion_of_width</code> ( <code>Angle/RightAngle</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Proposition 6.9,<br>(6.1) | <code>maximal_monotone_has_right_angle</code> ( <code>Main</code> ),<br><code>right_angle_motion_of_pinned_bounds</code> ( <code>AngleExtension</code> );<br>the monotone sofa $U$ : <code>maximal_envelope</code> , <code>own_cap_isMax</code><br>( <code>Main</code> ); (6.1): <code>maximizer_contained_in_gerver</code> ( <code>Main</code> )                                                                                                                                                                                                                                                       |
| Lemma 7.1                 | <code>inj_param_minus</code> , <code>inj_param_plus</code> ,<br><code>inj_param_minus_length</code> , <code>inj_param_plus_length</code> ,<br><code>inj_polygon_sigmaAt_eq</code> ( <code>Injectivity/DiscreteIneq</code> )                                                                                                                                                                                                                                                                                                                                                                             |
| Lemma 7.2                 | <code>polygon_tau_le_geom</code> ( <code>Curvature</code> ), with<br><code>inj_not_mem_qMinus</code> ( <code>Injectivity/DiscreteIneq</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Lemma 7.3                 | <code>polygon_curvature_with_defect</code> ( <code>Curvature</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Lemma 7.4                 | <code>polygon_arm_cell</code> ( <code>Curvature</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Lemma 7.5                 | <code>polygon_step_with_error</code> , <code>polygon_Ico_with_errors</code> ,<br><code>polygon_Ioo_with_errors</code> , <code>polygonGridError</code> ( <code>Curvature</code> )                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Lemma 7.6                 | <code>curvature_Ioo_limit</code> , <code>k0_integral_tendsto</code> ( <code>Curvature</code> ),<br>with <code>lemma6_4_2</code> , <code>inj_k0_lipschitz</code> ,<br><code>inj_dot_vplus_le_vint</code> , <code>inj_vint_le_dot_vminus</code> ,<br><code>inj_tendsto_vint</code> , <code>inj_tendsto_vint_right_dot</code> ,<br><code>inj_tendsto_vint_left_dot</code> ( <code>Injectivity/LimitIneq</code> )                                                                                                                                                                                           |
| Lemma 7.7                 | <code>firstCurvature_of_polygon_errors</code> ,<br><code>firstCurvature_of_Ioo</code> ( <code>Curvature</code> ),<br><code>inj_measure_le_of_Ioo</code> , <code>inj_intervalIntegrable_k0_gPlus</code><br>( <code>Injectivity/LimitIneq</code> )                                                                                                                                                                                                                                                                                                                                                        |
| Definition 7.8            | <code>FirstCurvatureBound</code> , <code>SecondCurvatureBound</code> ( <code>Curvature</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Proposition 7.9           | <code>curvature_of_maximal_positive</code> ,<br><code>firstCurvature_of_maximal_positive</code> , <code>diameter_le_box</code> ,<br><code>sampld_grid_error_le</code> ( <code>Curvature</code> ), <code>abs_supp_le_box</code><br>( <code>Basic/ConvexBody</code> ), <code>inj_gPlus_le_width</code> , <code>inj_k0_le</code><br>( <code>Injectivity/LimitIneq</code> ); the second bound:<br><code>secondCurvature_of_maximal_positive</code> ,<br><code>secondCurvature_of_mirror_first</code> , <code>sigma_eq_map_mirror</code> ,<br><code>gPlus_mirror_eq_fMinus</code> ( <code>Curvature</code> ) |
| Lemma 7.10                | <code>injCond1_of_curvature</code> , <code>first_arm_integral_lower</code> ,<br><code>second_arm_integral_lower</code> , <code>gPlus_sub_gPlus</code> ,<br><code>integral_fPlus</code> ( <code>Curvature</code> ); the product rule: <code>lemma5_1_3</code><br>( <code>Basic/LebesgueStieltjes</code> ), <code>inj_supp_sub_supp</code> ,<br><code>inj_product_coord</code> ( <code>Injectivity/ArmLengths</code> )                                                                                                                                                                                    |
| Lemma 7.11                | <code>lowerSeq_le_of_integral_bounds</code> ( <code>Curvature</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Proposition 7.12          | <code>injectivity_of_curvature</code> , <code>arms_strict_of_curvature</code><br>( <code>Curvature</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Corollary 7.13            | <code>isKi_of_maximal_area</code> , <code>sofaArea_pos_of_isMaxCap</code> ,<br><code>isMaxCap_of_area_eq</code> ( <code>Main</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Result           | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 8.1        | ki_maximizer_equality_conditions,<br>MamikonSegmentEquality,<br>mamikonSegmentEquality_of_isMax,<br>mamikonSegmentEquality_iff,<br>ki_upperQL_eq_gerver_of_sofaArea_eq,<br>ConvexDomain.eq_on_segment_of_isMax (Rigidity)                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (8.1), Lemma 8.2 | halfSquareIntegral_combo_gap, displacement_combo,<br>displacement_ae_eq_of_mamikon_eq,<br>displacement_eqOn_of_mamikon_eq,<br>displacement_continuousOn_arc (Rigidity), with<br>theorem7_1_2_vertices (Convex/ConvexDomain) and<br>theorem7_4_1 (Convex/Mamikon); (a)<br>tangentKernel_of_mamikon_eq,<br>tangent_displacement_formula,<br>tangentKernel_of_equation (Rigidity), with theorem8_3_1,<br>theorem8_3_2 (Optimality/Concavity); (b)<br>middleKernel_of_mamikon_eq,<br>outer_displacement_formula, integrated_middle_equation<br>(Rigidity), with opt_outerCorner_cbv<br>(Optimality/UpperBound) and inj_dot_outerCorner_uvec<br>(Injectivity/ArmLengths) |
| Proposition 8.3  | ki_supp_sub_gerver_eq (Main);<br>capKernel_of_triple_midpoint,<br>capKernel_of_mamikonS_eq, CapKernel,<br>CapKernel.eq_horizontal_translation (Rigidity)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Lemma 8.4        | (a) cap_eq_translate_of_upper_support,<br>mem_right_cap_iff, mem_cap_sub_horizontal_iff,<br>mem_right_niche_iff, mem_niche_sub_horizontal_iff,<br>sofa_eq_translate_of_upper_support (Rigidity); (b)<br>isCap_translate_horizontal, supp_translate_horizontal,<br>niche_translate_horizontal,<br>sofaArea_translate_horizontal (Rigidity)                                                                                                                                                                                                                                                                                                                           |
| Theorem 8.5      | sofaArea_eq_gerver_iff, cap_eq_gerver_translate,<br>translate_gerver_cap_sdiff_niche (Main); the maximizing<br>caps: isMaxCap_iff_translate_gerver_cap (Main); the<br>monotone sofas: right_angle_monotone_eq_gerver,<br>ki_sofa_eq_gerver_translate (Main)                                                                                                                                                                                                                                                                                                                                                                                                         |
| Lemma 9.1        | eq_of_subset_of_measure_eq,<br>eq_of_subset_of_null_sdiff,<br>interior_subset_of_null_sdiff (Rigid); its use in the proof<br>of Theorem 1.1: Rigid.recover, Rigid.isClosed_image,<br>Rigid.volume_image (Rigid)                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Lemma 9.2        | regularClosed_cap_sdiff_envelope, isCompact_envUnder,<br>outside_closed_envelope_subset,<br>mem_closed_of_positive_path (RegularClosed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Result          | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| Proposition 9.3 | <code>gerver_regularClosed, envelope_isCompact,</code><br><code>envelope_endpoint_order,</code><br><code>envelope_bounds_of_path_height, gerver_contactC_zero,</code><br><code>gerver_contactA_pi_div_two,</code><br><code>gerver_niche_eq_envUnderStrict (RegularClosed);</code> a cap<br>with rotation angle $\pi/2$ is closed downward above the floor:<br><code>opt_cap_down (Optimality/Domain)</code>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Corollary 9.4   | <code>isMaximal_iff_image_eq_gerver (Main)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Lemma 9.5       | (a) <code>gerver_corner_points, envUnderStrict_fst_mem_Ioo</code><br><code>(RegularClosed), GerverParams.gs_H_zero,</code><br><code>GerverParams.gs_H_pi, GerverParams.gs_K_bounds,</code><br><code>GerverParams.gs_InjCond3,</code><br><code>GerverParams.gs_antitoneOn_path_fst</code><br><code>(Gerver/StructureCap), GerverParams.gs_α_neg,</code><br><code>GerverParams.gs_β_pos (Gerver/Frame);</code> (b)<br><code>gerver_width_gt_one,</code> (c)<br><code>fst_eq_zero_of_gerver_width_le_one (RegularClosed)</code>                                                                                                                                                                                                                                                                                                             |
| Corollary 9.6   | (a) <code>translate_eq_gerver_of_volume_eq,</code><br><code>isMovingSofaWithAngle_pi_div_two_of_volume_eq,</code><br><code>snd_sub_le_one_of_isMovingSofa,</code><br><code>dot_sub_vvec_le_one_of_mem_image (Main),</code><br><code>Rigid.eq_translate_of_angle_eq_zero (Rigid);</code> (b)<br><code>sin_eq_zero_of_rot_gerver_mem_horizSide,</code><br><code>sin_eq_zero_of_isMovingSofa_rot_gerver,</code><br><code>not_isMovingSofa_rot_pi_div_two_gerver (Main)</code>                                                                                                                                                                                                                                                                                                                                                               |
| Fact 2.1, (2.1) | (a) <code>hasDerivWithinAt_supp_right,</code><br><code>hasDerivWithinAt_supp_left, tendsto_vplus_right,</code><br><code>tendsto_vminus_right, tendsto_vplus_left,</code><br><code>tendsto_vminus_left, tendsto_vint_right,</code><br><code>tendsto_vint_left (Basic/ConvexBody);</code> (b) <code>sigma_Ioc,</code><br><code>sigma_periodic, proposition2_1_2, vplus_sub_vplus,</code><br><code>theorem5_2_2 (Basic/SurfaceArea), theorem7_1_3</code><br><code>(Convex/ConvexDomain);</code> (c) <code>mpc_blaschke,</code><br><code>mpc_tendsto_area (Balanced/MaxPolygonCapExists);</code> (2.1):<br><code>leftLim_sigmaFun (Basic/SurfaceArea),</code><br><code>inj_sigma_Ioo_toReal (Injectivity/LimitIneq);</code> the<br>support function of a Minkowski combination: <code>supp_comb</code><br><code>(Convex/ConvexDomain)</code> |
| Definition 2.2  | <code>IsMovement, IsMovingSofaWithAngle, IsMovingSofa</code><br><code>(Sofa/Defs);</code> the definition of the statement of record:<br><code>Baek.IsMovingSofa (Challenge.lean),</code> and its equivalence<br>with the definition of the library: <code>Baek.isMovingSofa_iff_lib</code><br><code>(Solution.lean);</code> a moving sofa is bounded and compact:<br><code>isBounded_of_isMovingSofa, isCompact_of_isMovingSofa</code><br><code>(Monotone/SupportingHallway);</code> its translates:<br><code>mpc_isMovingSofaWithAngle_translate</code><br><code>(Balanced/BalancedMaximumSofa)</code>                                                                                                                                                                                                                                  |
| Fact 2.3        | (a) <code>theorem1_5_1 (Intro/RotationAngleBound);</code> (b)<br><code>proposition2_3_1_exists, proposition2_3_1_subset</code><br><code>(Monotone/MonotoneSofa)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Result          | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (2.2), Fact 2.4 | <p>the supporting hallways, their corners, walls and quarter-planes: <code>proposition2_2_2_hallway</code>, <code>proposition2_2_2_innerCorner</code>, <code>proposition2_2_2_outerCorner</code>, <code>proposition2_2_2_wallA</code>, <code>proposition2_2_2_wallB</code>, <code>proposition2_2_2_wallC</code>, <code>proposition2_2_2_wallD</code>, <code>proposition2_2_2_qPlus</code>, <code>proposition2_2_2_qMinus</code> (<code>Monotone/SupportingHallway</code>); (2.2): <code>IsCap.mem_iff</code> (<code>Monotone/CapContainsNiche</code>), and <math>K \subset P_\omega</math>: <code>IsCap.subset_para</code> (<code>Monotone/SupportingHallway</code>); (a) <code>theorem2_3_2</code>, <code>proposition2_3_3</code>, <code>IsMonotoneSofa.isMovingSofaWithAngle</code> (<code>Monotone/MonotoneSofa</code>); (b) <code>lemma2_3_5_supp</code> (<code>Monotone/MonotoneSofa</code>), <code>theorem2_4_1</code>, <code>theorem2_4_2</code> (<code>Monotone/CapNiche</code>); (c) <code>theorem2_4_3</code> (<code>Monotone/CapNiche</code>), <code>theorem2_5_9</code>, <code>theorem2_5_10</code> (<code>Monotone/CapContainsNiche</code>); (d) <code>proposition2_5_4_isCap</code>, <code>proposition2_5_4_supp</code>, <code>proposition2_5_4_vertices</code>, <code>proposition2_5_4_sets</code>, <code>proposition2_5_4_sigma</code>, <code>proposition2_5_4_hallway</code>, <code>cn_fan_mirror</code> (<code>Monotone/CapContainsNiche</code>), <code>sofaArea_mirror</code> (<code>Curvature</code>)</p> |
| Fact 2.7        | <p><code>theorem1_1_1</code> (<code>MovingSofaOptimality/Main.lean</code>), <code>area_le_gerver</code> (<code>Main</code>), <code>Baek.gerver_sofa_optimal</code> (stated in <code>Challenge.lean</code>, proved in <code>Solution.lean</code>); <math> G  &gt; 11/5</math>: <code>gerverSofa_area_mem</code> (<code>MovingSofaOptimality/Main.lean</code>); the bound for caps: <code>cap_area_le_gerver</code> (<code>Main</code>), with <code>theorem3_5_5</code>, <code>theorem3_5_6</code> (<code>Balanced/BalancedMaximumSofa</code>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Fact 2.8        | <p><code>proposition3_2_1</code>, <code>proposition3_2_1_fix</code>, <code>nef_supp_polyCap</code>, <code>proposition3_2_2</code>, <code>theorem3_2_3_le</code>, <code>theorem3_2_3</code> (<code>Balanced/PolygonCap</code>); a polygon cap is the intersection of its supporting half-planes: <code>nef_eq_setOf_supp</code> (<code>Balanced/PolygonCap</code>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Fact 2.9        | <code>lemma3_4_2</code> ( <code>Balanced/MaxPolygonCapExists</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Fact 2.10       | <p>the polyline <math>\mathbf{p}_K</math>: <code>theorem3_4_4</code> (<code>Balanced/Polyline</code>); (a) <code>lemma3_4_5_one</code> (<code>Balanced/Polyline</code>); (b) <code>lemma3_4_5_two</code> (<code>Balanced/Polyline</code>); (c) <code>mpc_walk</code>, <code>mpc_walk_total</code>, <code>mpc_sum_sigma_sin</code> (<code>Balanced/CapGeometry</code>), <code>mpc_tau_eq</code>, <code>mpc_sum_tau_sin</code> (<code>Balanced/Polyline</code>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Fact 2.11       | <p>the translates of polygon caps: <code>proposition3_3_1</code> (<code>Balanced/PolygonCap</code>); <math>C_\Theta(h) = K</math>, <math>\mathcal{N}_\Theta(h) = \mathcal{N}_\Theta(K)</math> and <math>\mathcal{A}_\Theta(h) = \mathcal{A}_\Theta(K)</math> for <math>h = h_K _{\Theta^\circ}</math>: <code>proposition3_3_4</code>, <code>proposition3_3_5</code> (<code>Balanced/PolygonCap</code>); (a) <code>lemma3_4_7</code> (<code>Balanced/MaximumPolygonCap</code>), <code>theorem3_1_2</code>, <code>IsSimpleNefPolygon</code> (<code>Balanced/NefPolygon</code>); (b) <code>lemma3_4_8</code> (<code>Balanced/MaximumPolygonCap</code>); (c) <code>proposition3_3_7</code>, <code>theorem3_3_6</code> (<code>Balanced/PolygonCap</code>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Definition 2.12 | <p><code>InjCond1</code>, <code>InjCond2</code>, <code>InjCond3</code>, <code>SatisfiesInjectivity</code> (<code>Injectivity/ArmLengths</code>); <math>\mathcal{K}^i</math>: <code>IsKi</code> (<code>Optimality/Domain</code>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Result    | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Fact 2.13 | <p>the arm lengths (before Definition 2.12): <code>proposition6_2_1</code> (<code>Injectivity/ArmLengths</code>), <code>inj_arm_nonneg</code> (<code>Injectivity/DiscreteIneq</code>), <code>inj_gPlus_sub_gMinus</code> (<code>Injectivity/LimitIneq</code>); (a) <code>theorem6_2_3_right</code>, <code>theorem6_2_3_left</code> (<code>Injectivity/ArmLengths</code>); (b) <code>theorem6_2_5</code>, <code>fPlus_sub_fPlus</code> (<code>Injectivity/ArmLengths</code>); (c) <code>proposition6_2_2</code> (<code>Injectivity/ArmLengths</code>); (d) <code>proposition6_4_5</code>, <code>proposition6_4_6_continuous</code>, <code>proposition6_4_6_deriv</code> (<code>Injectivity/LimitIneq</code>); its end values <math>f_K(0) = g_K(\pi/2) = 1</math>: <code>inj_fK_zero</code> (<code>Injectivity/BoundingArms</code>), <code>inj_cap_consecutive</code>, <code>inj_consecutive</code> (<code>Injectivity/DiscreteIneq</code>), <code>inj_fMinus_eq</code> (<code>Injectivity/ArmLengths</code>), <code>gK_end</code> (<code>Curvature</code>)</p>                                                                                                                                                                                                                                                                                                      |
| Fact 2.14 | <p><math>k_0</math>, <math>m_0</math> and <math>f_j</math>: <code>inj_k0_lipschitz</code> (<code>Injectivity/LimitIneq</code>), <code>inj_m0_mono</code>, <code>inj_m0_of_mem</code>, <code>inj_continuous_lowerSeq</code> (<code>Injectivity/BoundingArms</code>); (a) <code>lemma6_4_2</code> (<code>Injectivity/LimitIneq</code>); (b) <code>lemma6_5_5</code> (<code>Injectivity/BoundingArms</code>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Fact 2.15 | <p><math>\mathcal{J}</math> and <math>\mathcal{M}</math>: <code>curveArea</code>, <code>segArea</code> (<code>Convex/CurveArea</code>), <code>convexCurveArea</code> (<code>Convex/ConvexCurve</code>), <code>mamikon</code>, <code>theorem7_4_1</code> (<code>Convex/Mamikon</code>); <math>\mathcal{K}^i</math> and <math>\mathcal{T}</math> are closed under Minkowski combinations: <code>theorem8_1_1_convex</code>, <code>proposition8_1_2</code> (<code>Optimality/Domain</code>); (a) <code>theorem8_1_8</code>, with <code>lemma8_1_5</code> (<code>Optimality/Domain</code>), <code>theorem8_2_4</code> (<code>Optimality/UpperBound</code>); (b) <code>lemma8_3_3</code>, <code>lemma8_3_4</code>, <code>lemma8_3_7</code>, with <code>lemma8_3_5</code>, <code>lemma8_3_6</code>, and <code>theorem8_3_1</code>, <code>theorem8_3_2</code>, <code>theorem8_3_8</code> (<code>Optimality/Concavity</code>), <code>theorem7_1_2_supp</code>, <code>theorem7_1_2_vertices</code>, <code>theorem7_1_2_sigma</code> (<code>Convex/ConvexDomain</code>), <code>theorem7_4_1</code>, <code>theorem7_4_2</code> (<code>Convex/Mamikon</code>); (c) <code>corollary8_5_8</code> (<code>MovingSofaOptimality/Main.lean</code>), <code>theorem8_4_6</code>, with <code>theorem8_4_3_two</code>, <code>proposition8_4_4</code> (<code>Gerver/Properties</code>)</p> |

| Result    | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| Fact 2.16 | (a) <code>definition8_1_2_exists</code> , <code>definition8_1_2_unique</code> , <code>gerverSofa_area_mem</code> ( <code>MovingSofaOptimality/Main.lean</code> ), <code>Baek.gerver_params_exists</code> , <code>Baek.gerver_params_unique</code> , <code>Baek.gerver_sofa_area</code> (stated in <code>Challenge.lean</code> , proved in <code>Solution.lean</code> ); (b) <code>theorem8_4_1_monotone</code> , <code>theorem6_1_2</code> ( <code>Gerver/Properties</code> ), <code>theorem8_1_1_gerver</code> ( <code>MovingSofaOptimality/Main.lean</code> ), <code>GerverParams.gm_sofaArea_cap</code> , <code>GerverParams.gm_isKi</code> ( <code>Gerver/Properties</code> ); (c) <code>theorem8_4_1_niche</code> , <code>theorem8_4_1_tangents</code> ( <code>Gerver/Properties</code> ), <code>env_D1_strictMono</code> , <code>env_x1_strictAnti</code> , <code>env_B1_strictMono</code> , <code>env_niche_eq</code> , <code>env_niche_subset_strict</code> , <code>env_subset_niche</code> , <code>env_exists_curve_fst</code> ( <code>Gerver/Envelope</code> ), <code>GerverParams.gn_envHyp</code> , <code>GerverParams.gn_niche_eq</code> ( <code>Gerver/Niche</code> ), <code>gerver_niche_eq_envUnderStrict</code> ( <code>RegularClosed</code> ); (d) <code>GerverParams.gs_path_pi_div_two_snd</code> , <code>GerverParams.gs_beta_pi_div_two</code> ( <code>Gerver/Frame</code> ), <code>GerverParams.gs_C_pi_div_two</code> ( <code>Gerver/StructureCap</code> ); the contacts with the walls $b$ and $d$ described in Appendix A: <code>theorem8_4_1_walls</code> ( <code>Gerver/Properties</code> ) |
| Fact 2.17 | <code>GerverParams.path_snd_lt_one</code> ( <code>RegularClosed</code> ), <code>GerverParams.gb_x_pos</code> ( <code>Gerver/NicheBounds</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Fact 6.5  | (a) <code>lemma4_2_2</code> , (b) <code>lemma4_2_4</code> ( <code>Angle/RightAngle</code> ); $o_\omega - v_0 = cu_0$ and $o_\omega - u_\omega = cv_\omega$ : <code>proposition4_2_1</code> ( <code>Angle/RightAngle</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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